A Computationally Feasible Framework for Causal Probabilistic Explanation

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Abstract

Explaining why a specific outcome occurred—and which inputs deserve the blame or credit—is central to philosophical, scientific, and policy analysis. The available tools fall into two camps that do not meet. On one side, the Halpern–Pearl theory of *actual causality* gives principled answers to the question "did variable X cause outcome Y in this case?", but its verdicts are computed by enumerating counterfactual scenarios and so only work on small, toy-sized models. On the other side, attribution methods that do scale to modern machine-learning models—SHAP, LIME, and the like—assign importance scores without reference to the causal structure that generated the data, and so can give answers that conflict with what a careful causal analysis would conclude. We close this gap with Probabilistic Causal Impact (PCI).

PCI builds on actual causality and on Pearl's notions of probability of necessity and sufficiency, but recasts the question "which variables explain this outcome?" as an estimation problem on a probabilistic causal model that one can solve by Monte Carlo sampling. To compute a PCI score, the analyst specifies three ingredients: a distribution Γ over which subsets of variables to consider as candidate explanations, a distribution Δ over the counterfactual values to which those variables might be changed, and a function ci that measures how much such a change shifts the outcome. The result is a tractable, causally grounded explanation procedure that applies to large models. A correspondence theorem shows that, on the small models where actual causality is well-defined, PCI's verdicts agree with it.

We evaluate PCI in a sequence of increasingly demanding settings. On a synthetic structural causal model featuring redundant and preempting causes—the textbook stress tests for theories of causation—PCI recovers the verdicts that the actual-causality literature treats as correct, for both binary and continuous variables. A scaled-up benchmark with many binary candidate causes shows that PCI remains tractable in regimes where exhaustively enumerating actual causes becomes computationally infeasible. A dynamical SIR (epidemiological) benchmark with continuous outcomes and asymmetrically interacting causes extends the comparison to settings where actual causality offers no verdict at all. Finally, we demonstrate feasibility on a production causal spatio-temporal model with deep

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Preprint. Under review.

machine-learned components trained on millions of data points, where PCI and SHAP attributions diverge in the same qualitative ways as in the synthetic setting.

Keywords: Probabilistic Causal Models, Explanation, Responsibility, Actual Causality, Probability of Necessity and Sufficiency.

Contents

1 Introduction

Causal attribution is a central reasoning task in many important problems, such as gauging the efficacy of medical treatments [Shepherd et al., 2024], pinpointing drivers of economic growth [Cheshire and Magrini, 2009], root cause analysis in automated systems [Papageorgiou et al., 2022], or assessing the impact of public policy [Heckman and Pinto, 2022]. We may want to understand why an adverse outcome occurred so as to prevent recurrence, or how a predicted positive outcome could be brought about.

In the absence of scalable, formal causal attribution tools, analysts often resort to ad-hoc methods — strong correlation, historical comparison, domain expertise, or non-causal attribution like Granger causality [Granger, 1969] and SHAP [Lundberg and Lee, 2017]. Even researchers with an advanced command of statistical and experimental methods may not possess a thorough understanding of how to ascribe cause [Shepherd et al., 2024], and would benefit from scalable automation. Here we present a tool for accurate and tractable causal attribution applicable to a wide array of causal, probabilistic, machine-learned models.

Current methods for attributing causes to specific outcomes, such as actual causality [Halpern, 2016], are intractable in general, with no clear approximation [Eiter and Lukasiewicz, 2002]. Explanation methods that *do* scale are either non-causal [Lundberg and Lee, 2017] or amount to local, gradient-based sensitivity analyses [Butler et al., 2022]. SHAP [Lundberg and Lee, 2017] grounds attribution in observational correlations, so a feature can receive high attribution merely for being correlated with a true cause. Causal SHAP [Heskes et al., 2020] addresses this to some extent, but some failure modes persist (Section 4). Butler et al. [2022]'s Differential Causal Effect (DCE) relies on gradients, which requires model differentiability and does not capture effects from non-infinitesimal changes in potential causes.[6]

We introduce *probabilistic causal impacts* (PCI), an attribution framework that combines the context-sensitivity of actual causality with the scalability of probability-of-causation methods. PCI generalises Pearl's probability of necessary and sufficient causation [Pearl, 2022] from binary to arbitrary variable domains, and incorporates the *witness* mechanism from actual causality [Halpern, 2016] via integration over a distribution of witness sets rather than existential quantification.

PCI also shares an intuition with the literature on *descriptive normality* [Halpern and Hitchcock, 2015, Icard et al., 2017]: attributions are computed as expectations against a distribution over counterfactual worlds conditioned on upstream factual context, so factual upstream values play the role of a default against which alternatives are graded. We do not aim to resolve the specific counterexamples that motivate the normality literature; the connection is at the level of why a non-uniform alternative-value distribution matters, not in handling explicit normality orderings. Several components have precedents — expectations over exogenous noise [Halpern, 2016], weighting by alternative-value probabilities [Beckers and Vennekens, 2015], aggregation over witness sets [Beckers and Vennekens, 2015, Halpern, 2016] — but no prior construction combines them into a single tractable estimator handling arbitrary variable domains.

PCI presumes access to a trained probabilistic causal model from which counterfactual outcomes can be sampled — a stronger requirement than methods that work from observational data and a causal graph alone (e.g., PN/PS/PNS bounds, Causal SHAP). Within that requirement, however, we place no strong restrictions on the form of the model, distributions, or types of variables. Such models can comprise standard machine learning components such as neural networks or Gaussian processes.

With this breadth of application, and a measure that surfaces its design choices — the variable selection distribution, alternative value distribution, and causal impact function — as explicit user-facing components rather than hidden defaults, PCI hopes to provide causal attribution and explanation tooling for a wide array of users and applications.

Paper structure.

- **Section 2** works through a running example that surfaces what actual causality and the probability of necessity/sufficiency each get right and miss, and motivates the design choices behind PCI.

[6] DCE is, fairly, not designed for responsibility attribution: Butler et al. target rates of change of treatment.

- **Section 3** defines PCI: a kernel built from a sufficiency check at the factual configuration and a necessity check against alternatives, integrated against a distribution over suspect/witness sets. Three components are exposed as user choices: the variable selection distribution Γ, the alternative-value distribution Δ, and the causal impact function.
- **Section 4** compares PCI with SHAP and Causal SHAP on two small examples — OBCB (two binary features) and Signal (with mediation) — and tabulates which desiderata each method satisfies or fails, including cases where SHAP and Causal SHAP attributions track correlation rather than counterfactual dependence.
- **Section 5** reports a synthetic evaluation built around overdetermination and undercutting, the patterns on which pre-AC accounts most often diverge from intuition; PCI recovers the literature's verdicts on these cases.
- **Section 6** contrasts PCI with gradient-based attribution, instantiated by Differential Causal Effect [Butler et al., 2022], on a continuous credit-limit example, showing where the two diverge and why PCI better tracks responsibility under unit rescaling and outside the steep region of the response surface.
- **Section 7** shows that under specific choices of Γ and Δ the PCI kernel is positive on a configuration exactly when that configuration is an actual cause in Halpern's sense, so AC verdicts can be read off the kernel without enumerating witnesses.
- **Section 9** compares PCI against explicit actual-cause computation on benchmark models, reporting runtime, search-space size, and verdict recovery as model size grows. Exact subset enumeration times out at 17 variables, while the necessity-only PCI estimator continues to roughly 73–145 variables (depending on sample budget) and sustains above 0.85 correct-attribution rate up to about 60 variables, visiting several orders of magnitude fewer cause–witness configurations.
- **Section 10** extends the benchmarking to a continuous-outcome dynamical-SIR (Susceptible-Infected-Recovered) setting, exercising the joint necessity-sufficiency machinery on threshold events (peak overshoot) where AC has no comparable verdict. Where but-for analysis cannot separate the two policies, lockdown and masking, in their responsibility for a high overshoot level, PCI correctly assigns lockdown roughly twice the responsibility of masking, with the gap driven entirely by the necessity term under witness pinning.
- **Section 11** applies PCI to a deployed automated valuation model with machine-learned components trained on millions of points, a regime where actual causality is intractable; SHAP returns qualitatively different attributions, assigning virtually all responsibility to downstream variables, effectively ignoring the other features.

2 Causal Impact: motivations

The central difficulty in causal attribution is not computing counterfactuals — it is knowing which counterfactuals to compute, and recognising when a cause is blocked by context rather than absent altogether. The closest predecessors of this approach, taken individually, fail to capture this: actual causality handles context sensitivity but is computationally intractable and not general enough; the probability of necessity and sufficiency scales well, but lacks context sensitivity. This section builds intuition for both limitations through a running example, setting up the design choices that motivate PCI in Section 3.

2.1 Actual Causality

Consider the following (deterministic, discrete) model:

Example 1 (Bob at Old Boys' Club Bank)**.** *Bob applied for a home mortgage loan with the local branch of Old Boys' Club Bank (OBCB), and his application was denied. The OBCB policy requires applicants to pass a credit score check, but it only performs this check if the applicant is male; all* *non-male applicants are rejected outright. Thus, the procedure is:*

$$check = (gender = male)$$
$$check\text{-}failed = check \wedge (credit = bad)$$
$$loan = \neg check\text{-}failed \wedge (gender = male)$$

Why was Bob's loan application denied? An intuitive explanation is that his credit was bad—had it been good, the outcome would have been different. In contrast, his gender does not appear to be responsible for the rejection: had it been different, his application still would have been denied (albeit this time due to $check = F$).

This suggests a counterfactual approach to feature responsibility attribution: perhaps we can identify the features responsible for a particular outcome by asking counterfactual questions—would the outcome have been different if a specific feature had a different value? This is called the *but-for clause*: Bob wouldn't have been denied the loan but for his bad credit.

Concretely, consider a model M with *exogenous variables* $\mathbf{U}$, which are the source of stochasticity, and *endogenous variables* $\mathbf{V}$, which are influenced by other variables in the model. Let $\mathbf{S} \subseteq \mathbf{V}$ be the variables we suspect to be causally responsible (call them *causal suspects*), and $Y \in \mathbf{V}$ be an outcome of interest. Suppose we consider only one suspect $S \in \mathbf{S}$ and mark factual values with *. Under a setting $\mathbf{U} = \mathbf{u}$ of the exogenous variables, M produces $S = s^\star$ and $Y = y^\star$, denoted $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$. We call this the *factual* scenario. An intervened model where a variable S is set to an alternative value s' is denoted $(M, \mathbf{u})[S \leftarrow s']$.

A first attempt at defining a counterfactual notion of causal responsibility is:

Definition 2 (But-for, single)**.** *$S = s^\star$ is causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$ and $(M, \mathbf{u})[S \leftarrow s'] \not\models Y = y^\star$ for some $s' \neq s^\star$.*

In a given context, S is causally responsible for an outcome only when intervening on S with some alternative value would change that outcome. Bob's credit score satisfies this condition: had his credit score been higher, the loan might have been approved.

However, this cause test is too coarse. Consider another case:

Example 3 (Alice at OBCB)**.** *Alice is a woman with bad credit. She applies for a loan with OBCB and is rejected.*

Why was Alice denied the loan? Under the but-for clause for single cause candidates, neither her gender nor her credit score are responsible. No matter how high her credit score, her application would still have been denied because the bank does not loan to women. Had her gender been different, she would still have been denied for having bad credit.

To address this example, we could consider sets of factors as causes:

Attempt 1 (But-for, sets)**.** *Variables $\mathbf{C} \subseteq \mathbf{S}$ with factual values $\mathbf{C} = \mathbf{c}^\star$ are causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star, Y = y^\star$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}'] \not\models Y = y^\star$ for some $\mathbf{c}'$ everywhere different from $\mathbf{c}^\star$, where no proper subset of $\mathbf{C}$ has this property.*

The minimality condition reflects our preference for explanations that include only relevant properties.

Under this generalization, the set $\mathbf{c}^\star = \{gender = \text{female}, credit = \text{bad}\}$ is jointly responsible for Alice's application being denied because there exists $\mathbf{c}' = \{gender = \text{male}, credit = \text{good}\}$ that would change the decision and $\mathbf{c}^\star$ is minimal. But what can we say about the individual features? We could say that the whole set bears responsibility but individual features do not, or that any feature in a cause set is partially and equally accountable. Both approaches have issues, as they imply the features are equally responsible. This conflicts with our intuition: OBCB did not evaluate Alice's credit score, so her denial was caused by her gender.[7]

[7]Examples of this sort, involving both over-determination and undercutting of causal impact, abound. Suppose Jim is both stabbed and poisoned and dies as a result. Because stabbing works quickly, it kills Jim before the poison takes effect. In this case, it's the stabbing that caused his death. Yet analogous but-for clauses fail to break the symmetry here as well.

Prior work suggests a way out of this difficulty: hold some of the context in which counterfactuals are evaluated fixed, calling such fixed variables *witnesses*. Let us focus on the mediator $\textit{check-failed}$ between $\textit{gender}$ and $\textit{loan}$. If we fix this variable at the factual value by intervening, then counterfactually, Alice would have been granted the loan if she were male. However, there is no context fixed at the actual value such that, had her credit score been higher, she would have been granted the loan. This motivates the following formalization:

Definition 4 (Actual cause)**.** *Let $\mathbf{C} \subseteq \mathbf{S}$ be a cause set, $\mathbf{W} \subseteq \mathbf{V}$ be a set of possible witnesses. $\mathbf{C} = \mathbf{c}^\star$ is causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ just in case there is a $\mathbf{T}$ (an actual witness set to be held fixed) s.t. $\mathbf{T} \subseteq \mathbf{W}, \mathbf{T} \cap \mathbf{C} = \emptyset$ and $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star, Y = y^\star$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^\star] \not\models Y = y^\star$, for some $\mathbf{c}'$ distinct everywhere from $\mathbf{c}^\star$, where no proper subset of $\mathbf{C}$ satisfies this property.*

This definition extends but-for causality with a witness set $\mathbf{T}$ that is fixed to its factual value by an intervention. Under this definition, to determine the cause of an outcome, we need to find a cause set $\mathbf{C}$, alternative values $\mathbf{c}'$ (trivial if the variables are binary and less so if they are categorical or continuous), and a witness set $\mathbf{T}$. $\mathbf{C}$ and $\mathbf{T}$ are drawn from sets of size at most $2^{|\mathbf{V}|}$, so a brute-force computation is expensive.

Putting computational difficulties aside, we have so far considered deterministic models, but would like to capture probabilistic models as well, so we now consider a stochastic variant of the OBCB problem.

2.2 Probability of Necessity and Sufficiency

Example 5 (OBCB, stochastic)**.** *The bank performs credit checks for a randomly selected 20% of women and skips checks for a randomly selected 10% of men. Unchecked applicants are rejected. Furthermore, if a man's credit check fails, he is nevertheless accepted 5% of the time and if a woman's credit check succeeds, she is denied 10% of the time. We assume that the population is evenly divided into male and female and good and bad credit. Formally, the model, using 1 for male and 1 for good credit, is:*

$$\textit{loan-prob} = \begin{array}{r|cc} & \textit{check-failed}{=}0 & \textit{check-failed}{=}1 \\ \hline \textit{gender}{=}0 & 0.9 & 0 \\ \textit{gender}{=}1 & 1 & 0.05 \end{array}$$

$$\begin{aligned}
\textit{credit} &\sim \mathrm{Bern}(0.5) \\
\textit{gender} &\sim \mathrm{Bern}(0.5) \\
\textit{check} &\sim \mathrm{Bern}((1 - \textit{gender}) \cdot 0.2 + \textit{gender} \cdot 0.9) \\
\textit{check-failed} &= \textit{check} \cdot (1 - \textit{credit}) \\
\textit{loan-if-checked} &\sim \mathrm{Bern}(\textit{loan-prob}[\textit{gender}, \textit{check-failed}]) \\
\textit{loan} &= \textit{loan-if-checked} \cdot \textit{check}
\end{aligned}$$

The witness mechanism works well for deterministic models, but actual causality in its standard form offers no straightforward way to handle probabilistic ones. Let's shelve that problem for now and instead ask: what would a probabilistic approach to causal attribution look like, and what would it miss without witnesses?

Before extending the witness idea to the stochastic setting, it is instructive to examine a probabilistic approach that scales more naturally: Pearl's probabilities of necessity and sufficiency [Pearl, 2022]. This will let us identify what probabilistic machinery is needed, and what context-sensitivity is lost when witnesses are absent — motivating the synthesis that PCI provides. The following definitions allow for probabilistic models, but assume that all variables are Boolean (PCI will relax this assumption, allowing for continuous variables). Pearl considers two ways that a cause may relate to an effect: it may be necessary or sufficient for the effect. A cause is necessary if, counterfactually absent the cause, the effect is unlikely to occur; it is sufficient if the effect is counterfactually likely in the presence of the cause. We now formalize these ideas and discuss them in the context of the running example.

Definition 6 (Probability of Necessity)**.** *Let the probability of necessity be the probability that the outcome $Y = y$ would not have occurred in the absence of $C = c^\star$, given that both occurred:*

$$PN(C = c^\star, Y = y^\star) = P(Y_{c'} \neq y^\star \mid C = c^\star, Y = y^\star)\,,$$

where $Y_{c'}$ is the value of Y after intervening on C with $c' \neq c^\star$.[8]

Strictly speaking, the probability of necessity is less specific to a particular situation than the actual cause test discussed above. For example, we can consider the probability that being male causes someone to fail to get a loan, but we can't consider the probability that being male causes Bob specifically (who we know has bad credit) to fail to get a loan. We might try to patch this up by conditioning, for the time being (while also reporting population-level values that follow the definitions more closely).

We illustrate the computation for Alice's gender. Alice is factually $gender =$ female, $credit =$ bad, $loan = F$; the counterfactual intervenes to set $gender =$ male while leaving $credit =$ bad. Tracing through the model:

- Under the intervention, $check \sim \mathrm{Bern}(0.9)$ — men are checked 90% of the time.
- If checked, $check\text{-}failed = 1$ (Alice's credit is bad), and from the loan-probability table $P(loan\text{-}if\text{-}checked = 1 \mid gender = \text{male}, check\text{-}failed = 1) = 0.05$.
- If not checked, $loan = 0$ deterministically since $loan = loan\text{-}if\text{-}checked \cdot check$.

Combining the two branches:

$$\begin{aligned}&PN(gender = \text{female}, loan = F \mid \text{Alice})\\ &\qquad = P(loan_{gender=\text{male}} = T \mid gender = \text{female}, credit = \text{bad})\\ &\qquad\qquad = 0.9 \cdot 0.05 = 0.045.\end{aligned}$$

Computing the analogous quantities for the remaining (variable, person) pairs gives the individual-level values shown in Table 1; the table also lists population-level values for comparison.

The PN scores partially align with intuition: Bob's bad credit is high ($PN = 0.90$), and his gender score is zero, matching the reading that his gender played no role. But Alice's PN values do not separate gender (0.045) from credit (0.18) particularly well, and Alice's gender PN is only marginally above Bob's — not the sharp distinction we would want, given that the bank never even evaluated Alice's credit. PN cannot register this asymmetry: it asks counterfactually what would happen under intervention without any context about which causal paths were active. For the full picture we need a second quantity, the probability of sufficiency.

Definition 7 (Probability of Sufficiency)**.** *Let the probability of sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$, given that neither $C = c^\star$ nor $Y = y^\star$ occurred:*

$$PS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star \mid C = c',\; Y \neq y^\star)$$

where $c' \neq c^\star$ and $Y_{c^\star}$ is the value of Y after intervening on C with $c^\star$.

To query whether the factual cause was sufficient, we condition on the counterfactual: we assume the cause did *not* take its factual value and the outcome did *not* occur, then intervene to restore the factual cause value and ask whether the outcome now occurs. Intuitively, this asks: if the cause had been absent and things had gone differently, would reinstating the cause have brought the outcome about? Let's look at the resulting values.

We illustrate PS for Alice's gender. Following the same convention as for PN, the individual-level computation additionally conditions on Alice's credit ($credit =$ bad). The PS conditional then has Alice's credit fixed but the cause and outcome flipped to their non-factual values ($gender =$ male, $loan = T$); we intervene to restore $gender =$ female and ask whether the loan is again denied:

- With $gender =$ female and $credit =$ bad, $check \sim \mathrm{Bern}(0.2)$.

[8] Read in twin-network fashion: one draws an exogenous noise $\mathbf{u} \sim P_{\mathbf{U}}$, computes the factual Y under the unintervened structural equations and the counterfactual $Y_{c'}$ under the intervened equations, and conditions on the joint event $\{C = c^\star, Y = y^\star\}$. Both Y and $Y_{c'}$ live on the same probability space and share the same $\mathbf{u}$.

- If checked, $\textit{check-failed} = 1$, and from the loan-probability table $P(\textit{loan-if-checked} = 1 \mid \textit{gender} = \text{female}, \textit{check-failed} = 1) = 0$.
- If not checked, $\textit{loan} = 0$ deterministically.

Either branch gives $\textit{loan} = F$, so

$$\begin{aligned} &PS(\textit{gender} = \text{female}, \textit{loan} = F \mid \text{Alice}) \\ &\qquad = P(\textit{loan}_{\textit{gender}=\text{female}} = F \mid \textit{gender} = \text{male}, \textit{loan} = T, \textit{credit} = \text{bad}) = 1. \end{aligned}$$

The remaining individual values, along with population-level values for comparison, appear in Table 1. Two cases warrant comment: Bob's gender PS is undefined because the required conditioning event ($\textit{gender} = \text{female}$, $\textit{credit} = \text{bad}$, $\textit{loan} = T$) has probability zero in the model; Bob's credit PS is 0.955 rather than 1 because the model includes a 5% exception rate for males who fail the credit check.

To somehow combine insights provided by the two scores, we would like to find causes that are both necessary and sufficient. This notion is formalized as follows:

Definition 8 (Probability of necessity and sufficiency)**.** *Let the probability of necessity and sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$ and that it would not have occurred under the intervention $C = c'$:*

$$PNS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star)$$

The population- and individual-level PNS values appear in the rightmost column group of Table 1. These values capture the relative causal impact of $\textit{gender}$ and $\textit{credit}$ but are not entirely satisfying. Table 1 reveals the problem directly: for Alice, $PNS(\textit{gender}) = 0.045 < PNS(\textit{credit}) = 0.18$, so PNS ranks credit as more causally responsible than gender. Yet the bank never evaluated Alice's credit—the $\textit{check}$ variable was F throughout. PNS has no mechanism to register this: it asks counterfactually what would happen if credit were different, without fixing $\textit{check-failed}$, the mediating variable that blocked credit's causal path.

As an attempt to help Alice better understand her particular situation, we might consider a modification of PNS with additional conditioning:

$$PNS_c(C = c^\star, Y = y^\star \mid F = f^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star \mid F = f^\star)$$

Using this definition, we can compute $PNS_c(\textit{gender} = \text{female}, \textit{loan} = F \mid \textit{credit} = \text{bad}) = 0.045$ and $PNS_c(\textit{credit} = \text{bad}, \textit{loan} = F \mid \textit{gender} = \text{female}) = 0.18$. Conditioning does not solve the problem described above—Alice's credit was in reality never evaluated—because the interventions override the mediating variable that tracks whether her credit has been checked. When we compute $PNS_c(\textit{credit} = \text{bad}, \textit{loan} = F \mid \textit{gender} = \text{female})$, conditioning on gender being female is consistent with the factual—but when we then intervene to set credit to good, this intervention propagates downstream and changes $\textit{check-failed}$, overriding what we learned from conditioning on Alice's factual outcome. The context we fixed is undone by the very intervention meant to probe it. We will not consider this use of conditioning further, and this observation motivates the need for intervention-based (as opposed to conditioning-based) context-sensitivity, which is what the witness mechanism provides.

2.3 Path forward

What should we conclude from this investigation of actual causality and the probability of necessity and sufficiency? First, we need a way to treat probabilistic models, and the probability of necessity and sufficiency gives us a tool for doing that. However, PNS is not as specific as we would like because it does not capture relevant features of individual units that are not part of the cause. Actual causality offers more specificity, but requires modification to deal with probability and continuous variables, and is computationally expensive to compute.

PCI addresses both limitations by combining the two approaches. To see the intuition, consider Alice's case once more. When PCI evaluates gender's causal impact on her denial, it considers counterfactual worlds in which her gender is changed, while drawing witness sets from a distribution that includes $\textit{check-failed}$ at its factual value of F — the credit check was never performed. In those worlds, changing gender to male, with $\textit{check-failed}$ held fixed at F (by intervention, so unlike

Table 1: PN, PS, and PNS for Alice and Bob in Example 5, with $loan = F$ as the outcome. "Population" values condition only on the cause–outcome pair prescribed by the definition; "Individual" values additionally condition on the subject's other factual variable (e.g. for Alice's gender, on $credit$ = bad). A dash marks an entry that is undefined because the conditioning premises are inconsistent with the model.

		Alice			Bob		
Variable	Level	PN	PS	PNS	PN	PS	PNS
$gender$	Population	0.43	0.83	0.39	0.02	0.10	0.01
$gender$	Individual	0.045	1.00	0.045	0	—	0
$credit$	Population	0.53	0.96	0.52	0.53	0.96	0.52
$credit$	Individual	0.18	1.00	0.18	0.90	0.955	0.86

in the case of patched-up PNS, the mediator value does not get overridden), leads to loan approval: a large counterfactual shift, and so high attribution. When PCI evaluates credit's causal impact, it searches for witness sets that, held fixed, would make a change to good credit matter for Alice — but no such context exists, because the credit check was never run. The distribution over witness sets therefore assigns low weight to any world in which credit is relevant for Alice, and her credit receives low attribution. Unlike actual causality, PCI does not ask whether *some* witness set validates a counterfactual dependency; it integrates over the distribution of all witness sets, weighting each by its probability. Unlike PNS, it inherits the context-sensitivity of actual causality through this integration. The formal definitions in Section 3 make this expectation precise — but the intuition is the one we have just seen: PCI inherits the witness mechanism of actual causality and the probabilistic language of PN/PNS, and generalizes and combines them into a tractable estimation problem even when variables are continuous.

Before moving to the formal development, we flag the closest neighbour of PCI in the existing literature, so that the comparison does not come out of nowhere later. Pearl [Pearl, 2009, Ch. 10] defines the *probability of actual causation* as

$$P(\text{caused}(x, y \mid e)) \;=\; \frac{P(U_{xy} \cap U_e)}{P(U_e)} \qquad \text{(Def. 10.3.5)},$$

where U_{xy} is the set of noise states $\mathbf{u}$ in which the chosen actual-causation predicate "$X{=}x$ is an actual cause of $Y{=}y$" holds, and U_e is the set of states compatible with the observed evidence e. The actual-causality predicate plugged in is Pearl's causal-beam definition [Pearl, 2009, Ch. 10] or the Halpern–Pearl AC1–AC3 [Halpern, 2016]. The formula is Bayes' rule applied to actual causation: each $\mathbf{u}$ determines a deterministic world through the structural equations, so "is $X{=}x$ an actual cause of $Y{=}y$?" has a yes/no answer once $\mathbf{u}$ is fixed; U_{xy} collects the yes-states. The ratio rewrites cleanly to $P(\mathbf{u} \in U_{xy} \mid \mathbf{u} \in U_e)$, i.e. the posterior probability, given the evidence, that we are in a noise state for which the actual-cause verdict holds. This is Pearl's bridge from actual causality to *probability of* actual causality. As PCI proposes a seemingly similar bridge, and as Pearl's notion gives the same rankings in typical simple cases, we devote Section 8 to a side-by-side comparison once enough technical machinery is on the page to make the discussion precise.

3 Causal Impact: definitions

Explanation and causal attribution seek to connect a potentially causal *event* $X = x$ to an outcome event $Y = y$.[9] In this section, we present a formal development. To ground it, we return to Example 5 and use Alice and Bob as running illustrations throughout this section. Both are rejected for a loan; the question is which of their attributes—$gender$ and $credit$—was causally responsible for each rejection, and to what degree. While the example is discrete, the machinery applies to continuous and mixed-type variables; a continuous illustration is given in Sections 4.3 and 5.

Let $R(V \rightsquigarrow loan \mid \text{person})$ denote the causal responsibility of variable V for the loan outcome at the individual level (R is a placeholder for any attribution score and should not be identified

[9] This is quite different from type-level causal queries, where one is interested in some group-level attribution, as when one estimates average treatment effect or conditional average treatment effect.

with a specific formal quantity; PCI interprets R as $\mathbb{E}[ci(Y^s, Y^n, y^\star)]$, to be defined below, but the desiderata are method-agnostic). Attribution methods may produce signed values; the desiderata are stated in terms of absolute magnitudes $|R(\cdot)|$, allowing for directionality:

$$\begin{aligned}
&\textbf{D-A1} && |R(\textit{gender} \rightsquigarrow \textit{loan} \mid \text{Alice})| > 0 \\
&\textbf{D-A2} && |R(\textit{credit} \rightsquigarrow \textit{loan} \mid \text{Alice})| > 0 \\
&\textbf{D-A-rank} && |R(\textit{gender} \rightsquigarrow \textit{loan} \mid \text{Alice})| > |R(\textit{credit} \rightsquigarrow \textit{loan} \mid \text{Alice})| \\
&\textbf{D-B1} && |R(\textit{gender} \rightsquigarrow \textit{loan} \mid \text{Bob})| > 0 \\
&\textbf{D-B2} && |R(\textit{credit} \rightsquigarrow \textit{loan} \mid \text{Bob})| > 0 \\
&\textbf{D-B-rank} && |R(\textit{credit} \rightsquigarrow \textit{loan} \mid \text{Bob})| > |R(\textit{gender} \rightsquigarrow \textit{loan} \mid \text{Bob})| \\
&\textbf{D-comp} && |R(\textit{gender} \rightsquigarrow \textit{loan} \mid \text{Alice})| > |R(\textit{gender} \rightsquigarrow \textit{loan} \mid \text{Bob})|
\end{aligned}$$

D-A1 and **D-A2** both require non-zero attribution for Alice because two distinct causal routes were available: gender filtered her out at the checking stage without her credit ever being evaluated, while bad credit would have mattered had she been checked. Neither is causally inert. **D-A-rank** is the substantive constraint: because Alice never underwent a credit evaluation, gender's causal role is immediate, whereas credit's is purely hypothetical. A method that ranks credit above gender for Alice is misattributing the cause of her rejection.

For Bob, **D-B2** reflects that the credit check occurred and directly produced the rejection: the causal chain $\textit{credit} \to \textit{check-failed} \to \textit{loan}$ was activated. **D-B1** requires strictly positive attribution for $\textit{gender}$: being male gave Bob a 90% probability of being checked, so gender opened the evaluation that rejected him. This indirect causal role is non-zero; a method returning exactly zero fails to capture it. **D-B-rank** corresponds to the intuition that credit is the proximate cause. **D-comp** captures the cross-individual asymmetry that is hardest for naive methods to express: gender is more causally responsible for Alice's rejection than for Bob's, because for Alice $\textit{gender}$=female was sufficient on its own to produce the rejection — no evaluation took place, so credit was never a factor — while for Bob $\textit{gender}$=male merely opened the evaluation that $\textit{credit}$=bad then caused. We will verify at the end of this section that PCI satisfies all seven desiderata, while PN, PS, PNS fail some of them. Later, we will also revisit this example and its desiderata in the context of SHAP and Causal SHAP (Section 4).

We work within the context of probabilistic structural causal models [Pearl, 2009].

Definition 9 (Probabilistic Structural Causal Model (PSCM))**.** *A probabilistic structural causal model is a tuple*

$$M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle,$$

where $\mathbf{V} = \{V_1, \ldots, V_n\}$ *is a finite set of endogenous variables,* $\mathbf{U} = \{U_1, \ldots, U_n\}$ *is a set of exogenous variables, and* $\mathbf{F} = \{f_1, \ldots, f_n\}$ *is a set of structural assignments*

$$V_i := f_i(\mathbf{Pa}_i, U_i), \qquad i = 1, \ldots, n,$$

with $\mathbf{Pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ *denoting the parents of* V_i*. The distribution* $P_{\mathbf{U}}$ *is a probability measure on* $\operatorname{dom}(\mathbf{U})$*, and the assignments in* $\mathbf{F}$ *are assumed to be acyclic and to have a unique solution for* $\mathbf{V}$ *given* $\mathbf{U}$*.*[10]

A PSCM encodes which variables influence which others, through what mechanisms, and subject to what randomness. The structural equations specify each variable as a deterministic function of its causal parents and local noise; the exogenous distribution captures all irreducible uncertainty in the system. Together they make every counterfactual question well-posed and computable.

The OBCB model (Example 5) is a PSCM with endogenous variables

$$\mathbf{V} = \{\textit{gender}, \textit{credit}, \textit{check}, \textit{check-failed}, \textit{loan}\},$$

[10]Working within the PSCM framework commits the practitioner to fully specified structural assignments $\mathbf{F}$ and an exogenous distribution $P_{\mathbf{U}}$ — analogous to how Bayesian inference requires a likelihood and a prior. This is precisely what enables PCI to yield individual-level counterfactual attributions rather than population-level bounds, and once $\mathbf{F}$ and $P_{\mathbf{U}}$ are specified, sampling from $P_{\mathbf{U}}$ is a routine probabilistic inference operation.

and exogenous variables

$$\mathbf{U} = \{U_{gender}, U_{credit}, U_{check}, U_{loan}\}$$

which induce independent Bernoulli noise.

The structural assignments in the case at hand can be written as:

$$\begin{aligned} \textit{gender} &\sim \mathrm{Bernoulli}(0.5), \qquad \textit{credit} \sim \mathrm{Bernoulli}(0.5), \\ \textit{check} &\sim \mathrm{Bernoulli}\big(0.2(1-\textit{gender}) + 0.9\textit{gender}\big), \\ \textit{check-failed} &:= \textit{check}(1-\textit{credit}), \\ \textit{loan} &\sim \mathrm{Bernoulli}\big(\text{loan-prob}[\textit{gender}, \textit{check-failed}]\big) \cdot \textit{check}. \end{aligned}$$

Definition 10 (Interventions)**.** *Given a model $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, an intervention on variables $\mathbf{X} = \{X_1, \ldots, X_k\} \subseteq \mathbf{V}$ with values $\mathbf{x} = (x_1, \ldots, x_k)$ replaces the structural assignments for X_i by $X_i := x_i$, where $i = 1, \ldots, k$. We denote this intervention by either $[\mathbf{X} \leftarrow \mathbf{x}]$ or $do(\mathbf{X} = \mathbf{x})$.*

An intervention surgically replaces a variable's structural equation with a fixed constant, disconnecting it from its usual causes. The rest of the model propagates downstream as normal under the new assignment; the exogenous noise is unchanged.

In the OBCB model, $\mathrm{do}(\textit{gender} = 1)$ replaces Alice's gender assignment with male, leaving all other structural equations intact. The downstream effect propagates: $\textit{check}$ now draws from $\mathrm{Bern}(0.9)$ rather than $\mathrm{Bern}(0.2)$, so Alice would be checked in 90% of noise realizations instead of 20%. This is the counterfactual world in which we ask whether her loan outcome would have differed.

Because PCI supports arbitrary variable types, it will be helpful to define a measure theoretic object corresponding to interventional outcomes. Because outcomes are a deterministic function of exogenous noise, this will be a Dirac measure.

Definition 11 (Interventional Law)**.** *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and let $Y \in \mathbf{V}$. For an intervention $\mathrm{do}(\mathbf{X} = \mathbf{x})$, let $Y' : \mathrm{dom}\,(\mathbf{U}) \to \mathrm{dom}\,(Y)$ denote the measurable map induced by the modified structural assignments. The interventional law of Y given fixed $\mathbf{u} \in \mathrm{dom}\,(\mathbf{U})$ is the probability measure on $\mathrm{dom}\,(Y)$ defined by*

$$P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{X} = \mathbf{x})) \;:=\; \mathbb{I}\{Y'(\mathbf{u}) \in A\} \;=\; \delta_{Y'(\mathbf{u})}(A), \qquad A \subseteq \mathrm{dom}\,(Y).$$

The interventional law records the deterministic outcome of Y under $\mathrm{do}(\mathbf{X} = \mathbf{x})$ for a fixed noise realization $\mathbf{u}$: since the modified structural equations make Y a function of $\mathbf{u}$ alone, its distribution collapses to a point mass. Integrating over $P_{\mathbf{U}}$ then recovers the full interventional distribution.

We define a *causal set* to be a set of assignments to the variables of a probabilistic structural causal model.

Definition 12 (Causal set)**.** *Given a model $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, a causal set is a set of pairs $X = x$ where $X \in \mathbf{V}$ and $x \in dom(X)$.*

Informally, a causal set is a list of specific variable-value assignments — a precise statement of which variables are set to which values. It is the basic unit of intervention: from a causal set one can read off exactly what is being manipulated or held fixed.

In PCI, causal sets play four distinct roles — two on the cause side and two on the context side:

Suspects S. The full set of variables suspected to bear on the outcome. Subsets of $\mathbf{S}$ will be sampled and intervened on.

Active suspects $\mathbf{C} \subseteq \mathbf{S}$. The specific subset intervened on in a given search step — either to alternative values (necessity regime) or to factual values (sufficiency regime).

Witnesses W. The full set of context variables that may be held fixed at their factual values. Subsets of $\mathbf{W}$ will be sampled and held fixed.

Active witnesses $\mathbf{T} \subseteq \mathbf{W}$. The specific subset held fixed in a given search step. To avoid contradicting the active intervention, $\mathbf{T}$ is pruned so that $\mathbf{T} \cap \mathbf{C} = \emptyset$.

To evaluate the causal impact of a variable on an outcome, we consider two interventional regimes: the *necessity regime*, in which we intervene on the variables $\mathbf{C} \subseteq \mathbf{S}$ to have an alternative value, and the sufficiency regime, in which we intervene so that $\mathbf{C}$ takes on factual values. Moreover, we hold some elements $\mathbf{T} \subseteq \mathbf{W}$ of the context fixed (i.e., intervened on to retain factual values, despite intervention on $\mathbf{C}$). This is exactly what makes the necessity component *context-sensitive* in Halpern's sense [Halpern, 2016]: the alternative cause is evaluated against a held-fixed witness configuration, not in isolation. We will reuse this terminology throughout: "context-sensitive necessity" refers to the necessity regime evaluated under an active witness set $\mathbf{T}$. When looking for an explanation, we perform a search across possible cause sets and possible contexts. Complete model samples in these regimes will be called *necessity worlds* and *sufficiency worlds*, respectively.

The intuition is that the suspect set $\mathbf{S}$ names every variable that could plausibly have caused the outcome; the witness set $\mathbf{W}$ names context variables to be selectively held fixed, isolating the causal channels under investigation. Intervening on suspects tests what would have happened under an alternative cause in the necessity worlds and under the factual values in the sufficiency worlds; fixing witnesses blocks confounding paths that would otherwise absorb the effect. Responsibility attribution increases when the outcome is far from the factual value in the necessity world and decreases when it is far from the factual value in the sufficiency world.

For both Alice (female, bad credit, loan denied) and Bob (male, bad credit, loan denied) we set $\mathbf{S} = \{gender, credit\}$ and $\mathbf{W} = \{check\text{-}failed\}$. The witness $check\text{-}failed$ records whether the applicant was checked and had bad credit; holding it at its factual value isolates the causal path from $gender$ or $credit$ to $loan$ independently of the credit-check bottleneck. Without this witness, intervening on $credit$ for Alice would leave 80% of noise realizations unaffected — those in which she is never checked — making credit's causal role invisible. The witness makes the bottleneck explicit and manipulable.

Moving on with the general framework, we will be searching for causes by intervening on subsets of suspects and holding fixed subsets of witnesses. To make this search tractable, we need a way to prioritize which subsets to consider.

Definition 13 (Variable Selection Distribution)**.** *Let* $\mathbf{X}$ *be a finite set of random variables. A* variable selection distribution Γ *is a probability distribution on* $2^{\mathbf{X}}$*. We write* $\mathbf{Y} \sim \Gamma$ *to mean* $\mathbf{Y}$ *is a random subset of* $\mathbf{X}$ *drawn from* Γ*, with probability mass function* $p^{\Gamma}(\mathbf{y}) = \mathbb{P}(\mathbf{Y} = \mathbf{y})$ *for* $\mathbf{y} \subseteq \mathbf{X}$*.*

The variable selection distribution Γ encodes prior beliefs about which subsets of suspects are worth examining. A uniform distribution treats all suspect subsets as equally plausible; a cardinality-constrained choice focuses the search on interactions of a particular order, analogous to choosing which interaction terms to include in a regression. More complex distributions could encode domain knowledge about which variables are more likely to interact, or could be learned from data. The choice of Γ is a design decision that shapes the search process and the resulting attributions. As a default, a uniform Γ over the full powerset is the least-informative starting point, assigning equal weight to all subset sizes, analogous to including all interaction orders in a regression. There is no universally principled criterion for restricting the cardinality range; the honest advice is to treat J and K as explicit design choices. Two practical considerations bear on this choice, however: with $K = 1$, attribution scores are based on singleton interventions and straightforward to communicate to stakeholders; higher K averages over joint interventions on X_k with other suspects simultaneously, which is harder to explain and, in large or structurally complex models, may amplify estimation noise rather than capturing genuine higher-order causal structure. For applications where attribution has serious consequences, we recommend a cardinality sensitivity analysis: if rankings are stable as K increases from 1 to $|\mathbf{S}|$, the simpler choice is supported; if they diverge, higher-order interactions are materially relevant and the full powerset should be used. The key advantage of PCI is that this commitment is made transparently rather than absorbed into an implicit default.[11]

In particular, we will be using variable selection distributions for the powerset of causal suspects $\mathbf{S}$ and for the powerset of witnesses $\mathbf{W}$. One simple pattern of suspect and witness selection can

[11] SHAP faces an analogous decision without flagging it as one: by averaging over all $2^{|N|}$ feature coalitions, plain SHAP and Causal SHAP commit to the maximum interaction order $K = |N|$ as an unstated default, with no diagnostic offered for whether that level of joint conditioning is right for a given model. Restricting Shapley to $K < |N|$ would break the efficiency axiom, so the choice is hardcoded into the framework rather than treated as a tunable. PCI surfaces the same decision as an explicit knob.

be derived by setting size limits on the number of activated suspects or witnesses, and sampling uniformly otherwise.

Example 14 (Cardinality-Constrained Uniform Selection)**.** *Let* $\mathbf{X}$ *be a candidate set of random variables and let* $k \sim \mathrm{Unif}\{J, \dots K\}$ *with* $1 \le J \le K \le |\mathbf{X}|$*. A variable selection* $\mathbf{Y} \sim \Gamma(2^{\mathbf{X}})$ *is sampled from a cardinality-constrained uniform selection distribution* $\Gamma(2^{\mathbf{X}})$ *with probability mass function*

$$p^{\Gamma}(\mathbf{Y}) \propto \mathbb{I}[|\mathbf{Y}| = k].$$

The parameters J *and* K *determine the cardinality range of the search; specific choices connect PCI to existing methods and enable efficient Monte Carlo approximation.*[12]

In PCI the variable selection distribution Γ is a probability mass function on $2^{\mathbf{S}} \times 2^{\mathbf{W}}$, governing the joint choice of an active suspect set $\mathbf{C} \subseteq \mathbf{S}$ and an active witness set $\mathbf{T} \subseteq \mathbf{W}$. The recommended construction samples suspects and witnesses separately and then enforces the constraint that no variable plays both roles in the same draw: pick a marginal Γ_s on $2^{\mathbf{S}}$ and a marginal Γ_w on $2^{\mathbf{W}}$, draw $\mathbf{C} \sim \Gamma_s$ and $\mathbf{T} \sim \Gamma_w$ independently, and *reject* any draw with $\mathbf{T} \cap \mathbf{C} \neq \emptyset$, resampling until the constraint holds. Equivalently,

$$p^{\Gamma}(\mathbf{C}, \mathbf{T}) \;=\; \frac{p^{\Gamma}_s(\mathbf{C})\, p^{\Gamma}_w(\mathbf{T})\, \mathbb{I}[\mathbf{T} \cap \mathbf{C} = \emptyset]}{Z}, \qquad Z \;=\; \sum_{(\mathbf{C},\mathbf{T})} p^{\Gamma}_s(\mathbf{C})\, p^{\Gamma}_w(\mathbf{T})\, \mathbb{I}[\mathbf{T} \cap \mathbf{C} = \emptyset].$$

Throughout, Γ_s and Γ_w are abbreviations for the sampling marginals of this construction; the formal primitive in Definition 17 is the joint Γ.

For the OBCB analysis, with $|\mathbf{S}| = 2$, we use a Γ built from Γ_s uniform over the three non-empty subsets of $\mathbf{S}$ ($\{\mathit{gender}\}$, $\{\mathit{credit}\}$, $\{\mathit{gender}, \mathit{credit}\}$, each with probability $1/3$) and Γ_w uniform over $\{\emptyset, \{\mathit{check\text{-}failed}\}\}$, each with probability $1/2$. Since no element appears in both $\mathbf{S}$ and $\mathbf{W}$, the rejection step is vacuous, $Z = 1$, and Γ reduces to the product of marginals. In larger suspect sets the choice of cardinality range becomes substantive: sampling uniformly from the full powerset yields an expected intervention size of $|\mathbf{S}|/2$, which spreads attribution across large coalitions and may make individual feature contributions harder to isolate; we return to this issue in Section 9, where the cardinality cap is treated as a tunable hyperparameter and a dynamic-set-sizes ablation measures the empirical cost of bounding it.

Causal attribution hinges on questions such as "if X had, contrary to fact, been fixed to $x' \neq x$ instead of x, what would the outcome have been?" For non-binary causes, this is ambiguous. To resolve this in a way that does not require committing to a single, contrastive causal event [Kawakami et al., 2024], we rely on a *distribution* over alternative values.

Definition 15 (Alternative Value Distribution)**.** *Let* $\mathbf{X}$ *be a set of random variables with domain* $\mathrm{dom}\,(\mathbf{X})$ *and let* $\mathbf{X} = \mathbf{x} \in \mathrm{dom}\,(\mathbf{X})$ *be a factual event. An* alternative value distribution $\Delta(\mathbf{x})$ *is a probability measure on* $\mathrm{dom}\,(\mathbf{X})$ *such that* $\mathbf{x} \notin \mathrm{supp}(\Delta(\mathbf{x}))$.

[12] When $J = 1$ and $K = |\mathbf{X}|$, this distribution coincides with the Shapley weighting kernel, which samples coalition size uniformly and then selects a coalition uniformly at that size [Shapley, 1953]. The parameters J and K allow the practitioner to restrict the search to cause sets of a specific cardinality range: setting $J = K = 1$ confines attribution to individual variables, while $J = 2, K = 3$ focuses on small multi-variable interactions. Moreover, whereas exact Shapley computation requires summing over all $2^{|\mathbf{X}|}$ coalitions, PCI treats Γ as a generative model: approximation proceeds by sampling subsets from Γ and outcomes from $P_{\mathbf{U}}$, making Monte Carlo estimation a native feature of the framework rather than an external algorithmic layer. We note that the weighting alone does not distinguish PCI from SHAP at $J = 1$, $K = |\mathbf{X}|$: the subset weights are identical. The distinction lies in the integrand, and it matters for what is being evaluated and what results follow. SHAP averages marginal contributions using the observational conditional $\mathbb{E}[f(\mathbf{X}_{\mathcal{S}} = \mathbf{x}_{\mathcal{S}}, \mathbf{X}_{\bar{\mathcal{S}}})]$, marginalising over the empirical distribution of non-suspect features; this means a variable that is merely correlated with a cause — but not itself causally active — can receive positive attribution. PCI instead evaluates counterfactual outcomes $P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{t}^{\star}))$ computed via the structural model, so attribution flows only along genuine causal paths. The witness mechanism adds a further distinction: by holding intermediate variables fixed, PCI can resolve overdetermination and undercutting scenarios where SHAP assigns equal or incorrect attribution. The choice of J and K remains a design decision; practitioners can set $J = K = 1$ for single-variable attribution or allow higher-order interactions as discussed above. A detailed comparison of PCI, SHAP, and Causal SHAP on concrete examples is given in Section 4.

Informally, $\Delta(\mathbf{x})$ specifies what counts as a genuinely different value for a suspect variable. The support condition ensures no mass falls on the factual value $\mathbf{x}$ itself, so every sampled alternative is a real counterfactual departure rather than a noisy copy of what actually happened.

When $\Delta(\mathbf{x})$ is a posterior conditioned on upstream factual values (possibly with some excised regions around the factual values, read on for details)— the construction we use throughout — alternatives are weighted by their plausibility under the structural model with upstream context held fixed. The upstream-only conditioning is itself a normality-respecting choice: factual upstream values play the role of the *default* against which counterfactual alternatives are graded, while downstream evidence (which would leak the consequence back into the counterfactual) is deliberately excluded. This is the operational link to the descriptive-normality literature [Halpern and Hitchcock, 2015, Icard et al., 2017] mentioned in the introduction; PCI does not aim to resolve the counterexamples motivating that line of work, only to share its core intuition that causal attribution depends on what counts as a plausible counterfactual.

Three alternatives to upstream-only conditioning were considered and not adopted as the default. *Uniform* alternatives over $\operatorname{dom}(\mathbf{X})$ assign equal density to every non-factual value regardless of context, including values the structural model itself rules near-zero — unrealistic and potentially incompatible with M. *Fully-conditional* alternatives, conditioned on every observed value including downstream evidence, leak the consequence back into the counterfactual: $\Delta(\mathbf{x})$ would be tightened by knowledge of the very outcome whose causes we are interrogating. *Marginal* (empirical) alternatives drop covariate dependence entirely, producing alternatives that need not be plausible given the rest of the factual instance. The framework permits any of these — Δ is user-pluggable — but we recommend upstream-only as the principled default.

One natural alternative value distribution results from taking the one currently embodied in the model and excising it around the factual setting. Then, the semantics for "alternative value" means "a sufficiently different value, following the original distribution otherwise". In the discrete case, of course, the analogue is simply the probability mass function conditioned on the factual value not holding.

Example 16 (ϵ-Excised Distribution)**.** *Assume* $\mathbf{X}$ *is* $\mathbb{R}^d$*-valued with density* p*. Given a factual event* $\mathbf{X} = \mathbf{x}$*, define the* ϵ*-excised alternative value distribution* $\Delta_\epsilon(\mathbf{x})$ *as the probability measure on* $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ *with density*

$$p_{\Delta_\epsilon}(\mathbf{x}' \mid \mathbf{x}) \propto \begin{cases} 0, & \mathbf{x}' \in B_\epsilon(\mathbf{x}), \\ p(\mathbf{x}'), & \textit{otherwise}, \end{cases}$$

where $B_\epsilon(\mathbf{x})$ *denotes the* ϵ*-ball centered at* $\mathbf{x}$*. In particular,* $\mathbf{x} \notin \operatorname{supp}(\Delta_\epsilon(\mathbf{x}))$*.*

Informally, the alternative value distribution Δ operationalises what "counterfactually different" means for each suspect variable: it specifies which alternative values are considered when asking whether the outcome would have changed under an alternative cause.

In the OBCB model both $gender$ and $credit$ are binary, so Δ is trivial. For Alice, the only alternative to $gender = 0$ (female) is $gender = 1$ (male), and the only alternative to $credit = 0$ (bad) is $credit = 1$ (good); for Bob, the only alternative to $gender = 1$ is $gender = 0$. Accordingly, $\Delta(gender = 0)$ is a point mass on $gender = 1$ and $\Delta(credit = 0)$ a point mass on $credit = 1$ — no distributional choice is required. The binary case is exactly where Δ is unambiguous; it is for continuous suspects such as income or loan amount that the ϵ-excised distribution becomes essential, forcing an explicit commitment to what counts as a meaningfully different value.

Why ϵ-excision? Any method that asks "what would happen if $\mathbf{X}$ had been different" must operationalise what *different* means. Three natural choices for Δ are available. The observational marginal $p(\mathbf{x}')$ — as in kernel SHAP [Lundberg and Lee, 2017] — places mass near the factual value $\mathbf{x}$, conflating the necessity question ("what if $\mathbf{x}$ had genuinely differed?") with evaluating essentially the same setting. The fully conditional posterior $p(x'_k \mid \mathbf{x}_{-k})$ conditions on all other observed features — potentially including descendants of X_k — which imposes structural constraints that may rule out otherwise plausible counterfactuals, and still concentrates mass near x_k when features are correlated. The ϵ-excised distribution occupies the middle ground: it draws from the marginal informed by upstream variables only, excising the factual neighbourhood to guarantee genuine departure without downstream conditioning.

The choice parallels the Region of Practical Equivalence (ROPE) in Bayesian hypothesis testing [Kruschke, 2018]: just as ROPE forces an explicit decision about what difference is large enough to matter, ϵ-excision forces an explicit decision about what counterfactual is sufficiently far from the factual to count as genuinely alternative. The practitioner makes this commitment regardless of method; the only question is whether it is made transparently or absorbed into a default. When X is standardised to $\mathcal{N}(0,1)$ the choice acquires a direct probabilistic interpretation: the excised mass $\Phi(x^\star+\epsilon)-\Phi(x^\star-\epsilon)$ is the fraction of the distribution ruled out as insufficiently different from $x^\star$. At $x^\star=0$, $\epsilon=0.2,\ 0.5,\ 0.8$ excise approximately $16\%, 38\%, 58\%$ of the distribution, corresponding to small, medium, and large effect thresholds in the sense of Cohen [1988]. Alternatively, ϵ can reflect measurement precision or a decision-relevant threshold; sensitivity analysis across ϵ values is advisable when the appropriate scale is unclear. For concrete examples of how SHAP's implicit choice leads to problematic attributions, see Section 4.

We are now ready to state the main definitions of PCI. The core idea is to measure two complementary things for a variable of interest X_k: (a) how much X_k's factual value contributes to producing $y^\star$ — the *sufficiency* question — and (b) how much replacing X_k's value with alternatives would shift the outcome away from $y^\star$ — the *necessity* question. We capture both by constructing two counterfactual outcome distributions conditioned on the same noise realization $\mathbf{u}$, then marginalising their product over $P_\mathbf{U}$. Because PCI is not constrained to particular variable types, we work in general measure-theoretic terms supporting combinations of continuous and discrete variables.

Definition 17 (Joint Necessity and Sufficiency Measure)**.** *Let $M=\langle \mathbf{U},\mathbf{V},\mathbf{F},P_\mathbf{U}\rangle$ be a probabilistic structural causal model and Y an outcome variable with domain $\mathrm{dom}\,(Y)$. Let p^Γ be the mass function of a variable selection distribution on $2^\mathbf{S}\times 2^\mathbf{W}$, and let $\Delta(\mathbf{s}^\star)$ be an alternative value distribution on $\mathrm{dom}\,(\mathbf{S})$. Let $(\mathbf{S},\mathbf{W})=(\mathbf{s}^\star,\mathbf{w}^\star)$ be a factual event.*

For a variable of interest X_k, let $2^\mathbf{S}_k := \{\mathbf{C}\subseteq\mathbf{S} : X_k\in\mathbf{C}\}$, and for each $\mathbf{C}\subseteq\mathbf{S}$ let $\Delta_\mathbf{C}(\mathbf{s}^\star)$ denote the pushforward of $\Delta(\mathbf{s}^\star)$ under the restriction map $\mathbf{s}'\mapsto\mathbf{s}'\,|_\mathbf{C}$.[13] For any measurable $A\subseteq\mathrm{dom}\,(Y)$ and exogenous realization $\mathbf{u}$, define three sub-probability measures on $\mathrm{dom}\,(Y)$ (the first two) and $\mathrm{dom}\,(Y)\times\mathrm{dom}\,(Y)$ (the third).

(i) Sufficiency measure. *$P^s_k(\cdot\mid\mathbf{u})$ intervenes on each active suspect set $\mathbf{C}$ to its factual values $\mathbf{s}^\star\,|_\mathbf{C}$, holds the active witnesses $\mathbf{T}$ fixed, and takes the Γ-weighted sum of $P(Y\in A\mid\mathbf{u},\mathrm{do}(\cdot))$ under factual cause restoration:*

$$P^s_k(A\mid\mathbf{u}) = \sum_{\substack{(\mathbf{C},\mathbf{T})\\ X_k\in\mathbf{C}}} P(Y\in A\mid\mathbf{u},\mathrm{do}(\mathbf{C}=\mathbf{s}^\star\,|_\mathbf{C},\mathbf{T}=\mathbf{w}^\star\,|_\mathbf{T}))\;p^\Gamma(\mathbf{C},\mathbf{T}). \tag{1}$$

(ii) Necessity measure. *$P^n_k(\cdot\mid\mathbf{u})$ instead integrates over alternative values $\mathbf{c}'$ for $\mathbf{C}$ (with $\mathbf{T}$ still pinned at factual values), and takes the Γ-weighted average over alternative cause values:*

$$P^n_k(A\mid\mathbf{u}) = \sum_{\substack{(\mathbf{C},\mathbf{T})\\ X_k\in\mathbf{C}}} \int P(Y\in A\mid\mathbf{u},\mathrm{do}(\mathbf{C}=\mathbf{c}',\mathbf{T}=\mathbf{w}^\star\,|_\mathbf{T}))\,\Delta_\mathbf{C}(\mathbf{s}^\star)(d\mathbf{c}')\,p^\Gamma(\mathbf{C},\mathbf{T}). \tag{2}$$

(iii) Joint necessity-sufficiency measure. *On product rectangles $A\times B\subseteq\mathrm{dom}\,(Y)\times\mathrm{dom}\,(Y)$, set*

$$P^{s,n}_k(A\times B)\;:=\;\int P^s_k(A\mid\mathbf{u})\,P^n_k(B\mid\mathbf{u})\,P_\mathbf{U}(d\mathbf{u}). \tag{3}$$

This rectangle-by-rectangle specification extends uniquely (by Carathéodory's extension theorem) to a sub-probability measure on the product σ-algebra of $\mathrm{dom}\,(Y)\times\mathrm{dom}\,(Y)$, with total mass $\Pr_\Gamma[X_k\in\mathbf{C}]^2\leq 1$.

Informally: $P^s_k(\cdot\mid\mathbf{u})$ is the distribution of Y in the *sufficiency world* — restore the factual value of a candidate cause set $\mathbf{C}$ containing X_k, hold the witness set $\mathbf{T}$ at its factual values, and observe where the outcome lands. $P^n_k(\cdot\mid\mathbf{u})$ is the distribution of Y in the *necessity world* — replace $\mathbf{C}$'s values with alternatives drawn from Δ, hold $\mathbf{T}$ fixed, and ask how the outcome shifts. The sum

[13] For an assignment $\mathbf{x}$ to a collection of variables $\mathbf{X}$ and any subset $\mathbf{Y}\subseteq\mathbf{X}$, the restriction $\mathbf{x}\,|_\mathbf{Y}$ denotes the sub-assignment of $\mathbf{x}$ to the variables in $\mathbf{Y}$.

over suspect-containing pairs $\{(\mathbf{C}, \mathbf{T}) : X_k \in \mathbf{C}\}$ weights each configuration by its Γ-mass — more focused interventions receive whatever weight the practitioner assigns them; configurations with $X_k \notin \mathbf{C}$ would contribute 0 to the attribution and are absent from the sum, so P_k^s and P_k^n are sub-probability measures with total mass $\Pr_\Gamma[X_k \in \mathbf{C}] \leq 1$ rather than renormalized expectations. In the OBCB example, for $X_k = gender$ under a noise realization in which Alice is credit-checked: the sufficiency world restores $gender =$ female while holding $check\text{-}failed$ fixed; the necessity world substitutes $gender =$ male under the same noise.

The product decomposition in $P_k^{s,n}$ is legitimate because, conditional on $\mathbf{u}$, the sufficiency and necessity interventions operate in separate counterfactual regimes with no causal connection: $\mathrm{do}(\mathbf{C} = \mathbf{c}^\star)$ and $\mathrm{do}(\mathbf{C} = \mathbf{c}')$ act on the same noise but in distinct hypothetical worlds, so Y^s and Y^n are conditionally independent given $\mathbf{u}$, and $P(Y^s \in A,\, Y^n \in B \mid \mathbf{u}) = P_k^s(A \mid \mathbf{u})\, P_k^n(B \mid \mathbf{u})$.

With the necessity-sufficiency measure fully defined, we are ready to characterize the causal impact of an event X_k. In particular, we will proceed by defining an arbitrary *causal impact function*, whose expectation over the necessity-sufficiency measure determines the causal impact. We first proceed with a formal definition of the causal impact measure, followed by several intuitive candidates for the causal impact function.

Definition 18 (Causal Impact Function and Its Expectation)**.** *Let $y^\star \in \mathrm{dom}\,(Y)$ denote the factual outcome value, and let $Y^s, Y^n \sim P_k^{s,n}$. For any measurable function $ci : \mathrm{dom}\,(Y)^3 \to \mathbb{R}$, the* (expected) causal impact *of X_k is defined as*

$$\mathbb{E}[\, ci(Y^s, Y^n, y^\star)] \;:=\; \iint ci(y^s, y^n, y^\star)\, P_k^{s,n}(dy^s, dy^n). \tag{4}$$

The integral is taken against the sub-probability $P_k^{s,n}$ directly, not against its renormalisation; the symbol $\mathbb{E}[\cdot]$ is shorthand for this sub-probability integral throughout. Suspect configurations with $X_k \notin \mathbf{C}$ contribute zero mass to P_k^s and P_k^n (and hence to $P_k^{s,n}$), correctly capturing that those configurations make no claim on X_k's attribution.

Informally, ci is a user-defined scoring rule that translates the joint sufficiency–necessity measure into a single scalar. The choice is not arbitrary: the Absolute Difference example below is an instance of L1 distance between outcomes, the same metric underlying Average Treatment Effect, Conditional Treatment Effect, and SHAP. The binary indicator score recovers this as a special case when outcomes are binary — indicators are L1 on $\{0, 1\}$. Practitioners with reasons to prefer L2 or another distance measure may substitute it freely; the framework places no restriction on ci beyond measurability.

For the OBCB analysis we instantiate ci as the PNS binary score defined in the following example, with $y^\star = loan = 0$. This is the choice that produces the values in Table 2.

Example 19 (PNS for Binary Outcomes)**.** *Assume the outcome Y is binary with $\mathrm{dom}\,(Y) = \{0, 1\}$, and let $y^\star = 1$ denote the factual outcome. Let Y^s and Y^n denote the sufficiency and necessity outcome variables. Define the causal impact score*

$$ci(y^s, y^n, y^\star) = \mathbb{I}\{y^n \neq y^\star\}\, \mathbb{I}\{y^s = y^\star\}.$$

Then the expected causal impact reduces to

$$\mathbb{E}[ci(Y^s, Y^n, 1)] = \mathbb{P}(Y^n = 0,\, Y^s = 1),$$

which conceptually coincides with Pearl's probability of necessity and sufficiency (PNS). The alignment is exact *when $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\mathrm{dom}\,(X_k) = \{0, 1\}$ where, factually, $x_k^\star = 1$.*

We now illustrate the full computation for Alice's gender attribution using this ci. With $\mathbf{S} = \{gender, credit\}$, $\mathbf{W} = \{check\text{-}failed\}$, Γ_s and Γ_w uniform, Δ a point mass on the alternative binary value, and $y^\star = loan = 0$, the exogenous noise U_{check} falls into three regions:

- u_1 (prob. 0.2): Alice is checked regardless of gender; $check\text{-}failed = 1$ factually.
- u_2 (prob. 0.7): Alice is not checked as female but would be as male; $check\text{-}failed = 0$ factually.
- u_3 (prob. 0.1): Alice is not checked regardless of gender.

In all regions, restoring Alice's factual values always yields a denial, so $P^s_k(\{0\} \mid \mathbf{u}) = \frac{2}{3}$ everywhere (four suspect-containing (C, T) combinations each with $p^\Gamma(C, T) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$ — since $\mathbf{S} \cap \mathbf{W} = \emptyset$ here, rejection is vacuous and Γ is just the product of marginals — each contributing probability 1). The necessity measure varies.

In u_1, the witness holds $\textit{check-failed} = 1$. The four (C, T) contributions, each weighted $\frac{1}{6}$, are:

- $\mathbf{C} = \{\textit{gender}\}$, $T = \emptyset$: male, $\textit{check-failed} = 1$ propagates structurally $\Rightarrow P(\textit{loan} = 1) = 0.05$.
- $\mathbf{C} = \{\textit{gender}\}, T = \{\textit{check-failed}\}$: male, witness holds failure $\Rightarrow P(\textit{loan} = 1) = 0.05$.
- $\mathbf{C} = \{\textit{gender}, \textit{credit}\}, T = \emptyset$: male and good credit, check passes $\Rightarrow P(\textit{loan} = 1) = 1.0$.
- $\mathbf{C} = \{\textit{gender}, \textit{credit}\}$, $T = \{\textit{check-failed}\}$: male and good credit, but witness holds failure $\Rightarrow P(\textit{loan} = 1) = 0.05$.

$$P^n_k(\{1\} \mid u_1) = \tfrac{1}{6}(0.05 + 0.05 + 1.0 + 0.05) \approx 0.192.$$

In u_2, the witness holds $\textit{check-failed} = 0$. The four contributions:

- $\mathbf{C} = \{\textit{gender}\}, T = \emptyset$: male would be checked; $\textit{check-failed} = 1$ propagates structurally $\Rightarrow P(\textit{loan} = 1) = 0.05$.
- $\mathbf{C} = \{\textit{gender}\}, T = \{\textit{check-failed}\}$: male, witness holds $\textit{check-failed} = 0$, check passes $\Rightarrow P(\textit{loan} = 1) = 1.0$.
- $\mathbf{C} = \{\textit{gender}, \textit{credit}\}, T = \emptyset$: male and good credit, check passes $\Rightarrow P(\textit{loan} = 1) = 1.0$.
- $\mathbf{C} = \{\textit{gender}, \textit{credit}\}$, $T = \{\textit{check-failed}\}$: male and good credit, witness holds $0 \Rightarrow P(\textit{loan} = 1) = 1.0$.

$$P^n_k(\{1\} \mid u_2) = \tfrac{1}{6}(0.05 + 1.0 + 1.0 + 1.0) \approx 0.508.$$

In u_3, male Alice is also not checked ($\textit{check} = 0$ regardless of gender), so all necessity worlds yield denial: $P^n_k(\{1\} \mid u_3) = 0$.

Combining, with $ci(y^s, y^n, y^\star) = \mathbb{I}\{y^n \neq y^\star\}\, \mathbb{I}\{y^s = y^\star\}$ and $y^\star = \textit{loan} = 0$:

$$\begin{aligned}
\mathrm{PCI}_{\textit{gender}}(\text{Alice}) &= \mathbb{E}[ci(Y^s, Y^n, y^\star)] = \mathbb{P}(Y^n = 1,\, Y^s = 0) \\
&= \sum_u p(u)\, P^s_k(\{0\} \mid u)\, P^n_k(\{1\} \mid u) \\
&= 0.2 \times \tfrac{2}{3} \times 0.192 + 0.7 \times \tfrac{2}{3} \times 0.508 + 0.1 \times 0 \\
&\approx 0.026 + 0.237 = 0.263.
\end{aligned}$$

Table 2 gives the full results for Alice and Bob under these parameter choices: Pearl's PN/PS/PNS, PCI without witnesses ($\mathbf{W} = \emptyset$), and PCI with the *check-failed* witness active — with the latter further decomposed into its necessity and sufficiency marginals.

	Pearl			PCI	PCI ($\mathbf{W} = \{\textit{check-failed}\}$)		
Person, Variable	PN	PS	PNS	($\mathbf{W} = \emptyset$)	Necessity	Sufficiency	Total
Alice (*gender*)	0.045	1.000	0.045	0.210	0.394	0.667	**0.263**
Alice (*credit*)	0.180	1.000	0.180	0.240	0.298	0.667	**0.199**
Bob (*gender*)	0.000	—	0.000	0.038	0.030	0.637	**0.019**
Bob (*credit*)	0.900	0.955	0.860	0.228	0.188	0.637	**0.119**

Table 2: Per-individual responsibility scores for *gender* and *credit* on $\textit{loan} = 0$ in the stochastic OBCB model. Pearl's PN/PS/PNS, PCI without witnesses ($\mathbf{W} = \emptyset$), and PCI with the *check-failed* witness shown side by side; the rightmost three columns decompose PCI($\mathbf{W} = \{\textit{check-failed}\}$) into its necessity and sufficiency marginals (their joint expectation is the bold total). Bob's PS for gender is undefined because the conditioning event (female, bad, $\textit{loan} = 1$) has probability zero in the OBCB model; the PCI marginals are well-defined throughout, since they integrate over $P_{\mathbf{U}}$ rather than conditioning on the factual outcome.

For Alice, PNS and PCI without witnesses both fail **D-A-rank** (p. 10): they rank $credit$ above $gender$ (0.180 vs. 0.045; 0.240 vs. 0.210). PCI with witnesses reverses this (0.263 vs. 0.199), satisfying **D-A-rank** as well as **D-A1** and **D-A2** (both values positive). Holding $check\text{-}failed$ at its factual value blocks the credit-check bottleneck in noise realizations where Alice was checked, isolating gender's structural role. The $check\text{-}failed$ witness is the decisive ingredient.

For Bob, **D-B2** and **D-B-rank** hold under all three methods: $credit$ is ranked above $gender$ throughout. **D-B1** requires strictly positive attribution for $gender$: PNS returns exactly 0.000 and therefore fails, because $P(loan = 1 \mid gender = \text{female}, credit = \text{bad}) = 0$ makes the counterfactual necessity calculation zero, missing gender's indirect role in opening the credit evaluation. PCI with witnesses returns 0.019, correctly capturing this indirect contribution. The cross-individual desideratum **D-comp** holds under both PCI ($0.263 > 0.019$) and PNS ($0.045 > 0.000$). A full verification of all seven desiderata is given at the end of this section.

The PNS and PCI (no witnesses) columns differ even though neither uses witnesses, because PCI still Γ-averages over candidate suspect sets. With $|\mathbf{S}| = 2$ PCI averages over $\{gender\}$, $\{credit\}$, and $\{gender, credit\}$ (each weighted $\frac{1}{3}$), whereas PNS evaluates a single counterfactual on one variable at a time. Property (i) below establishes that the two agree exactly when $|\mathbf{S}| = 1$ (single suspect, no witnesses, binary outcome) — the OBCB setup uses $|\mathbf{S}| = 2$ deliberately, to demonstrate joint-subset attribution.

The causal impact function ci is a user-defined component of PCI, and different choices encode different causal questions. In the benchmarking experiments of Section 9, we instantiate ci using only the necessity component, since Halpern's actual causality has no sufficiency counterpart — necessity-only attribution allows a direct parallel comparison. The full framework incorporating both Y^s and Y^n is instantiated in the PNS binary example above and in the Absolute Difference example below; the OBCB analysis of this section (Tables 2–3) provides a concrete validation, since the PNS binary ci jointly conditions on $Y^s = y^\star$ and $Y^n \neq y^\star$ and it is precisely this combination that produces the correct **D-A-rank** ordering. Further experiments using the full ci on continuous outcomes are presented in the signal-with-mediation example (Section 4.3), the synthetic evaluation (Section 5), and the SIR benchmark (Section 10).

When the outcome variable Y is continuous rather than binary, indicators are replaced by a distance-based score that measures how far the necessity and sufficiency worlds land from the factual value $y^\star$. The natural choice, paralleling L1-based scores such as ATE and SHAP, is the absolute difference.

Example 20 (Absolute Difference Impact Score)**.** *We can develop an analogous measure for continuous outcomes where we increase attribution with distance from the factual value $y^\star$ in the necessity world and decrease impact with distance to the factual value in the sufficiency world.*

$$ci(y^s, y^n, y^\star) = |y^n - y^\star| - |y^s - y^\star|.$$

With the full machinery in place — structural model, variable selection, alternative value distribution, joint necessity-sufficiency measure, and causal impact function — PCI delivers the following properties.

(i) **Generalisation of PN/PS/PNS.** When $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\mathrm{dom}\,(X_k) = \{0, 1\}$, the PNS binary ci recovers Pearl's probability of necessity and sufficiency exactly; PN and PS are recovered by the corresponding one-sided ci choices.

(ii) **Connection to actual causality.** For suitable Γ and Δ, PCI recovers Halpern's actual causality judgements (Section 7).

(iii) **Necessity–sufficiency decomposition.** P^s_k and P^n_k are defined and interpretable independently; practitioners may choose a ci that uses either alone or both, depending on whether the causal question concerns necessity, sufficiency, or both.

(iv) **Witness-mediated symmetry breaking.** Holding intermediate variables fixed via $\mathbf{W}$ disambiguates overdetermination and undercutting scenarios where necessity-only methods assign equal or zero attribution, as demonstrated above for Alice and Bob, and further developed in Section 4 (SHAP and Causal SHAP comparison), Section 5 (synthetic overdetermination and undercutting archetypes), and Section 9 (witness ablation on the scaled throwing problem).

(v) **Composability.** Γ, Δ, and ci are probability distributions and a measurable function respectively; they slot naturally into Bayesian pipelines, accept posteriors as inputs, and admit Monte Carlo estimation without modification to the core formalism.

Table 3 verifies the seven desiderata introduced at the start of this section (p. 10) against the values in Table 2. PCI with witnesses satisfies all seven; PCI without witnesses fails **D-A-rank**; PNS fails **D-A-rank** and **D-B1**. (The table uses the shorthand $|R(V \mid \text{person})|$ for $|R(V \rightsquigarrow loan \mid \text{person})|$.)

Desideratum	PNS	PCI (no wit.)	PCI (with wit.)
D-A1: $\lvert R(gender \mid \text{Alice})\rvert > 0$	✓ 0.045	✓ 0.210	✓ 0.263
D-A2: $\lvert R(credit \mid \text{Alice})\rvert > 0$	✓ 0.180	✓ 0.240	✓ 0.199
D-A-rank: $\lvert R(gender \mid \text{Alice})\rvert > \lvert R(credit \mid \text{Alice})\rvert$	× 0.045<0.180	× 0.210<0.240	✓ 0.263>0.199
D-B1: $\lvert R(gender \mid \text{Bob})\rvert > 0$	× 0.000	✓ 0.038	✓ 0.019
D-B2: $\lvert R(credit \mid \text{Bob})\rvert > 0$	✓ 0.860	✓ 0.228	✓ 0.119
D-B-rank: $\lvert R(credit \mid \text{Bob})\rvert > \lvert R(gender \mid \text{Bob})\rvert$	✓ 0.860>0.000	✓ 0.228>0.038	✓ 0.119>0.019
D-comp: $\lvert R(gender \mid \text{Alice})\rvert > \lvert R(gender \mid \text{Bob})\rvert$	✓ 0.045>0.000	✓ 0.210>0.038	✓ 0.263>0.019

Table 3: Verification of the seven desiderata (p. 10) against the values in Table 2. PCI with witnesses satisfies all seven; PCI without witnesses fails **D-A-rank**; PNS fails **D-A-rank** and **D-B1**.

*PNS fails **D-A-rank** and **D-B1**. The D-A-rank failure reflects that without a* $\mathit{check\text{-}failed}$ *witness, necessity-based attribution cannot distinguish Alice's gender (which blocked the evaluation entirely) from her credit (which would only have mattered had she been evaluated). The D-B1 failure reflects that PNS assigns Bob's gender exactly zero: because* $P(loan = 1 \mid female, bad) = 0$*, the counterfactual necessity calculation returns zero, missing gender's indirect role in opening the credit evaluation. The* $\mathit{check\text{-}failed}$ *witness resolves both.*

To close this section, it is worth comparing PCI's decomposition with the one already present in Pearl's framework. Pearl's framework defines three related but distinct binary scores: the probability of necessity $\text{PN} = P(Y_{x'} = y' \mid X = x^*, Y = y^*)$, the probability of sufficiency $\text{PS} = P(Y_{x^*} = y^* \mid X = x', Y = y')$, and their joint $\text{PNS} = P(Y_{x^*} = y^*, Y_{x'} = y')$. The first two are conditional probabilities; PNS is unconditional, and the three satisfy $\text{PNS} = \text{PN} \cdot P(Y_{x^*} = y^*) = \text{PS} \cdot P(Y_{x'} = y')$ rather than $\text{PNS} = \text{PN} \cdot \text{PS}$ (which would require $Y_{x^*} \perp Y_{x'}$, an assumption typically violated since the two potential outcomes share exogenous noise). PCI admits a related split: with $ci = \mathbb{I}\{y^n \neq y^\star\}\, \mathbb{I}\{y^s = y^\star\}$,

$$\mathbb{E}[ci(Y^s, Y^n, y^\star)] = \int P_k^s(\{y^\star\} \mid \mathbf{u})\, P_k^n(\{1{-}y^\star\} \mid \mathbf{u})\, dP_{\mathbf{U}},$$

giving a *sufficiency marginal* and a *necessity marginal* that record how often, on average, restoring the factual values of a suspect set containing X_k reproduces the factual outcome, and how often substituting alternatives overturns it. The two marginals are independent given $\mathbf{u}$.

The components are not identical to Pearl's: PN and PS condition on the factual outcome and consider a single counterfactual flip; PCI's marginals integrate over $P_{\mathbf{U}}$, average over suspect sets weighted by Γ, and hold the witness configurations fixed when active. Table 2 (presented earlier) places both decompositions side by side for $gender$ and $credit$.

For Alice's gender, $\text{PN} = 0.045$ is the bare probability that a male with bad credit would have been approved; PCI's necessity marginal is 0.394, nearly an order of magnitude higher, because the $\mathit{check\text{-}failed}$ witness keeps the credit-check bottleneck active when the male counterfactual would otherwise have been checked and might still have failed — structural information that the unconditional PN cannot register. For Bob's gender, PN is exactly 0 (a female with bad credit cannot be approved in this model) and PS is undefined; PCI's marginals remain well-defined at 0.030 and 0.637, recovering the indirect role of gender in opening the evaluation.

The next section turns from comparing PCI with Pearl's family of counterfactual scores to comparing it with the SHAP family — methods grounded in cooperative game theory rather than in structural counterfactual reasoning.

4 Examples and comparison to SHAP and Causal SHAP

Shapley values have become one of the most widely used tools for explaining the predictions of complex machine learning models. Grounded in cooperative game theory, they provide a principled decomposition of a model's prediction into additive contributions from individual features. Janzing et al. [2020] argued that the conditional expectation in plain SHAP is the conceptually wrong way to model feature removal — a dropped feature should be marginalised under a do-intervention, not under conditioning. Heskes et al. [2020] propose Causal SHAP, which implements this prescription by replacing the observational marginal used for features outside the active coalition with an interventional distribution: when a dropped feature is causally downstream of a coalition member, its distribution under the intervention differs from its marginal, and Causal SHAP corrects for this using Pearl's do-calculus.[14] Formal definitions of both are given in Section 4.1.

Despite this repair, both methods share deeper limitations. First, both plain and Causal SHAP can assign non-zero responsibility to variables that play no causal role with respect to the target — attribution can run against the direction of the causal graph, a failure the interventional correction does not address. Second, both methods explain a model's predicted output averaged over the population, not the individual factual outcome of a specific person: when the goal is *causal responsibility attribution*, this produces further failures that are not always made explicit in the literature. The first limitation will surface in our signal example as violations of D-YX, D-YM, and D-MX; the second will surface in OBCB as the D-A-rank overdetermination effect, and in the signal example as Instance 2's budget collapse when the prediction equals its baseline.

We trace these failures through two examples, comparing plain SHAP, Causal SHAP, and PCI against the desiderata of Section 2 and some further desiderata related to the example we will employ. We first return to the OBCB model: it permits exact analytic computation and the desiderata are already known. In OBCB the feature set contains only exogenous roots, so plain and Causal SHAP coincide — the comparison is short. We then turn to a continuous mediation example where downstream variables are observable model inputs; there the two methods diverge, and we examine what Causal SHAP fixes and what it does not.

4.1 SHAP and Causal SHAP

In this section $\mathcal{S} \subseteq N$ denotes a SHAP coalition (a subset of features held to their factual values during evaluation), playing the same role here that $\mathbf{C} \subseteq \mathbf{S}$ played in Section 3. We use the calligraphic $\mathcal{S}$ to keep the SHAP coalition typographically distinct from PCI's suspect set $\mathbf{S}$ on the printed page; the SHAP literature's plain S is the same object.

Definition 21 (SHAP). *A cooperative game is a pair (N, v), where $N = \{1, \ldots, n\}$ is the set of players and $v : 2^N \to \mathbb{R}$ is the* characteristic function *mapping each subset (coalition) $\mathcal{S} \subseteq N$ to the value that coalition can produce. The Shapley value [Shapley, 1953] is the unique attribution satisfying four axioms:*

1. ***Efficiency**: $\sum_i \phi_i = v(N) - v(\emptyset)$*
2. ***Symmetry**: If $v(\mathcal{S} \cup \{i\}) = v(\mathcal{S} \cup \{j\})$ for all $\mathcal{S}$, then $\phi_i = \phi_j$*
3. ***Dummy**: If $v(\mathcal{S} \cup \{i\}) = v(\mathcal{S})$ for all $\mathcal{S}$, then $\phi_i = 0$*
4. ***Linearity**: For two games v_1, v_2: $\phi_i(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \phi_i(v_1) + \alpha_2 \phi_i(v_2)$*

[14] A different counterfactual line is taken by Sharma et al. [2022], whose CF-Shapley values use a structural causal model to attribute the change in a metric from a single fixed reference x^{ref} to the observed x: $\phi_j \propto \sum_{\mathcal{S} \subseteq N \setminus \{j\}} [f(x_{\mathcal{S}}, x_j^{\text{ref}}) - f(x_{\mathcal{S}}, x_j)]$. This is the closest neighbour to PCI in spirit — single contrastive scenario, do-style intervention — and differs in three ways. *(i)* CF-Shapley commits to one reference x^{ref}, while PCI integrates over an alternative-value distribution $\Delta(\mathbf{x})$ and need not pick a single contrast. *(ii)* CF-Shapley returns a one-axis contrast $f(\cdot, x_j^{\text{ref}}) - f(\cdot, x_j)$, while PCI reports a joint (Y^s, Y^n) measure and applies a user-chosen ci, decomposing necessity and sufficiency separately. *(iii)* CF-Shapley has no witness mechanism, so it cannot break the overdetermination and undercutting symmetries that PCI's witness set $\mathbf{W}$ resolves on the OBCB and signal walkthroughs of this section.

The unique solution is:

$$\phi_i = \sum_{\mathcal{S} \subseteq N \setminus \{i\}} \frac{|\mathcal{S}|!\,(|N| - |\mathcal{S}| - 1)!}{|N|!} \big[v(\mathcal{S} \cup \{i\}) - v(\mathcal{S}) \big] \tag{5}$$

The weight $\frac{|\mathcal{S}|!(|N|-|\mathcal{S}|-1)!}{|N|!}$ *equals the probability that, in a uniformly random ordering of all players, exactly the members of* $\mathcal{S}$ *precede* i*. Equivalently,* ϕ_i *is the average marginal contribution of* i *across all orderings. Lundberg and Lee [2017] apply Shapley values to explain model predictions by taking players to be input features, and the characteristic function to be the expected model output given a subset of features:*

$$v(\mathcal{S}) = \mathbb{E}_{X_{\bar{\mathcal{S}}}} \big[f(X_{\mathcal{S}} = x_{\mathcal{S}},\, X_{\bar{\mathcal{S}}}) \big] \tag{6}$$

where $\bar{\mathcal{S}} = N \setminus \mathcal{S}$ *and missing features are marginalized over their marginal distribution.*

Causal SHAP replaces the characteristic function with an interventional expectation [Heskes et al., 2020]:

Definition 22 (Causal SHAP)**.**

$$v_{\textit{causal}}(\mathcal{S}) = \mathbb{E}\big[f(X) \mid do(X_{\mathcal{S}} = x_{\mathcal{S}}) \big] = \int P(X_{\bar{\mathcal{S}}} \mid do(X_{\mathcal{S}} = x_{\mathcal{S}}))\, f(X_{\bar{\mathcal{S}}}, x_{\mathcal{S}})\, dX_{\bar{\mathcal{S}}} \tag{7}$$

By the rules of do-calculus, intervening on $X_{\mathcal{S}}$ *only affects the distribution of descendants of* $X_{\mathcal{S}}$*. For non-descendants,* $P(X_j \mid do(X_{\mathcal{S}} = x_{\mathcal{S}})) = P(X_j)$*. Writing* $X_{\bar{\mathcal{S}},\,\mathrm{nd}}$ *for the features in* $\bar{\mathcal{S}}$ *that are not descendants of any member of* $\mathcal{S}$*, and* $X_{\bar{\mathcal{S}},\,\mathrm{d}}$ *for those that are, equation (7) is equivalent to:*

$$v_{\textit{causal}}(\mathcal{S}) = \mathbb{E}_{X_{\bar{\mathcal{S}},\,\mathrm{nd}} \sim P(\cdot),\; X_{\bar{\mathcal{S}},\,\mathrm{d}} \sim P(\cdot \mid do(X_{\mathcal{S}} = x_{\mathcal{S}}))} \big[f(X_{\bar{\mathcal{S}}}, x_{\mathcal{S}}) \big] \tag{8}$$

The Shapley formula (5) is otherwise unchanged. The method retains all four Shapley axioms.

At the level of subset weighting alone, PCI (with $J = 1$, $K = |\mathbf{S}|$) and SHAP are not distinguishable: Γ's subset weights match the Shapley kernel. The distinction lies in the integrand. SHAP averages marginal contributions using the observational conditional $\mathbb{E}[f(\mathbf{X}_{\mathcal{S}} = \mathbf{x}_{\mathcal{S}}, \mathbf{X}_{\bar{\mathcal{S}}})]$, marginalising over the empirical distribution of non-coalition features; a feature merely correlated with a true cause can therefore receive positive attribution. PCI evaluates counterfactual outcomes $P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{t}^{\star}))$ computed via the structural model, so attribution flows only along genuine causal paths. The witness mechanism adds a further distinction: by holding intermediate variables fixed, PCI can resolve overdetermination and undercutting scenarios where SHAP assigns equal or incorrect attribution. The remainder of this section makes both points concrete on two examples.

4.2 OBCB: SHAP and Causal SHAP

To apply the SHAP and Causal SHAP definitions of Section 4.1 to OBCB we must specify both a feature set N and a function f on the joint feature space; the value functions and Shapley sums are derived from these. The natural choice for f is the conditional expectation of the outcome, $f(x_N) = \mathbb{E}[\textit{loan} \mid X_N = x_N]$. But the OBCB PSCM has five variables — $\textit{gender}$, $\textit{credit}$, $\textit{check}$, $\textit{check-failed}$, $\textit{loan\_if\_checked}$ — so the choice of N is non-trivial. Two options are natural:

- **Option A: two upstream roots.** $N = \{\textit{gender}, \textit{credit}\}$. Both features are exogenous; the three internal variables ($\textit{check}$, $\textit{check-failed}$, $\textit{loan\_if\_checked}$) are integrated out inside f. This is the smallest sufficient feature set and matches the way the model is normally described: predict approval from gender and credit.
- **Option B: two roots plus the mediator.** $N' = \{\textit{gender}, \textit{credit}, \textit{check-failed}\}$. The mediator becomes an observable model input, the same status it has in the signal-with-mediation example (§4.3). This is the comparison PCI's witness mechanism naturally invites: PCI accesses $\textit{check-failed}$ as a witness, so allowing SHAP to access it as a feature is the fair foil.

We work through both. Option A is simpler and exposes the qualitative failures of plain SHAP. Option B then asks whether admitting the mediator rescues Causal SHAP.

Option A: two upstream roots. Here $f(g, c) = P(loan = 1 \mid g, c)$ takes values $f(\mathrm{F}, \mathrm{bad}) = 0.000$, $f(\mathrm{F}, \mathrm{good}) = 0.180$, $f(\mathrm{M}, \mathrm{bad}) = 0.045$, $f(\mathrm{M}, \mathrm{good}) = 0.900$, with the three internal variables integrated out. We apply SHAP to the rejection probability $g(x) = 1 - f(x)$ for Alice and Bob from Example 5, attributing responsibility for the denial outcome.[15] By the SHAP definition (and equivalently by Definition 22 once we note that no dropped feature is a descendant of any coalition member), the value function on the 2-feature game is

$$v(\mathcal{S}) \;=\; \mathbb{E}\big[f(X) \,\big|\, X_{\mathcal{S}} = x^{\star}_{\mathcal{S}}\big] \;=\; \sum_{x_{N\setminus S} \in \{0,1\}^{|N\setminus S|}} P(X_{N\setminus S} = x_{N\setminus S})\, f(x^{\star}_{\mathcal{S}},\, x_{N\setminus S}).$$

Under independent uniform marginals $P(gender) = P(credit) = \frac{1}{2}$ this specialises to

$$\begin{aligned}
v(\emptyset) &= \tfrac{1}{4} \sum_{g,c\in\{0,1\}} f(g, c), && [\text{both dropped: weight } \tfrac{1}{2}\cdot\tfrac{1}{2}] \\
v(\{gender\}) &= \tfrac{1}{2}[f(g^{\star}, 0) + f(g^{\star}, 1)]\,, && [\text{only credit dropped: weight } P(c) = \tfrac{1}{2}] \\
v(\{credit\}) &= \tfrac{1}{2}[f(0, c^{\star}) + f(1, c^{\star})]\,, && [\text{only gender dropped: weight } P(g) = \tfrac{1}{2}] \\
v(\{gender, credit\}) &= f(g^{\star}, c^{\star}). && [\text{nothing dropped: no average}]
\end{aligned}$$

Plugging in the four values of f listed at the start of Option A yields Table 4; pushing those through the closed-form Shapley above gives the magnitudes in Table 5, which compares against PNS and PCI.

$\mathcal{S}$	$v(\mathcal{S})$ for Alice	$v(\mathcal{S})$ for Bob
$\emptyset$	0.281	0.281
$\{gender\}$	0.090	0.473
$\{credit\}$	0.023	0.023
$\{gender, credit\}$	0.000	0.045

Table 4: Coalition values $v(\mathcal{S}) = \mathbb{E}[f(X) \mid X_{\mathcal{S}} = x^{\star}_{\mathcal{S}}]$ for the 2-feature game ($N = \{gender, credit\}$). $v(\emptyset) = \mathbb{E}[f(X)] = 0.281$ is the population baseline; $v(N)$ is the factual prediction $f(x^{\star})$. Plain SHAP and Causal SHAP coincide on this game (neither feature is downstream of the other), so a single column suffices per individual.

Person	Feature	PNS	SHAP	PCI (no witnesses)	PCI (with witnesses)
Alice	*gender*	0.045	0.107	0.210	**0.263**
Alice	*credit*	0.180	0.174	0.240	**0.199**
Bob	*gender*	0.000	−0.107	0.038	**0.019**
Bob	*credit*	0.860	0.343	0.228	**0.119**

Table 5: Responsibility attributions for Alice and Bob. SHAP values are signed (ϕ_i^g for $P(loan = 0 \mid x)$); desiderata compare magnitudes $|\cdot|$. PNS and PCI values from Table 2.

SHAP fails two desiderata. The primary failure is **D-A-rank**: Alice's credit outranks her gender ($0.174 > 0.107$). This is an overdetermination effect: *credit*=bad correlates with the *check-failed* mechanism that blocked Alice's credit evaluation, so SHAP's observational marginalization inflates credit attribution above gender's. PCI includes the factual *check-failed* value directly, isolating gender's upstream causal role and recovering the correct ranking. PNS fails **D-A-rank** for the same reason; it also fails **D-B1** by returning exactly zero for Bob's gender (see Table 3).

The second SHAP failure is **D-comp**: $|\phi^{g}_{gender}| = 0.107$ for both Alice and Bob. But gender played a strictly larger causal role in Alice's rejection than in Bob's. For Alice, *gender*=female

[15] Because $g = 1 - f$, the characteristic functions satisfy $v_g(\mathcal{S}) = 1 - v_f(\mathcal{S})$ and $\phi_i^g = -\phi_i^f$; we compute ϕ^f and flip signs. With $|N| = 2$ and uniform marginals $P(gender) = P(credit) = \frac{1}{2}$, equation (5) reduces to $\phi_i = \frac{1}{2}[v(\{i\}) - v(\emptyset)] + \frac{1}{2}[v(N) - v(N\setminus\{i\})]$.

was sufficient on its own to produce rejection — it blocked the credit evaluation entirely, so her credit was never a factor. For Bob, $gender$=male was a facilitating condition: it triggered the credit check, but $credit$=bad was the actual driver of the denial. SHAP cannot distinguish these because it measures how far each feature value deviates from the population prediction mean — female and male are equidistant deviations in opposite directions, so they receive equal magnitude regardless of their individual causal weight. This is an instance of a deeper limitation: SHAP explains predicted outputs, not individual factual outcomes, which we examine further in Section 4.3.

Causal SHAP coincides exactly with plain SHAP under Option A: both features are exogenous roots, so $P(credit \mid do(gender = g)) = P(credit)$ and vice versa, no dropped feature is a descendant of any coalition member, and $v_{\text{causal}}(\mathcal{S}) = v_{\text{plain}}(\mathcal{S})$ for every coalition. Option A therefore reduces to a single SHAP analysis, and the failures above belong to both methods.

Option B: two roots plus the mediator. Throughout this subsection we abbreviate $gender$, $credit$, and $check\text{-}failed$ as g, c, and cf in mathematical expressions and table cells; running prose retains the full names. Now we admit $check\text{-}failed$ as a third model input and apply both methods to the 3-feature game $N' = \{gender, credit, check\text{-}failed\}$. This requires defining f on $\{0,1\}^3$ rather than on $\{0,1\}^2$. The natural observational extension is again the conditional expectation $\tilde{f}(g, c, cf) = \mathbb{E}[loan \mid gender = g,\ credit = c,\ check\text{-}failed = cf]$. This is well defined for three of the four (c, cf) combinations, but not the fourth: a failed check requires that a check happened *and* the credit was bad, so $check\text{-}failed = 1$ together with $credit =$ good is impossible under the PSCM and has no observational data to condition on. We leave $\tilde{f}$ undefined on this cell.

Both methods need a value function $v(\mathcal{S})$: the expected value of $\tilde{f}$ when the features in $\mathcal{S}$ are held at their factual values $x^\star_{\mathcal{S}}$ and the remaining features are averaged out. The methods differ in how that average is taken. Plain SHAP averages over each dropped feature using its own observational marginal $P(X_i)$, with the dropped features treated as independent of one another. Causal SHAP averages over the joint distribution the PSCM produces under the intervention $do(X_{\mathcal{S}} = x^\star_{\mathcal{S}})$, so the dropped features keep whatever joint dependencies the PSCM imposes on them. Whether either average ever lands on the impossible cell (c=1, cf=1) — and so whether $v(\mathcal{S})$ is finite — depends on both the method and on which features are held fixed at which factual values. Table 6 works out all eight coalition values for both individuals.

	Alice (g, c, cf)=$(0, 0, 0)$		Bob (g, c, cf)=$(1, 0, 1)$	
$\mathcal{S}$	v_{plain}	v_{causal}	v_{plain}	v_{causal}
$\emptyset$	**NaN**	0.281	**NaN**	0.281
$\{g\}$	**NaN**	0.090	**NaN**	0.473
$\{c\}$	0.007	0.023	0.007	0.023
$\{cf\}$	0.270	0.270	**NaN**	**NaN**
$\{g, c\}$	0.000	0.000	0.014	0.045
$\{g, cf\}$	0.090	0.090	**NaN**	**NaN**
$\{c, cf\}$	0.000	0.000	0.025	0.025
N'	0.000	0.000	0.050	0.050

Table 6: Coalition values for the 3-feature game $N' = \{gender, credit, check\text{-}failed\}$. Bold **NaN** entries flag coalitions whose value function evaluates $\tilde{f}$ at the impossible cell (c=1, cf=1) with positive weight. Plain SHAP and Causal SHAP coincide whenever neither *credit* nor *check-failed* is dropped, since the only structural coupling in the model is between these two; they diverge whenever one is in $\mathcal{S}$ and the other is not.

Plain SHAP. Plain SHAP treats $credit$ and $check\text{-}failed$ as independent, so whenever both are dropped from the coalition the average runs over all four (c, cf) combinations, with the impossible (c=1, cf=1) combination receiving weight $P(c{=}1) \cdot P(cf{=}1) = 0.5 \cdot 0.275 \approx 0.140$ — the same nonzero weight as any other corner. For Alice, both $credit$ and $check\text{-}failed$ are dropped at $\mathcal{S} \in \{\emptyset, \{gender\}\}$, the first two NaN entries in Table 6. For Bob the same two coalitions fail, and two more break for an additional reason: at $\mathcal{S} = \{check\text{-}failed\}$ and $\mathcal{S} = \{gender, check\text{-}failed\}$ the value $cf^\star = 1$ is held fixed while $credit$ is sampled from its marginal, so the c=1 branch enters with weight 0.5 and again hits the impossible cell. Plain SHAP gives undefined Shapley values for both individuals.[16]

Causal SHAP for Alice. Causal SHAP samples cf following its structural equation $cf = check \cdot (1-c)$, with $check \sim \mathrm{Bern}(p_{\text{check}}[g])$, given whatever (g, c) values appear in the integrand. When $c = 1$ the equation forces $cf = 0$ regardless of $check$, so the $(c{=}1, cf{=}1)$ combination is never produced. Every Alice entry in the v_{causal} column of Table 6 is therefore finite. Plugging Alice's column into the Shapley formula gives magnitudes $|\phi_{gender}| = 0.097$, $|\phi_{credit}| = 0.176$, $|\phi_{check\text{-}failed}| = 0.007$, so $|\phi_{credit}| > |\phi_{gender}|$ and **D-A-rank** fails: credit outranks gender, exactly the same pathology plain SHAP exhibited on the 2-feature game.

Causal SHAP for Bob. Two coalitions break for Bob: $\mathcal{S} = \{check\text{-}failed\}$ and $\mathcal{S} = \{gender, check\text{-}failed\}$. Both hold $check\text{-}failed$ at Bob's factual value $cf^\star = 1$ and average $credit$ over its marginal. To see why this is fatal, apply Definition 22 to the first:

$$v_{\text{causal}}(\{cf\}) \;=\; \mathbb{E}\Big[\tilde{f}(X) \,\Big|\, do(cf = 1)\Big] \;=\; \tfrac{1}{4} \sum_{g \in \{0,1\}} \sum_{c \in \{0,1\}} \tilde{f}(g, c, 1).$$

The uniform $\frac{1}{4}$ weights come from $gender$ and $credit$ sitting upstream of $check\text{-}failed$ in the causal graph: intervening on $check\text{-}failed$ overrides the structural equation that would normally determine it from credit and check, but does not alter the marginal distributions of the upstream features. Two of the four summands evaluate $\tilde{f}$ at $(g, c{=}1, cf{=}1)$ — the impossible cell — with weight $\frac{1}{4}$ each. So $v_{\text{causal}}(\{cf\})$ is undefined, and so is $v_{\text{causal}}(\{g, cf\})$ for the same reason. These two coalitions appear in the Shapley sum for every feature, so all three of Bob's Causal SHAP values are NaN (Bob's columns in Table 6).

Why Bob breaks and Alice does not. The same two coalitions arise for Alice; the difference is the value of $cf^\star$ being held fixed. With Alice's $cf^\star = 0$, the $c{=}1$ summand in the formula above evaluates $\tilde{f}(g, 1, 0) = p_{\text{check}}[g] \cdot \text{loan\_prob}[g, 0]$, which is defined because the PSCM does allow $(c{=}1, cf{=}0)$: when credit is good, cf is forced to 0 deterministically, so the conditional expectation has a value. With Bob's $cf^\star = 1$, the $c{=}1$ summand evaluates $\tilde{f}(g, 1, 1)$, the impossible cell — "good credit and a failed check" — which the PSCM rules out, since failing the credit check requires bad credit by the structural equation $cf = check \cdot (1-c)$.

The mechanism producing this asymmetry is the intervention itself. When the PSCM runs forward, $credit$ and $check\text{-}failed$ are linked by $cf = check \cdot (1-c)$, which keeps $c{=}1$ and $cf{=}1$ from co-occurring. When Causal SHAP intervenes via $do(cf = cf^\star)$, this link is overridden: $check\text{-}failed$ is fixed externally and $credit$ is sampled freely. The override always produces a $(c{=}1, cf^\star)$ combination with positive weight; whether $\tilde{f}$ has a value for that combination depends on which $cf^\star$ value Causal SHAP is intervening on.

Promoting the mediator to a feature therefore does not rescue Causal SHAP: it fails the desideratum on Alice and breaks down outright on Bob.

The OBCB analysis exposed two structural failure modes: SHAP's overdetermination effect on the 2-feature game, and Causal SHAP's breakdown once the mediator is admitted as a feature under Option B. The second of these depends on a structural feature of the OBCB PSCM — the equation $cf = check \cdot (1-c)$ ties $credit$ and $check\text{-}failed$ together, so Option B's natural feature set contains one combination of input values ("good credit and a failed check") that the model is not defined on. To see whether the same families of failure appear without this complication, we now turn to the signal-with-mediation chain $X \to M \to Y$, where the mediator is naturally an observable input, $\tilde{f}$ is defined everywhere, and no impossible-cell issue arises. Causal SHAP's failures there are not artifacts of an undefined $\tilde{f}$ but properties of the method itself.

[16] The underlying issue is that plain SHAP does not see that $credit$ and $check\text{-}failed$ are linked by the structural equation $cf = check \cdot (1-c)$, and so cheerfully samples a combination the PSCM rules out.

4.3 Signal with mediation

Example 23 (Signal with mediation)**.** *We work with the following* ***generative model****: X sends a signal, M mediates it with some noise, and Y is the noisy final outcome:*

$$X \sim \mathcal{N}(0.5,\ 0.25) \tag{9}$$

$$M = X + \varepsilon_M, \quad \varepsilon_M \sim \mathcal{N}(0,\ 0.1) \tag{10}$$

$$Y = M + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,\ 0.1) \tag{11}$$

The causal graph is $X \to M \to Y$, a simple mediation chain. All noise terms are independent of each other and of X. Since everything is jointly Gaussian, all conditional expectations are linear functions of the conditioning variables — no approximations are required.

Let $R(X \rightsquigarrow Y)$ denote the causal responsibility of X for Y. Following the convention of Section 3, the desiderata are stated in terms of absolute magnitudes $|R(\cdot)|$, allowing for directionality. The ***qualitative causal responsibility desiderata*** *for this example are:*

$$\textbf{\textit{D-XY}} \quad |R(X \rightsquigarrow Y)| > 0$$

$$\textbf{\textit{D-MY}} \quad |R(M \rightsquigarrow Y)| > 0$$

$$\textbf{\textit{D-XM}} \quad |R(X \rightsquigarrow M)| > 0$$

$$\textbf{\textit{D-YX}} \quad |R(Y \rightsquigarrow X)| = 0$$

$$\textbf{\textit{D-YM}} \quad |R(Y \rightsquigarrow M)| = 0$$

$$\textbf{\textit{D-MX}} \quad |R(M \rightsquigarrow X)| = 0$$

A somewhat more subtle desideratum is:

$$\textbf{\textit{D-MXY}} \quad |R(M \rightsquigarrow Y)| > |R(X \rightsquigarrow Y)|$$

This captures the intuition that responsibility should be higher for variables that are closer to the outcome in the causal graph.

The key ***distributional facts*** *we will use are:*

$$\mathbb{E}[X] = \mathbb{E}[M] = \mathbb{E}[Y] = 0.5, \tag{12}$$

$$\big(\mathit{Var}(X),\ \mathit{Var}(M),\ \mathit{Var}(Y)\big) = (0.25,\ 0.35,\ 0.45), \tag{13}$$

$$\big(\mathit{Cov}(X,M),\ \mathit{Cov}(X,Y),\ \mathit{Cov}(M,Y)\big) = (0.25,\ 0.25,\ 0.35). \tag{14}$$

For each target variable, the optimal predictor under squared loss is the conditional expectation, computable exactly in this Gaussian model:[17]

$$f_Y(X,M) = \mathbb{E}[Y \mid X,M] = M \quad (\{X,M\} \Rightarrow Y) \tag{15}$$

$$f_M(X,Y) = \mathbb{E}[M \mid X,Y] = 0.5X + 0.5Y \quad (\{X,Y\} \Rightarrow M) \tag{16}$$

$$f_X(M,Y) = \mathbb{E}[X \mid M,Y] = 0.5 + \frac{5}{7}(M - 0.5) \quad (\{M,Y\} \Rightarrow X) \tag{17}$$

4.4 Computations and desiderata analysis

We analyze two instances. **Instance 1** sets $X^\star = M^\star = Y^\star = 1$: every variable lies above its mean, the prediction is non-trivial at every target, and SHAP, Causal SHAP, and PCI all return non-zero

[17]**Distributional facts.** From the structural equations: $\mathrm{Var}(M) = \mathrm{Var}(X) + \mathrm{Var}(\varepsilon_M) = 0.25 + 0.1 = 0.35$; $\mathrm{Var}(Y) = \mathrm{Var}(M) + \mathrm{Var}(\varepsilon_Y) = 0.35 + 0.1 = 0.45$. $\mathrm{Cov}(X,M) = \mathrm{Cov}(X, X + \varepsilon_M) = \mathrm{Var}(X) = 0.25$; $\mathrm{Cov}(X,Y) = \mathrm{Cov}(X, M + \varepsilon_Y) = \mathrm{Cov}(X,M) = 0.25$; $\mathrm{Cov}(M,Y) = \mathrm{Cov}(M, M + \varepsilon_Y) = \mathrm{Var}(M) = 0.35$. **Conditional expectations.** $\mathbb{E}[Y \mid X, M] = M$ since $Y = M + \varepsilon_Y$ with $\varepsilon_Y \perp (X, M)$. For $\mathbb{E}[M \mid X, Y]$: the covariance matrix of (X,Y) is $\Sigma_{XY} = \left(\begin{smallmatrix} 0.25 & 0.25 \\ 0.25 & 0.45 \end{smallmatrix}\right)$ with $\det(\Sigma_{XY}) = 0.05$, so $\Sigma_{XY}^{-1} = 20\left(\begin{smallmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{smallmatrix}\right)$; the regression coefficients are $[\mathrm{Cov}(M,X),\ \mathrm{Cov}(M,Y)]\,\Sigma_{XY}^{-1} = [0.25,\ 0.35] \cdot 20\left(\begin{smallmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{smallmatrix}\right) = [0.5,\ 0.5]$, giving $\mathbb{E}[M \mid X, Y] = 0.5 + 0.5(X - 0.5) + 0.5(Y - 0.5) = 0.5X + 0.5Y$. For $\mathbb{E}[X \mid M, Y]$: by d-separation, $X \perp Y \mid M$ in the chain $X \to M \to Y$, so $\mathbb{E}[X \mid M, Y] = \mathbb{E}[X \mid M] = 0.5 + \frac{\mathrm{Cov}(X,M)}{\mathrm{Var}(M)}(M - 0.5) = 0.5 + \frac{0.25}{0.35}(M - 0.5) = 0.5 + \frac{5}{7}(M - 0.5)$.

values. **Instance 2** sets $X^\star = M^\star = Y^\star = 0.5$: every variable sits at its mean, the prediction equals its baseline, and SHAP's attribution budget $f(x^\star) - \mathbb{E}[f(X)]$ collapses to zero. We work through Instance 1 first, then return to Instance 2 to isolate what each method is really tracking.

Plain SHAP. *Target Y, features $\{X, M\}$.* First, compute the coalition values:

$$v(\emptyset) = \mathbb{E}[M] = 0.5 \tag{18}$$
$$v(\{X\}) = \mathbb{E}[M \mid X = 1] = 1 \tag{19}$$
$$v(\{M\}) = \mathbb{E}[M \mid M = 1] = 1 \tag{20}$$
$$v(\{X, M\}) = f_Y(1, 1) = 1 \tag{21}$$

Next, apply equation (5) with $|N| = 2$:

$$\phi_X = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.25 \qquad \phi_M = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.25 \tag{22}$$

Target M, features $\{X, Y\}$. $v(\emptyset) = 0.5$; $v(\{X\}) = 1$; $v(\{Y\}) = 0.5\,\mathbb{E}[X \mid Y{=}1] + 0.5 = 0.889$ (using $\mathbb{E}[X \mid Y{=}1] = 0.5 + \frac{0.25}{0.45}(0.5) = 0.778$); $v(\{X, Y\}) = 1$.

$$\phi_X = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 0.889) = 0.306, \tag{23}$$
$$\phi_Y = \tfrac{1}{2}(0.889 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.194. \tag{24}$$

Target X, features $\{M, Y\}$. $v(\emptyset) = 0.5$; $v(\{M\}) = 0.5 + \frac{5}{7}(0.5) = 0.857$; $v(\{Y\}) = \mathbb{E}[X \mid Y{=}1] = 0.778$ (as above); $v(\{M, Y\}) = \mathbb{E}[X \mid M{=}1] = 0.857$ (since $X \perp Y \mid M$).

$$\phi_M = \tfrac{1}{2}(0.857 - 0.5) + \tfrac{1}{2}(0.857 - 0.778) = 0.219, \tag{25}$$
$$\phi_Y = \tfrac{1}{2}(0.778 - 0.5) + \tfrac{1}{2}(0.857 - 0.857) = 0.139. \tag{26}$$

Plain SHAP assigns $\phi_Y = 0.139 > 0$ here, even though Y does not cause X. This arises because $v(\{Y\}) = 0.778 > 0.5$: observing $Y{=}1$ is statistically informative about $X{=}1$ through the chain $X \to M \to Y$. SHAP cannot distinguish "Y is informative about X" from "Y causes X" because the value function is grounded in observational conditionals, which are symmetric under reversal of causal direction.

Causal SHAP. We now apply Causal SHAP (Definition 22) and compare against plain SHAP.

Target Y, features $\{X, M\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = \mathbb{E}[M \mid do(X{=}1)] = 1$ (M is downstream of X: $P(M \mid do(X{=}1)) = \mathcal{N}(1, 0.1)$); $v_{\text{causal}}(\{M\}) = 1$ ($do(M{=}1)$ sets $f_Y = M = 1$ directly, $P(X \mid do(M{=}1)) = P(X)$ since X has no causal parents); $v_{\text{causal}}(\{X, M\}) = 1$. All values match plain SHAP, so $\phi_X = \phi_M = 0.25$.

Target M, features $\{X, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = 1$ (Y is downstream of X, same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{X \sim P(X)}[0.5X + 0.5{\cdot}1] = 0.75$ (Y does not cause X, so $P(X \mid do(Y{=}1)) = P(X)$; differs from plain SHAP's 0.889); $v_{\text{causal}}(\{X, Y\}) = 1$.

$$\phi_X = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 0.75) = 0.375, \tag{27}$$
$$\phi_Y = \tfrac{1}{2}(0.75 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.125. \tag{28}$$

Target X, features $\{M, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{M\}) = 0.857$ (Y is downstream of M but f_X does not depend on Y since $X \perp Y \mid M$, so $\mathbb{E}_{Y \sim P(Y \mid do(M=1))}[f_X(1, Y)] = f_X(1, \cdot) = 0.857$; same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{M \sim P(M)}[f_X(M, 1)] = 0.5$ (M is not a descendant of Y, so $P(M \mid do(Y{=}1)) = P(M)$ and $\mathbb{E}[f_X(M, 1)] = 0.5 + \frac{5}{7}(\mathbb{E}[M] - 0.5) = 0.5$; differs from plain SHAP's 0.778); $v_{\text{causal}}(\{M, Y\}) = 0.857$.

$$\phi_M = \tfrac{1}{2}(0.857 - 0.5) + \tfrac{1}{2}(0.857 - 0.5) = 0.357, \tag{29}$$
$$\phi_Y = \tfrac{1}{2}(0.5 - 0.5) + \tfrac{1}{2}(0.857 - 0.857) = 0. \tag{30}$$

PCI computation. We work at the factual instance $X^\star = M^\star = Y^\star = 1$ with the absolute-difference score (Example 20) $ci(y^s, y^n, y^\star) = |y^n - y^\star| - |y^s - y^\star|$. Both worlds run the PSCM with fresh exogenous noise drawn from $P_{\mathbf{U}}$: the necessity world replaces the suspect A with $A' \sim P(A)$ and propagates; the sufficiency world restores it to $a^\star$ and propagates. For target B, $\mathbf{S}$ is the two input features of B and $\mathbf{W}$ the remaining input; Γ_s is uniform over the three non-empty subsets of $\mathbf{S}$ and Γ_w uniform over $2^{\mathbf{W}}$.

$R(X \rightsquigarrow Y)$. $\mathbf{S} = \{X, M\}$, $\mathbf{W} = \{M\}$, $w^\star = m^\star = 1$, and $X' \sim \mathcal{N}(0.5, 0.25)$. Three of the four accepted $(\mathbf{C}, \mathbf{T})$ pairs intervene on X, each with weight $\frac{1}{4}$; the fourth, $(\mathbf{C}{=}\{M\}, \mathbf{T}{=}\emptyset)$, does not touch X and contributes nothing. The key observation is that the witness pair $(\mathbf{C}{=}\{X\}, \mathbf{T}{=}\{M\})$ pins M at $m^\star$ in both worlds, which equalizes their distributions and zeros out $\bar{c}$; the other two pairs let M propagate freely. Computing $\mathbb{E}[|Y^n - 1|]$ and $\mathbb{E}[|Y^s - 1|]$ in closed form via the folded-normal formula gives:

$(\mathbf{C}, \mathbf{T})$	$\mathbb{E}[\lvert Y^n - 1\rvert]$	$\mathbb{E}[\lvert Y^s - 1\rvert]$	$\bar{c}$
$(\{X\}, \emptyset)$	0.677	0.357	0.320
$(\{X\}, \{M\})$ *[witness]*	0.252	0.252	0.000
$(\{X, M\}, \emptyset)$	0.677	0.252	0.425

Combining at weight $\frac{1}{4}$ each,

$$R(X \rightsquigarrow Y \mid \mathbf{W}{=}\{M\}) = \tfrac{1}{4}(0.320 + 0 + 0.425) \approx 0.186.$$

$R(Y \rightsquigarrow X)$. X is exogenous: no *do*-intervention on $\{M, Y\}$ alters its structural equation, so necessity and sufficiency distributions coincide and $R(Y \rightsquigarrow X \mid \mathbf{W}{=}\{M\}) = 0$.

Other cells. The remaining entries of Table 7 (D-MY, D-XM, D-YM, D-MX, D-MXY, and the $\mathbf{W} = \emptyset$ column) follow the same closed-form machinery; full per-pair computations are in the companion notebook `signal_mediation_computations.ipynb`.

Desiderata analysis. Following the convention of Section 3, we use $R(A \rightsquigarrow B)$ for the method-agnostic responsibility and ϕ_A^B exclusively for SHAP/Causal SHAP values; prose statements about desiderata refer to magnitudes $|R(A \rightsquigarrow B)|$. Table 7 maps the desiderata from Section 4.3 to the values computed above.

Desideratum	Condition	Plain SHAP	Causal SHAP	PCI ($\mathbf{W}{=}\emptyset$)	PCI ($\mathbf{W}$=3rd)
D-XY	$\lvert R(X \rightsquigarrow Y)\rvert > 0$	0.250 ✓	0.250 ✓	0.249 ✓	0.186 ✓†
D-MY	$\lvert R(M \rightsquigarrow Y)\rvert > 0$	0.250 ✓	0.250 ✓	0.283 ✓	0.319 ✓
D-XM	$\lvert R(X \rightsquigarrow M)\rvert > 0$	0.306 ✓	0.375 ✓	0.253 ✓	0.284 ✓
D-YX	$\lvert R(Y \rightsquigarrow X)\rvert = 0$	0.139 ×	0.000 ✓	0.000 ✓	0.000 ✓
D-YM	$\lvert R(Y \rightsquigarrow M)\rvert = 0$	0.194 ×	0.125 ×	0.000 ✓	0.000 ✓
D-MX	$\lvert R(M \rightsquigarrow X)\rvert = 0$	0.219 ×	0.357 ×	0.000 ✓	0.000 ✓
D-MXY	$\lvert R(M \rightsquigarrow Y)\rvert > \lvert R(X \rightsquigarrow Y)\rvert$	0.25=0.25 ×	0.25=0.25 ×	0.283>0.249 ✓	0.319>0.186 ✓‡

Table 7: Desiderata satisfied (✓) and violated (×) by plain SHAP, Causal SHAP, and PCI on the signal-with-mediation example ($X \to M \to Y$), instance $X = M = Y = 1$. PCI uses $\bar{c} = \mathbb{E}[|B^n - B^\star|] - \mathbb{E}[|B^s - B^\star|]$ with Γ_s, Γ_w uniform over non-empty subsets and rejection on $\mathbf{C} \cap \mathbf{T} \neq \emptyset$. **W=3rd** sets the third variable (neither A nor B) as the witness when not in $\mathbf{C}$. † 0.186 weights three suspect-containing (C, T) configurations at $1/4$ each (the rejection-sampling normalizer $Z = 2/3$ also counts a fourth valid pair $(\{M\}, \emptyset)$ that does not contain X): one pins M as witness (contribution 0, necessity and sufficiency coincide) and two let M propagate freely (contributions 0.320 and 0.425). ‡ The two values use different witness sets ($\mathbf{W}{=}\{M\}$ for D-XY, $\mathbf{W}{=}\{X\}$ for D-MY); this column reports the amplified verdict under an asymmetric witness rule. The $\mathbf{W}{=}\emptyset$ column above is the corresponding single-game ranking and already passes D-MXY.

Positive results (D-XY, D-MY, D-XM). Both methods correctly assign positive attribution to every direct or upstream cause of the target. Causal prediction models are built from observational conditionals, which do preserve correlation from cause to effect, so these desiderata pose no challenge to either method.

Causal SHAP fixes D-YX, not D-YM. For target X, the interventional value $v_{\text{causal}}(\{Y\}) = 0.5$ equals the baseline because M is not downstream of Y: $P(M \mid do(Y{=}1)) = P(M)$, so fixing

$Y = 1$ provides no interventional leverage over f_X, and Causal SHAP correctly yields $\phi_Y^X = 0$. For target M, however, $\phi_Y^M = 0.125 > 0$ persists. The prediction model $f_M(X, Y) = 0.5X + 0.5Y$ contains Y as an input because Y is observationally predictive of M; setting $Y = 1$ raises the model output even under the interventional regime. Causal SHAP replaces the conditioning distribution but cannot remove a feature that the prediction model itself uses.

D-MX worsens under Causal SHAP. Plain SHAP assigns $\phi_M^X = 0.219$; Causal SHAP assigns 0.357. Fixing D-YX reduces $v_{\text{causal}}(\{Y\})$ from 0.778 to 0.5, driving ϕ_Y^X to zero, but reallocates the freed attribution to M. Since the structural noise that actually drives X is unobserved, all attribution concentrates on the only visible predictor, M, even though M does not cause X.

D-MXY fails for both. Since $f_Y = M$, the singleton values $v(\{X\}) = \mathbb{E}[M \mid X{=}1] = 1$ and $v(\{M\}) = 1$ are identical, giving $\phi_M^Y = \phi_X^Y = 0.25$ for both methods. The Shapley formula cannot distinguish that X reaches Y only through M: when $X = 1$, the expected prediction is already 1 regardless of whether M is observed or not, so both features receive the same marginal contribution.

PCI: structural intervention separates the paths; witnesses sharpen the gap. Table 7 reports PCI in two configurations: without witnesses ($\mathbf{W} = \emptyset$) and with the third variable as witness.

Without witnesses, PCI satisfies all six single-variable desiderata. D-YX, D-YM, and D-MX are zero exactly: intervening on a variable that does not structurally affect the target leaves both the necessity and sufficiency worlds with identical distributions, so $ci = 0$ identically. D-XY, D-MY, and D-XM are positive. D-MXY also **passes** at $\mathrm{PCI}(M \to Y) \approx 0.283 > \mathrm{PCI}(X \to Y) \approx 0.249$, but only by a margin of 0.034. The asymmetry comes from the sufficiency world: at $C{=}\{X\}$, X's path to Y accumulates $\varepsilon_M{+}\varepsilon_Y$ in $|Y^s{-}Y^\star|$, while at $C{=}\{M\}$, M's path accumulates only ε_Y. The necessity expectations match across paths ($\mathrm{Var}(X) + \mathrm{Var}(\varepsilon_M) + \mathrm{Var}(\varepsilon_Y) = \mathrm{Var}(M) + \mathrm{Var}(\varepsilon_Y) = 0.45$), so the gap is carried entirely by the sufficiency variance asymmetry $\mathrm{Var}(\varepsilon_M)$. A small structural margin like this is fragile: under any perturbation of the noise variances or the contrast function the $\mathbf{W} = \emptyset$ verdict could flip.

Admitting the third variable as a witness amplifies the gap by roughly fourfold. $\mathrm{PCI}(X \to Y \mid \mathbf{W}{=}\{M\}) \approx 0.186$ weights three suspect-containing (C, T) configurations at $1/4$ each: in two cases M propagates freely (contributing 0.320 and 0.425) and in one case M is pinned as witness in both worlds, equalizing them and contributing 0. $\mathrm{PCI}(M \to Y \mid \mathbf{W}{=}\{X\}) \approx 0.319$ weights three cases in which M takes an alternative value; since X does not appear in Y's structural equation, pinning X has no effect on Y^n or Y^s, and all three cases contribute the same 0.425. The gap widens from 0.034 at $\mathbf{W} = \emptyset$ to $0.319 - 0.186 = 0.133$ at $\mathbf{W} = \{\text{third}\}$ — D-MXY is now a *robust* attribution rather than a marginal one. The comparison relies on different witness sets for the two cells, so it remains illustrative rather than a single-game ranking.

Instance 2 (baseline). We now return to **Instance 2**: every variable sits at its mean, $X^\star = M^\star = Y^\star = 0.5$, and the model predicts the baseline at every target: $f_Y(0.5, 0.5) = 0.5$, $f_M(0.5, 0.5) = 0.5$, $f_X(0.5, 0.5) = 0.5$. At this instance, SHAP's attribution budget $f(x^\star) - \mathbb{E}[f(X)]$ vanishes, while PCI's machinery is unaffected.

	Plain / Causal SHAP		PCI ($\mathbf{W} = \emptyset$)		PCI ($\mathbf{W} = \{\text{third}\}$)	
desideratum	value	status	value	status	value	status
D-XY	0.000	×	0.154	✓	0.115	✓
D-MY	0.000	×	0.189	✓	0.212	✓
D-XM	0.000	×	0.146	✓	0.165	✓
D-YX	0.000	✓	0.000	✓	0.000	✓
D-YM	0.000	✓	0.000	✓	0.000	✓
D-MX	0.000	✓	0.000	✓	0.000	✓
D-MXY	$0.000 = 0.000$ ×		$0.189 > 0.154$ ✓		$0.212 > 0.115$ ✓	

Table 8: Desiderata at the baseline instance $X^\star = M^\star = Y^\star = 0.5$. SHAP collapses to zero on every cell — correctly assigning zero to the non-causal triples by accident, but with no signal to offer on the causal ones. PCI reports the same structural verdicts as at $X^\star = M^\star = Y^\star = 1$.

The pattern in Table 8 reflects the object each method is decomposing. SHAP attributes the quantity $f(x^\star) - \mathbb{E}[f(X)]$ across features. At the baseline this gap is zero on every target, so the budget being distributed is zero, and every cell collapses to zero — uniformly, regardless of how the realized outcome was actually produced. The non-causal triples (D-YX, D-YM, D-MX) are satisfied only by this collapse: SHAP is not isolating the absence of a causal path; it is reporting that there is nothing to attribute. The same collapse turns D-XY, D-MY, and D-XM into violations.

PCI never invokes a population expectation. With $\mathbf{S}$ and $\mathbf{W}$ defined relative to a fixed factual instance, the question is whether varying the suspect would move the realized outcome $y^\star$ away from itself — not whether $f(x^\star)$ deviates from $\mathbb{E}[f(X)]$. The structural mechanism is unchanged at the baseline, so the three causal pairs continue to register positive values; the three non-causal pairs continue to register zero exactly, because intervening on a variable that does not structurally affect the target leaves both worlds with identical distributions. The D-MXY verdict is preserved in both witness configurations, with margin 0.035 at $\mathbf{W} = \emptyset$ (vs. 0.034 at the $X^\star = M^\star = Y^\star = 1$ instance) and 0.097 at $\mathbf{W} = \{\text{third}\}$ (vs. 0.133). The gap is a property of the PSCM's noise variances, not of where the factual sits.

The two instances together make the contrast sharp. At Instance 1, SHAP's attribution budget $f(x^\star) - \mathbb{E}[f(X)]$ is non-zero and SHAP returns non-trivial numbers; at Instance 2 the budget vanishes and every cell collapses. PCI returns the same structural verdicts on all seven desiderata across the two instances. The lesson is not that SHAP gives the wrong answer at the baseline — it correctly reports that the factual prediction does not deviate from its mean. The point is that this is a different question from causal responsibility for $y^\star$: when the prediction equals its baseline, SHAP has no budget to distribute even when the PSCM's causal mechanism is perfectly intact. PCI asks the responsibility question directly, and answers it whether or not the prediction happens to coincide with its population mean.

4.5 Summary across both examples

The failures above are not artifacts of a particular feature set or parameter choice: each traces to a structural limitation of the SHAP/Causal SHAP framework. Causal SHAP strictly improves on plain SHAP for one desideratum (D-YX), by replacing observational conditioning with interventional distributions; the remaining failures share two sources — prediction models include non-causal features that the conditioning distribution cannot exclude (D-YM, D-MX), and Shapley averaging cannot distinguish direct from indirect causal paths (D-MXY). The OBCB analysis of Section 4.2 exhibits the same family in different surface form: plain SHAP fails **D-A-rank** on the 2-feature game, and Causal SHAP either inherits that failure (Alice) or becomes undefined outright (Bob) once the mediator is admitted as a feature. Across both examples, neither SHAP variant reliably satisfies the desiderata that motivated PCI in Section 2. PCI addresses these gaps through three ingredients the SHAP framework systematically lacks — the witness mechanism, the suspect-set distribution Γ_s, and the use of the realized outcome rather than the prediction baseline; the path-noise asymmetry in the sufficiency world produces a structural gap on D-MXY, which appears already in the witness-free single game and is amplified into a robust margin under a witness rule.

5 Synthetic evaluation with overdetermination and undercutting

To evaluate PCI in a setting that exhibits multiple distinct causal patterns simultaneously, we construct a synthetic structural causal model carrying three archetypes — linear necessary-and-sufficient, overdetermined-not-necessary, and preempted (gated) — together with an irrelevance control.[18]

[18] The classical motivating example is Sally and Billy throwing stones at a bottle: Sally's stone arrives first and shatters it; Billy's would have done the same had Sally's not preempted it. The case is canonical in the actual-causality literature [Halpern, 2016] and motivates the preempted archetype below; we use generic letters rather than personal names to keep variable labelling tight.

The model has six independent root variables and three deterministic branches feeding the outcome E:

$$\begin{aligned} L_1, L_2 \sim \mathcal{N}(0,1), \quad O_1, O_2 \sim \mathcal{N}(1,1), \quad P, D \sim \mathcal{N}(0,1), \\ \text{lin} &= 5L_1 + 10L_2, \\ \text{od} &= \max(5O_1,\, 5O_2), \\ \text{gate} &= \mathbb{1}\{|L_2| \le \tau\}, \\ \text{p\_branch} &= 5P \cdot \text{gate}, \\ E &= \text{lin} + \text{od} + \text{p\_branch}. \end{aligned}$$

We pick $\tau = 0.674$ so the gate is on / off about half the time ($\Pr(|L_2| \le 0.674) \approx 0.5$ for $L_2 \sim \mathcal{N}(0,1)$). The DAG is shown in Figure 1. The variable D is sampled but never enters E — it serves as an irrelevance control, so any non-zero score for D is the noise floor of the procedure.

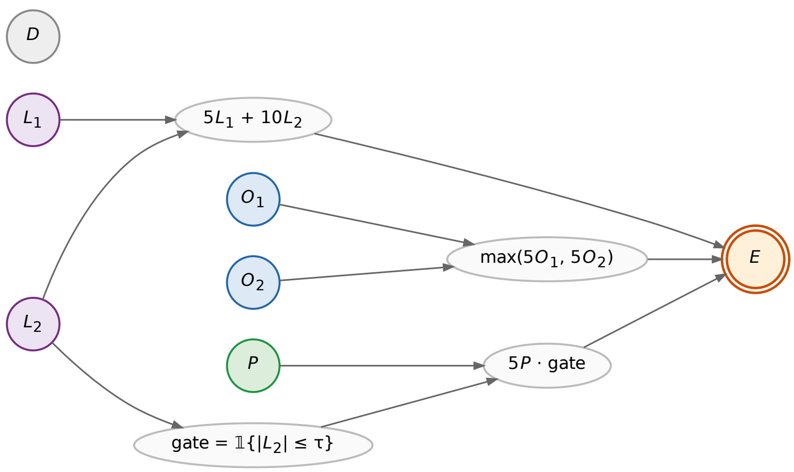


Figure 1: Synthetic archetype model. Roots (circles) are independent draws: $L_1, L_2 \sim \mathcal{N}(0,1)$ (linear, purple); $O_1, O_2 \sim \mathcal{N}(1,1)$ (overdetermined via max, blue); $P \sim \mathcal{N}(0,1)$ (gated by $|L_2| \le \tau$, green); $D \sim \mathcal{N}(0,1)$ (irrelevance control, grey). Mediators (white ellipses) carry the deterministic functions of their parents. Outcome E (orange double-circle) is the sum $\text{lin} + \text{od} + \text{p\_branch}$. The variable D has no outgoing edges by construction.

The PCI evaluation treats the six roots as the suspect set; deterministic mediators are not suspects. Restricting suspects to roots loses no causal information here since downstream values are determined by the roots (formal statement deferred to Section 3).

Causal impact, decomposed. We follow the impact function $ci(y^s, y^n, y^\star) = |y^n - y^\star| - |y^s - y^\star|$ defined in Section 3, but report its two components separately as ci_N and ci_S:

$$\begin{aligned} ci_N(V) &= \mathbb{E}\big[\,|Y^n - y^\star|\;\big|\;V \text{ intervened}\big] \quad \text{(necessity)}, \\ ci_S(V) &= -\,\mathbb{E}\big[\,|Y^s - y^\star|\;\big|\;V \text{ pinned at factual}\big] \quad \text{(sufficiency)}, \end{aligned}$$

with $ci(V) = ci_N(V) + ci_S(V)$. Higher ci_N means more necessary; higher ci_S (closer to zero) means more sufficient. The raw scores carry a co-intervention baseline that does not vanish for D, so we report *excess* over D within each case:

$$\Delta_N(V) = ci_N(V) - ci_N(D), \qquad \Delta_S(V) = ci_S(V) - ci_S(D).$$

D is the noise floor; any other variable's Δ is its causal signal above it. SHAP attributions [Lundberg and Lee, 2017], defined for a comparable prediction function, are introduced in Section 4.1 and recalled in the comparison block below.

Two factual cases. We instantiate the model with one batch of 500 events and select two factual observations to read all archetypes off.

Case 1 (preempted, contestable O pair). Filter: $|L_2| > \tau$ (gate off, so $\mathrm{p\_branch} = 0$ regardless of P) and $|O_1 - O_2| < \delta$ with $\delta = 1$ (the two overdetermined contributors are close — the winner is barely the winner; we call such a pair *contestable*, since a small perturbation to either contributor could flip which one wins $\max(5O_1, 5O_2)$). Linear N+S on the Ls, between-variable sufficiency asymmetry on a contestable O pair, and noise floor for both P and D should all be readable from this single observation.

Case 2 (unpreempted, dominant O winner). Filter: $|L_2| \le \tau$ (gate on, P branch active) and $|O_1 - O_2| > \delta$. The dominant winner is what makes the within-variable S-not-N gap visible: the winner is large enough that alternatives rarely beat it (high Δ_S), and the loser provides a partial floor when the winner is intervened (reduced Δ_N). A contestable pair cannot show this within-variable gap — pin either one and the partner can still flip the max.

The two cases together test the structural-role-not-identity claim: the same variable P should sit at the noise floor in Case 1 and mirror L_1 in Case 2.

Desiderata and measurements. Table 9 pairs each archetype expectation with the formal inequality on the excess scores Δ_N, Δ_S, the measured value at the production sample budget ($N = 20\,000$ Monte Carlo draws per intervention), and a bootstrap 95% confidence interval (4 000 resamples, joint across V and D so per-sample correlations are preserved). The noise-floor variable D itself drifts across the two cases: co-intervention variance is structurally larger in the gate-on regime than in the gate-off regime, so $\mathrm{ci}_N(D)$ shifts between regimes by a non-negligible amount. This is exactly why every archetype claim is stated on the excess scores Δ_N, Δ_S rather than raw ci values — subtracting D within each case normalises out the regime-dependent floor, so the archetype patterns can be read off stably regardless of D's absolute drift.

The threshold from the data, not from a guess. The directional desiderata of the form $\Delta > \epsilon$ or $|\Delta| < \epsilon$ all depend on a single tolerance ϵ. Rather than hand-pick it, we derive it from the Monte Carlo budget. For every Δ score appearing in any desideratum we bootstrap a standard error σ_V and take

$$\epsilon \;=\; 1.645 \cdot \sigma_{\max}, \qquad \sigma_{\max} \;=\; \max_{V,\text{case}} \sigma_V,$$

the one-sided 95% Z-quantile times the largest standard error across all measured Δs. Under the null hypothesis $\Delta = 0$, fewer than 5% of measurements would exceed ϵ; a Δ that clears ϵ is therefore not Monte Carlo chatter at the chosen budget. At $N = 20\,000$ samples we measure $\sigma_{\max} \approx 0.092$, giving $\epsilon \approx 0.151$. Smaller budgets give a noisier ϵ that defeats the point (e.g. at $N = 5\,000$, $\sigma_{\max} \approx 0.18$ would push ϵ into the signal range); this fixes the production sample size for the section. Ten of the twelve desiderata listed in earlier drafts survive this honest threshold; the two that do not (cross-regime P flip, and quantitative match of $\Delta_N(P)$ to $\Delta_N(L_1)$ in Case 2) are discussed below as diagnostics.

#	Formal desideratum	Measured	95% CI	Result
1	$\Delta_N(L_2)$ is the largest excess necessity, Case 1	+2.129	[+1.953, +2.300]	✓
2	$\Delta_N(L_2)$ is the largest excess necessity, Case 2	+0.642	[+0.476, +0.813]	✓
3	$\Delta_S(O_w) - \Delta_S(O_l) > \epsilon$, Case 1 (contestable)	+0.347	[+0.216, +0.473]	✓
4	$\Delta_S(O_w) - \Delta_S(O_l) > \epsilon$, Case 2 (dominant)	+0.529	[+0.387, +0.667]	✓
5	$\lvert\Delta_N(O_w) - \Delta_N(O_l)\rvert < \epsilon$, Case 1	−0.093	[−0.266, +0.081]	✓
6	$\Delta_S(O_w) - \Delta_N(O_w) > \epsilon$, Case 2 *(within-variable S-not-N signature)*	+0.487	[+0.256, +0.712]	✓
7	$\Delta_N(L_2) - \Delta_N(O_w) > \epsilon$, Case 2	+1.082	[+0.915, +1.251]	✓
8	$\lvert\Delta_N(P)\rvert < \epsilon$, Case 1 (gate off)	+0.112	[−0.058, +0.280]	✓
9	$\Delta_N(L_1) - \Delta_N(P) > \epsilon$, Case 1	+0.670	[+0.499, +0.843]	✓
10	$\Delta_N(P) > \epsilon$, Case 2 (gate on)	+0.250	[+0.079, +0.422]	✓

Table 9: Desiderata satisfaction on the synthetic archetype model. Threshold $\epsilon = 1.645 \cdot \sigma_{\max} \approx 0.151$ is the smallest Δ gap that, at $N = 20\,000$ samples, exceeds Monte Carlo noise at one-sided 95% confidence. Subscripts w/l denote the winner / loser of the $\max(5O_1, 5O_2)$ branch in the given case. Per-row check identifiers and the underlying numerical computations are listed in the companion notebook `responsibility_archetypes.ipynb`.

Figure 2 visualises the same ten claims as a forest plot: each row carries a plain-language statement of the structural expectation; the dot is the point estimate of the relevant Δ score (or difference of Δ scores); the horizontal bar is its bootstrap 95% CI; and the green band is the row's *pass region*, i.e. the set of Δ values that would confirm that row's claim given ϵ. The pass region depends on the claim type: $\Delta > \epsilon$ claims pass to the right of ϵ; $|\Delta| < \epsilon$ closeness claims pass inside the strip $(-\epsilon, +\epsilon)$; rank claims (#1 and #2) pass to the right of 0, since they compare $\Delta_N(L_2)$ against the largest competing suspect. A row is counted confirmed when the point estimate lies in its green band; the CI is shown separately so the reader can see when a measurement is statistically tight and when its CI crosses the pass boundary. All ten dots fall inside their pass bands. The tightest row is #10 ($\Delta_N(P) > \epsilon$ in Case 2, gate on): the point estimate $+0.250$ clears $\epsilon \approx 0.151$ by roughly one σ, with its CI lower edge at $+0.079$ still inside the band on the right side of zero.

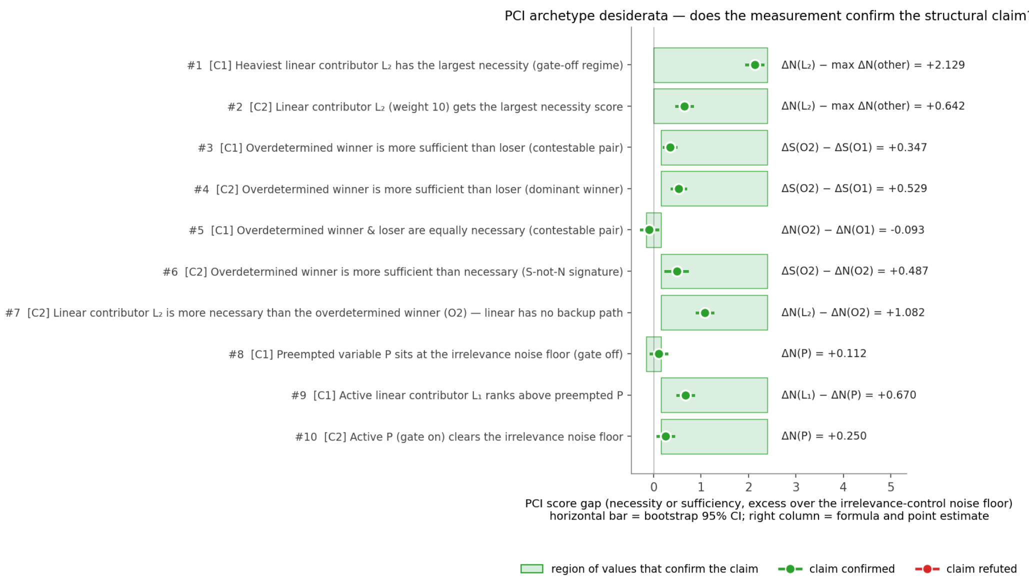


Figure 2: Forest plot of the ten numbered archetype desiderata. Each row carries a plain-language statement of the structural expectation (the case label C1 / C2 marks the factual observation it applies to); the dot is the bootstrap point estimate of the relevant Δ score, the horizontal bar is its 95% CI, and the green band is the row's *pass region* — the set of Δ values that would confirm that row's claim given the bootstrap-derived threshold $\epsilon \approx 0.151$. The right-hand column annotates the formula and its point estimate. All ten dots fall inside their pass bands. Underlying numerical computations are in the companion notebook `responsibility_archetypes.ipynb`.

Diagnostics that do not survive the bootstrap-derived threshold. Two claims from earlier drafts of this section, both about the preempted variable P, fail at the honest threshold and are reported as diagnostics rather than numbered desiderata. (D1) The cross-regime flip $\Delta_N(P, \text{Case 2}) - \Delta_N(P, \text{Case 1})$ measures $+0.139$ with CI $[-0.081, +0.365]$ — the CI straddles zero, so P's rise across regimes is not, on its own, statistically resolved at the production budget. The qualitative content of this claim is already carried by desiderata #8 (P at the noise floor in Case 1) and #10 (P above ϵ in Case 2): if both pass, the regime flip is a corollary, not an independent claim. (D2) The quantitative match $|\Delta_N(P) - \Delta_N(L_1)| < \epsilon$ in Case 2 measures -0.267 with CI $[-0.441, -0.106]$, whose magnitude is well above ϵ. At this sample budget, gated P does not quantitatively mirror ungated L_1; what we can claim is the weaker, qualitative statement that both clear the noise floor in Case 2, which is exactly desideratum #10 for P and built into Case 2 by construction for L_1.

6 Comparison to Differential Causal Effect

Having tested PCI against ground truth in Section 5, we now compare it with gradient-based attribution on a continuous-input credit-limit example.

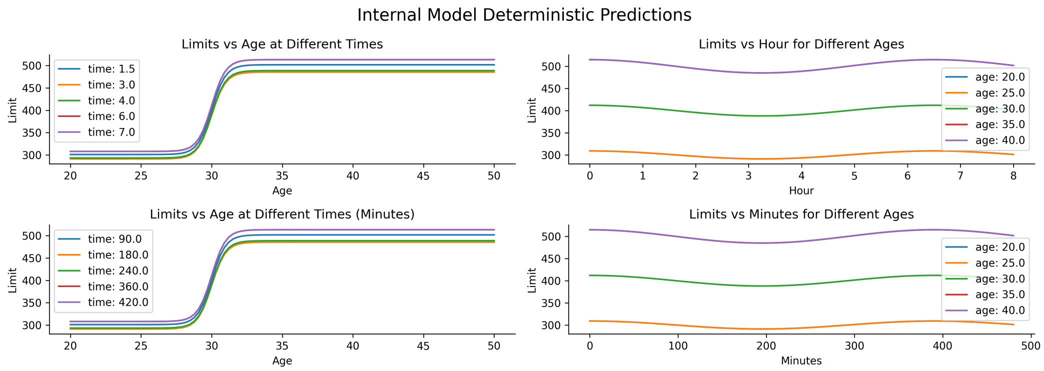


Figure 3: Deterministic component of the credit-limit model (Example 24). Top: time in hours; bottom: time in minutes. The two scales describe the same dependence modulo a linear transform; age dominates the limit, with a small sinusoidal modulation from time of day.

A natural baseline for continuous-input attribution is to read causal influence off the gradient of an interventional mean function. The motivation is direct: $\mathbb{E}\big[Y \mid do(X{=}1)\big] - \mathbb{E}\big[Y \mid do(X{=}0)\big]$ is the binary ATE, and for a continuous cause one "arrives at derivative calculus" [Ness, 2025, Section 7.4.3] by letting the contrast shrink: $\partial_x \, \mathbb{E}\big[Y \mid do(X{=}x)\big]$. Generalised to a vector $\mathbf{x} = (x_1, \ldots, x_d)$ with interventional mean $m(\mathbf{x}) = \mathbb{E}\big[Y_{\mathbf{X}=\mathbf{x}}\big]$, the *causal gradient* is

$$\nabla_{\mathbf{x}}\, m(\mathbf{x}) = \Big(\partial_{x_1} \mathbb{E}\big[Y_{\mathbf{X}=\mathbf{x}}\big],\ \ldots,\ \partial_{x_d} \mathbb{E}\big[Y_{\mathbf{X}=\mathbf{x}}\big]\Big),$$

and each partial is the *differential causal effect* (DCE) of a single variable at $\mathbf{x}$. Butler et al. [2022] introduced this quantity in the Gaussian Process regression setting, where, with the posterior expected mean expressible as $\sum_n k(x, x^{(n)})\alpha_n$, the DCE has the closed form $\partial_{x_i} \hat{F} = \sum_n \partial_{x_i} k(x, x^{(n)})\alpha_n$.

PCI and DCE are not direct competitors — DCE estimates a rate of change, PCI estimates token-level responsibility — but PCI inherits the territory of continuous attribution, so a contrast on a single example is informative. We highlight three differences:

1. DCE requires $y = F(\mathbf{X}, \epsilon)$ with F differentiable; PCI requires only that the causal model be sampleable, and the inference algorithm for the expanded model is the user's choice (we use Monte Carlo throughout).
2. DCE inherits the units of both $\mathbf{X}$ and Y, so cross-feature comparison is unit-sensitive. The PCI score integrates over the necessity- and sufficiency-world densities, so changes to input scale leave the score essentially unchanged.
3. DCE varies abruptly with the local geometry of F. The PCI score depends on where the factual value sits relative to the necessity/sufficiency densities and is therefore non-zero in flat regions while remaining locally informative.

Example 24 (credit-limit assignment)**.** *We model an applicant's approved credit limit as a function of* age *and the* time of application*. The deterministic component is a sigmoidal rise in age with a small sinusoidal modulation in time-of-day; intuitively, age should dominate. We consider four settings on two axes: time measured in hours vs. minutes (a units rescaling that should not affect attribution), and an age prior centred at* 30 *vs.* 40 *(a genuine change in what counts as unusual). The deterministic component is shown in Figure 3; marginals after the Bayesian wrap are in Figure 4. Computational details and code are in* `docs/source/gradient_based_attribution.ipynb`.

PCI scores. We instantiate ci as the absolute-difference score from Example 20, $ci(y^s, y^n, y^\star) = |y^n - y^\star| - |y^s - y^\star|$, and run the search of Section 3 on each setting, conditioning on age and on time of application as singleton suspects. Figure 5 shows that, when age is the active suspect, the necessity-world outcome tracks the deterministic curve while the sufficiency-world outcome stays near factual, yielding mean total scores of 100.45 and 97.95 for the age-30 prior (hours and minutes) and 177.87 and 176.56 for the age-40 prior. The dependence on prior is expected (more unusual factual age ⇒ larger search-space displacement); the near-equality across hours and minutes

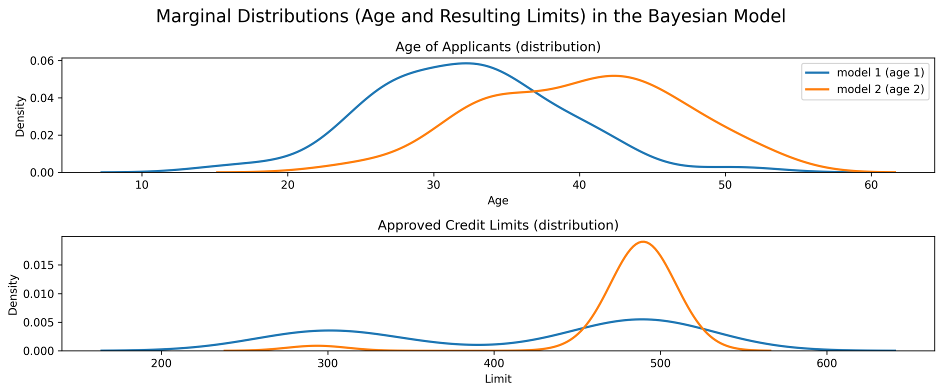


Figure 4: Marginals of age and approved limit for the two age priors in the Bayesian wrap. The shift in the age prior pushes the limit distribution to the upper plateau in the second model.

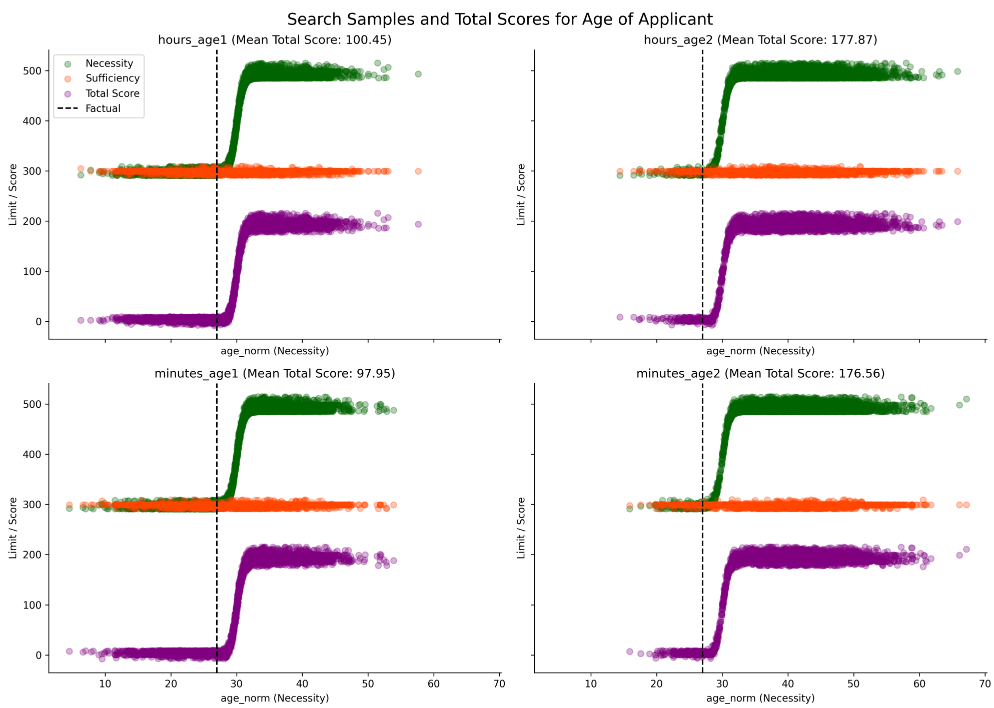


Figure 5: Necessity-world (green), sufficiency-world (orange) and total-score (purple) samples conditional on age being an active suspect, factual marked by the dashed line. Rows: hours vs. minutes; columns: age-30 vs. age-40 priors. Means in title.

reflects the unit-invariance above. Figure 6 shows the same picture for time of application, with means 68.61, 66.51, 117.97, 118.42 — systematically about two-thirds of the corresponding age scores in each row, reflecting that time is responsible but less so.

Score difference across the input grid. Plotting the difference (age score minus time-of-application score) on the input grid for the age-30 prior in hours, Figure 7 shows that age is consistently more responsible than time of application, with the gap modulated by how unusual the applicant is and how close the age falls to the steep region of the sigmoid. The other three settings give qualitatively the same picture; the second prior is in Figure 8.

DCE on the same grid. Figure 9 shows the DCE for both inputs across the same grid of factual values. Two observations make the contrast with PCI explicit. First, on the hour scale the DCE

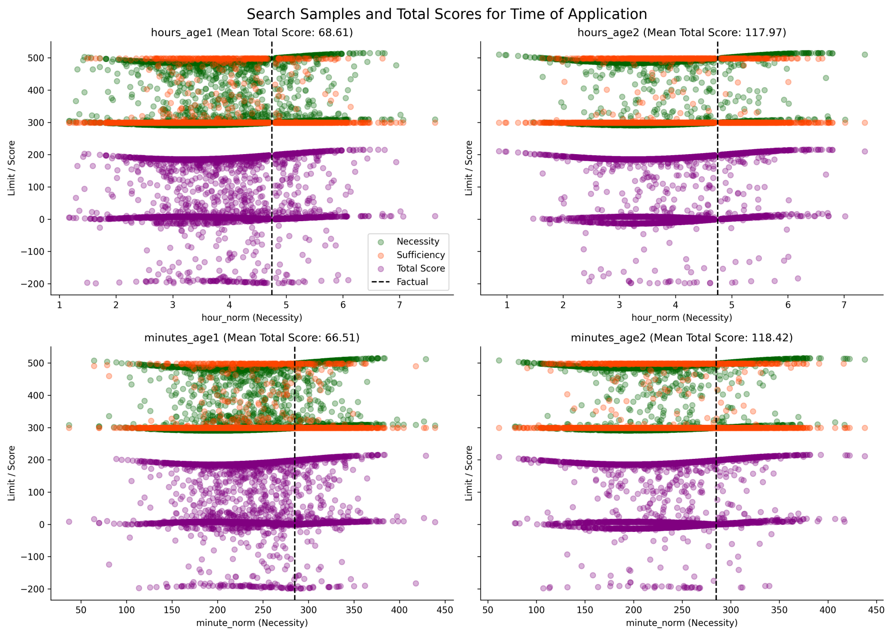


Figure 6: Same as Figure 5 for time of application. The sufficiency outcome is bimodal because the search occasionally explores age even with time as the named suspect; total-score variance reflects the smaller necessity-world displacement.

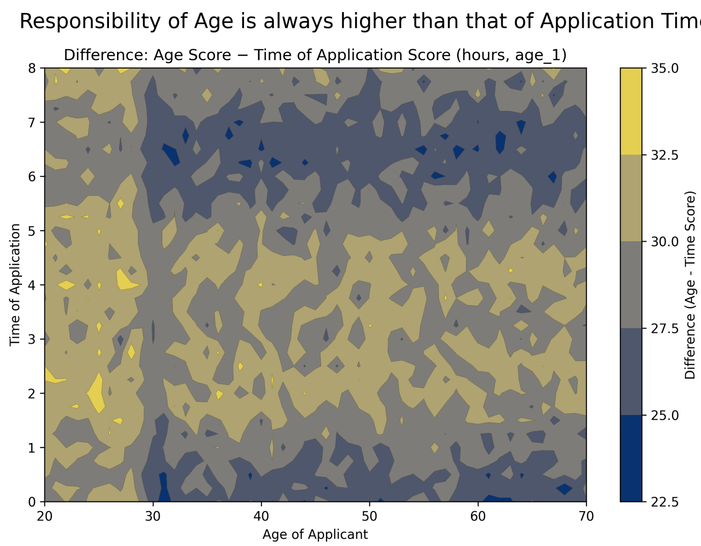


Figure 7: Difference between the PCI scores assigned to age and to time of application, on the hour scale and age-30 prior. Positive everywhere; the difference is largest near the steep region of the age-limit sigmoid and grows for ages further from the prior mean.

for time of application takes absolute values larger than the DCE for age over most of the grid, and exhibits a visible sine wave; PCI instead reports age as the more responsible variable everywhere, in line with the intuition that the age sigmoid moves the outcome by a much larger absolute amount than the daily modulation does. Second, switching the time unit from hours to minutes flattens the time DCE by roughly a factor of 60, illustrating the unit sensitivity discussed above. Both are direct consequences of DCE's role as a rate-of-change estimator rather than a responsibility score, but they make the score harder to compare across features and across reparameterisations.

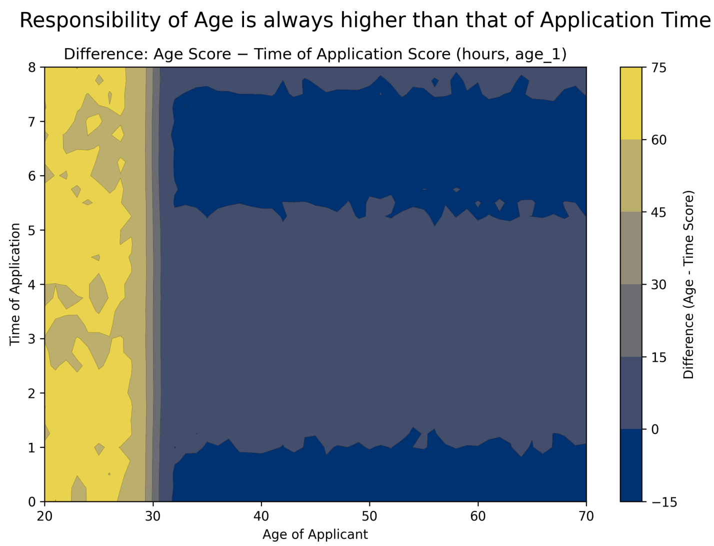


Figure 8: Same as Figure 7 on the age-40 prior.

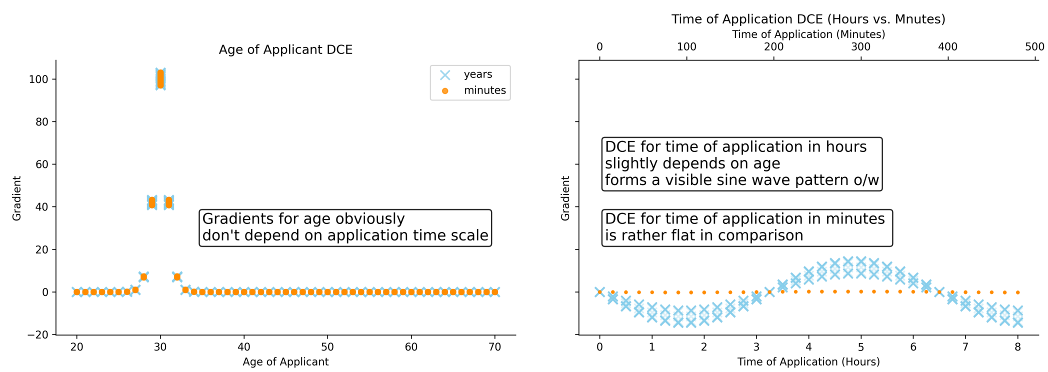


Figure 9: DCE for age (left) and time of application (right) over the input grid, on both unit scales. The age DCE is large only inside the narrow steep region of the sigmoid; outside it, time of application has the larger absolute DCE on the hour scale, and a comparatively flat DCE on the minute scale, illustrating both shape sensitivity and unit sensitivity.

7 Relation with Actual Causality

PCI is not meant to explicate Halpern's notion of actual causality; the goal is to combine the intuitions underlying Halpern's notion with those motivating Pearl's probabilities of necessity and sufficiency to provide a more general tool for causal attribution. Nevertheless, to show that we preserve the spirit of Halpern's definition, we now record the connection between actual causality and the joint necessity-sufficiency machinery of Section 3. The goal is to show that there is a natural choice of variable selection distribution Γ and alternative-value distribution Δ under which PCI verdicts agree with actual-causality verdicts. With one caveat: probability-of-causation maximisation will not identify subset-minimal causes that are not cardinality-minimal — a misalignment we characterise and remedy below.

Scope: a necessity-only correspondence. Halpern's definition involves factivity, context-sensitive necessity, and minimality, but no sufficiency clause analogous to Pearl's PS or PNS. The correspondence we establish in this section is therefore *necessity-only*: the sufficiency component of the kernel Φ defined below acts as a factivity check (restoring factual values must preserve φ under fixed noise), not as a substantive sufficiency-attribution component. The substantive sufficiency component of PCI — the Y^s factor in Definition 17 and the ci_S term in Section 5 — is exercised in the synthetic overdetermination and undercutting evaluation (Section 5), the differential causal effect comparison (Section 6), and the SIR benchmark (Section 10), all of which are settings where AC cannot adjudicate. The narrowness of this section's correspondence reflects what AC itself permits, not a limitation of PCI.

We first recall Halpern's definition.

Definition 25 (Actual Cause, after Halpern, 2016)**.** $\mathbf{C} = \mathbf{c}$ *is an actual cause of* φ *in the causal setting* $(M, \mathbf{u})$ *if the following three conditions hold:*

- ***Factivity****:* $(M, \mathbf{u}) \vDash (\mathbf{C} = \mathbf{c}\,)$ *and* $(M, \mathbf{u}) \vDash \varphi$*,*
- ***Context-sensitive Necessity****: There is a set of variables* $\mathbf{T} \subseteq \mathbf{V}$ *and an alternative setting* $\mathbf{c}'$ *of variables in* $\mathbf{C}$ *such that if* $(M, \mathbf{u}) \vDash \mathbf{T} = \mathbf{t}^\star$*, then*
$$(M, \mathbf{u}) \vDash [\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^\star]\neg\varphi,$$
- ***Minimality****:* $\mathbf{C}$ *is minimal; there is no strict subset* $\mathbf{C_{sub}} \subset \mathbf{C}$ *such that* $\mathbf{C_{sub}} = \mathbf{c_{sub}}$ *satisfies the above two conditions where* $\mathbf{c_{sub}}$ *is the restriction of* $\mathbf{c}$ *to the variables in* $\mathbf{C_{sub}}$*.*

To connect this to PCI, we instantiate the joint necessity-sufficiency machinery (Definition 17) at a single configuration $(\mathbf{C}, \mathbf{T})$ and a fixed noise realisation $\mathbf{u}$, and read off the configuration-level contribution. The resulting kernel combines the same three ingredients as Halpern's definition: a sufficiency check at factual values, a necessity check against alternatives, and a Γ-weight that records how much the configuration is being asked about.

Definition 26 (AC-aligned PCI kernel)**.** *Fix a PSCM* $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_\mathbf{U}\rangle$*,*[19] *a noise realisation* $\mathbf{u}$*, suspect and witness sets* $\mathbf{S}, \mathbf{W} \subseteq \mathbf{V}$ *with factual values* $\mathbf{s}^\star, \mathbf{w}^\star$*, a variable selection distribution* Γ *on* $2^\mathbf{S} \times 2^\mathbf{W}$ *with mass function* p^Γ*, and an alternative-value distribution* $\Delta(\mathbf{s}^\star)$*. For an outcome event* φ*, active suspect set* $\mathbf{C} \subseteq \mathbf{S}$ *and active witness set* $\mathbf{T} \subseteq \mathbf{W}$ *with* $\mathbf{T} \cap \mathbf{C} = \emptyset$*, define*

$$\begin{aligned}\Phi(\mathbf{C}, \mathbf{T}) \;:=\; & p^\Gamma(\mathbf{C}, \mathbf{T}) \;\cdot\; \mathbb{I}[(M, \mathbf{u}) \vDash [\mathbf{C} \leftarrow \mathbf{s}^\star|_\mathbf{C},\; \mathbf{T} \leftarrow \mathbf{w}^\star|_\mathbf{T}]\, \varphi] \\ & \cdot \int \mathbb{I}[(M, \mathbf{u}) \vDash [\mathbf{C} \leftarrow \mathbf{c}',\; \mathbf{T} \leftarrow \mathbf{w}^\star|_\mathbf{T}]\, \neg\varphi]\, \Delta_\mathbf{C}(\mathbf{s}^\star)(d\mathbf{c}'). \end{aligned} \tag{31}$$

Concretely, $\Phi(\mathbf{C}, \mathbf{T})$ is the matched-configuration slice of the joint necessity-sufficiency measure $P_k^{s,n}$ from Definition 17: integration over $P_\mathbf{U}$ is replaced by point evaluation at $\mathbf{u}$, and both the sufficiency and the necessity sums are pinned to the same $(\mathbf{C}, \mathbf{T})$ pair, mirroring AC's structure of testing factivity and the alternative against a single suspect/witness pair. The kernel is positive exactly when three conditions hold simultaneously: $(\mathbf{C}, \mathbf{T})$ has positive Γ-mass; restoring $\mathbf{C}$ to its factual values while holding $\mathbf{T}$ fixed preserves φ under $\mathbf{u}$ (the factivity-as-sufficiency component); and there exists a Δ-positive-density alternative $\mathbf{c}'$ for which the corresponding intervention overturns φ under $\mathbf{u}$ (the necessity component). The first factor takes the role played by the suspect/witness biases b_s, b_w in earlier work: regularity of Γ — positive p^Γ on every legal configuration — corresponds to the old $-0.5 < b_s, b_w < 0.5$ regularity. We will say that Γ is *regular* when $p^\Gamma(\mathbf{C}', \mathbf{T}') > 0$ for every $(\mathbf{C}', \mathbf{T}')$ with $\mathbf{C}' \subseteq \mathbf{S}, \mathbf{T}' \subseteq \mathbf{W}, \mathbf{C}' \cap \mathbf{T}' = \emptyset$.

We can now record the forward direction.

Theorem 27. *Suppose that* $\mathbf{C}$ *is an actual cause of* φ *with witness set* $\mathbf{T}$ *in* $(M, \mathbf{u})$*. Take any* $\{\mathbf{S} = \mathbf{s}\} \supseteq \mathbf{C}$ *as the potential cause set, subject to the constraint that no element of* $\mathbf{S}$ *is a deterministic function (under* M*) of any subset of the remaining elements of* $\mathbf{S}$*,*[20] *and any* $\mathbf{W} \supseteq \mathbf{T}$ *as the potential*

[19] The kernel and the theorems below take the full PSCM (structural equations $\mathbf{F}$ plus noise distribution $P_\mathbf{U}$) as given, since both the sufficiency and the necessity factors require sampling counterfactual outcomes under $\mathbf{u}$. As discussed in Section 1, this is a stronger commitment than approaches that bound $\mathrm{PN}/\mathrm{PS}/\mathrm{PNS}$ from observational data and a causal graph alone, but is the price of pointwise (not just bound) verdicts at the configuration level.

[20] Equivalently, $\mathbf{S}$ is closed under no nontrivial structural equation of M. The constraint excludes admitting both S and a deterministic descendant $H = f(S, \ldots)$ as candidates in the same suspect set, but covers all candidate sets standardly considered in the AC literature, where structural descendants of one candidate are not admitted as separate candidates — their contribution is mediated by the equation, and AC's minimality clause

witness set. Assume Γ is regular and that $\Delta(\mathbf{s})$ assigns positive density to every $\mathbf{s}' \in \operatorname{dom}(\mathbf{S}) \setminus \{\mathbf{s}\}$ (positive PMF in the discrete case, positive Lebesgue density in the continuous case). In the continuous case, assume additionally:

(i) the structural equations of M downstream of $\mathbf{C}$ are continuous in $\mathbf{c}'$ under fixed $\mathbf{u}$;

(ii) $\neg\varphi$ is open in $\operatorname{dom}(Y)$ (or, more weakly, the boundary of $\neg\varphi$ has measure zero under the pushforward of Y).

Then

$$\Phi(\mathbf{C}, \mathbf{T}) > 0, \tag{32}$$

and therefore

$$\sum_{\mathbf{T}' \subseteq \mathbf{W}} \Phi(\mathbf{C}, \mathbf{T}') \; > \; 0.$$

Proof. Fix $\mathbf{u}$, $\mathbf{c}'$, and $\mathbf{T}$ as witnesses to the satisfaction of the conditions in Definition 25. The context-sensitive-necessity clause forces $\mathbf{T} \cap \mathbf{C} = \emptyset$, since otherwise the joint intervention $[\mathbf{C} \leftarrow \mathbf{c}', \, \mathbf{T} \leftarrow \mathbf{t}^\star]$ would be ill-defined. It remains to show that each of the three factors in $\Phi(\mathbf{C}, \mathbf{T})$ is strictly positive.

- *Selection weight.* By regularity of Γ, $p^\Gamma(\mathbf{C}, \mathbf{T}) > 0$.
- *Sufficiency indicator.* With exogenous noise $\mathbf{u}$ held constant, factivity gives $(M, \mathbf{u}) \models \varphi$. Restoring $\mathbf{C}$ to $\mathbf{s}^\star\,|_{\mathbf{C}}$ and $\mathbf{T}$ to $\mathbf{w}^\star\,|_{\mathbf{T}}$ leaves the model agreeing with $(M, \mathbf{u})$ on every variable, so the indicator evaluates to 1.
- *Necessity integral.* Context-sensitive necessity provides a specific $\mathbf{c}'$ witnessing $(M, \mathbf{u}) \models [\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^\star]\neg\varphi$. The closure condition on $\mathbf{S}$ and the positive-density assumption on $\Delta(\mathbf{s})$ imply that $\mathbf{c}'$ lies in the support of $\Delta_{\mathbf{C}}(\mathbf{s}^\star)$. We treat the two cases separately. *Discrete case*: the singleton $\{\mathbf{c}'\}$ has positive PMF $\Delta_{\mathbf{C}}(\{\mathbf{c}'\}) > 0$ by hypothesis, so the integral is at least $\Delta_{\mathbf{C}}(\{\mathbf{c}'\}) > 0$. *Continuous case*: hypothesis (i) gives continuity of the intervention map $\mathbf{c}' \mapsto Y$ at fixed $\mathbf{u}$; hypothesis (ii) places $Y(\mathbf{c}')$ in the interior of $\neg\varphi$; pulling back an open neighbourhood within $\neg\varphi$ yields an open neighbourhood of $\mathbf{c}'$ on which the indicator equals 1, and this neighbourhood has positive Lebesgue measure and hence positive $\Delta_{\mathbf{C}}$-mass by the positive-density hypothesis. In either case the integral is strictly positive.

The product of three strictly positive factors is strictly positive, giving $\Phi(\mathbf{C}, \mathbf{T}) > 0$. Summing over $\mathbf{T}' \subseteq \mathbf{W}$ adds non-negative terms to a positive one, preserving positivity. The proof uses factivity and context-sensitive necessity but not minimality: the same conclusion holds whenever $\mathbf{C}$ satisfies factivity and context-sensitive necessity in $(M, \mathbf{u})$, a fact we will use in Proposition 33. ☐

Remark (when (i)–(ii) bite). In settings with continuous outcomes and an open event such as a threshold $\{Y > y_0\}$ — for instance the SIR overshoot benchmark of Section 10 — both hypotheses are satisfied trivially. Binary-output models, where Y takes values in $\{0, 1\}$, satisfy (ii) automatically (every subset of a discrete codomain is open) but may not satisfy (i) when the binary output comes

already prefers the ancestor alone. The same closure condition discharges a joint-support concern: the next assumption requires $\Delta(\mathbf{s})$ to place positive density on every *joint* alternative configuration $\mathbf{s}' \in \operatorname{dom}(\mathbf{S}) \setminus \{\mathbf{s}\}$, not merely positive marginal density on each component. In a chain $S \to H$ with $H = f(S)$, joint assignments such as $(S{=}0, H{=}1)$ are impossible under the structural equations, so any Δ that placed mass on them would be incompatible with M. Closure rules out exactly this case: when no element of $\mathbf{S}$ is a deterministic function of the remaining elements, the joint domain $\operatorname{dom}(\mathbf{S})$ contains no rectangles forced to zero measure by M, and a Δ with positive PMF or Lebesgue density on every $\mathbf{s}' \neq \mathbf{s}$ is consistent with M on the suspect set. In the stone-throwing benchmark of Section 9, the suspect set is $\{A_i, B_i\}_i$ and the deterministic mediators $\{W_i^a, W_i^b\}_i$ live in the witness set, not the suspect set — exactly to keep the suspect set closure-respecting. Witnesses are pinned at their factual values and are not subject to a joint-support requirement on alternatives, so they need no analogous condition.

from thresholding a continuous score. In such cases the proof falls back to the discrete branch, which only requires positive PMF on the witnessing $\mathbf{c}'$.

The opposite direction follows by reading the proof in reverse on the necessity factor.

Theorem 28. *Under the same hypotheses on $\{\mathbf{S} = \mathbf{s}\}$, Γ, and Δ as in Theorem 27, if*

$$\sum_{\mathbf{T}' \subseteq \mathbf{W}} \Phi(\mathbf{C}, \mathbf{T}') \;>\; 0,$$

then both factivity ($(M, \mathbf{u}) \models \varphi$) and the context-sensitive-necessity condition in Definition 25 hold for $\mathbf{C}$ in $(M, \mathbf{u})$.

Proof. Since Φ is non-negative, the assumption forces $\Phi(\mathbf{C}, \mathbf{T}') > 0$ for at least one $\mathbf{T}' \subseteq \mathbf{W}$. By Definition 26, each factor of $\Phi(\mathbf{C}, \mathbf{T}')$ is strictly positive. The sufficiency indicator equals 1, i.e. $(M, \mathbf{u}) \models [\mathbf{C} \leftarrow \mathbf{s}^\star|_{\mathbf{C}},\, \mathbf{T}' \leftarrow \mathbf{w}^\star|_{\mathbf{T}'}]\varphi$; since pinning variables to their factual values under fixed $\mathbf{u}$ leaves every variable unchanged, this is equivalent to $(M, \mathbf{u}) \models \varphi$, giving factivity. The necessity integral being positive provides a $\Delta_{\mathbf{C}}$-positive-density set of values $\mathbf{c}'$ for which $(M, \mathbf{u}) \models [\mathbf{C} \leftarrow \mathbf{c}',\, \mathbf{T}' \leftarrow \mathbf{t}^{\star\prime}]\neg\varphi$; any such $\mathbf{c}'$, together with $\mathbf{T}'$, witnesses the existential in the context-sensitive-necessity clause of Definition 25. ☐

Remark (deterministic intermediates and joint support). The hypothesis on $\{\mathbf{S} = \mathbf{s}\}$ is what rules out the following obstruction. Suppose the suspect set were closed under the structural equations of M — e.g., both Sally's throw S and the bottle's hit $H = f(S, \ldots)$ admitted as candidates. Then the joint assignment $(S{=}0, H{=}1)$ is structurally impossible, and any alternative-value distribution consistent with M assigns it density zero, so the joint full-support condition required to recover the witnessing $\mathbf{c}'$ from $\Phi > 0$ fails. Restricting suspects to variables that are not deterministic functions of the others under M removes this obstruction. The constraint is also conservative with respect to the AC literature: in the canonical stone-throwing setup, the candidate set is $\{S_{Sally}, S_{Bill}\}$, with H as the outcome rather than a candidate, and AC's own minimality clause already prefers ancestor candidates over their deterministic descendants when both are admitted. Sections 5 and 9 adopt the same convention experimentally — the continuous synthetic benchmark uses the six roots as suspects and the discrete scaled-throwing benchmark uses $\{A_i, B_i\}_i$ as suspects with the deterministic mediators $\{W_i^a, W_i^b\}_i$ as witnesses — in both cases deterministic intermediates are excluded from the suspect set. Cases where a user genuinely wants to attribute responsibility separately to a deterministic intermediate (rather than to its structural ancestors) lie outside the scope of these theorems and require an alternative reading of intervention semantics; we treat that as a separate problem.

With minimality, the situation is somewhat more complicated. Our choice of Γ will have an impact on which configurations maximise Φ. Moreover, Φ compares configurations by Γ-weight rather than by the subset relation, while Definition 25 formulates minimality strictly in terms of subsets, so some misalignment is to be expected. Some useful observations can be made nonetheless.

Definition 29 (Necessity dilution)**.** *A pair (M, Δ) satisfies* necessity dilution *on $\mathbf{S}$ if, writing $N(\mathbf{C}, \mathbf{T})$ for the necessity integral in Equation (31),*

$$N(\mathbf{C}^\sharp, \mathbf{T}^\sharp) \;\geq\; N(\mathbf{C}^\dagger, \mathbf{T}^\dagger)$$

for every $\mathbf{C}^\sharp \subsetneq \mathbf{C}^\dagger \subseteq \mathbf{S}$ and witnesses $\mathbf{T}^\sharp, \mathbf{T}^\dagger$ disjoint from the corresponding suspect set.

The first is that there is a simplified setting in which a positive value of Φ, under factivity, entails minimality with respect to the same parameters. Three conditions suffice:

- Γ_w is uniform over $2^{\mathbf{W}}$;
- Γ_s is *strictly cardinality-decreasing*, i.e. $p^{\Gamma_s}(\mathbf{C}_1) > p^{\Gamma_s}(\mathbf{C}_2)$ whenever $\mathbf{C}_1 \subsetneq \mathbf{C}_2$ (the modern counterpart of the old null witness bias and positive suspect preemption bias);
- the pair (M, Δ) satisfies necessity dilution on $\mathbf{S}$ (Definition 29).

The strategy used in the proof entails that we can safely marginalise over witnesses as well.

Theorem 30. *Consider the set of optimisers* $\Gamma^\star = \arg\max_{\mathbf{C}\subseteq\{\mathbf{S}=\mathbf{s}\},\,\mathbf{T}\subseteq\mathbf{W}} \Phi(\mathbf{C},\mathbf{T})$. *Assume* $(\mathbf{C}^\dagger,\mathbf{T}^\dagger) \in \Gamma^\star$ *with* $\Phi(\mathbf{C}^\dagger,\mathbf{T}^\dagger) > 0$*, that factivity is satisfied for* $\mathbf{C}^\dagger$ *(as in Definition 25), and that no element of* $\{\mathbf{S}=\mathbf{s}\}$ *is a deterministic function under* M *of the others. Let* Γ_w *be uniform on* $2^{\mathbf{W}}$ *and let* Γ_s *be strictly cardinality-decreasing, and assume that* (M,Δ) *satisfies necessity dilution on* $\mathbf{S}$*. Then* $\mathbf{C}^\dagger$ *is an actual cause of* φ *according to Definition 25.*

Proof. Factivity is assumed; by Theorem 28 the context-sensitive-necessity condition holds for $\mathbf{C}^\dagger$ in $(M,\mathbf{u})$. So the only way $\mathbf{C}^\dagger$ can fail to be an actual cause is by failing minimality. Suppose for contradiction that minimality fails: there is $\mathbf{C}^\sharp \subsetneq \mathbf{C}^\dagger$ satisfying factivity and context-sensitive necessity for φ with some witness $\mathbf{T}^\sharp$. By Theorem 27 (its proof requires only factivity and necessity, not minimality), $\Phi(\mathbf{C}^\sharp,\mathbf{T}^\sharp) > 0$ as well, with the sufficiency indicator and the necessity integral both equal to 1 and a strictly positive value, respectively. Both pairs therefore satisfy $\Phi(\mathbf{C},\mathbf{T}) = p^\Gamma(\mathbf{C},\mathbf{T})\cdot N(\mathbf{C},\mathbf{T})$.

By the construction of Γ from independent marginals (Section 3),

$$p^\Gamma(\mathbf{C},\mathbf{T}) \;\propto\; p^{\Gamma_s}(\mathbf{C})\, p^{\Gamma_w}(\mathbf{T})\, \mathbb{I}[\mathbf{T}\cap\mathbf{C}=\emptyset].$$

The $\mathbb{I}[\cdot]$ factor is 1 for both pairs (otherwise the contradiction would have been immediate). Uniformity of Γ_w on $2^{\mathbf{W}}$ makes $p^{\Gamma_w}(\mathbf{T}) = 2^{-|\mathbf{W}|}$ identically, so the witness factor cancels. Strict cardinality-decrease of Γ_s together with $\mathbf{C}^\sharp \subsetneq \mathbf{C}^\dagger$ gives $p^{\Gamma_s}(\mathbf{C}^\sharp) > p^{\Gamma_s}(\mathbf{C}^\dagger)$, and necessity dilution gives $N(\mathbf{C}^\sharp,\mathbf{T}^\sharp) \geq N(\mathbf{C}^\dagger,\mathbf{T}^\dagger)$. Multiplying these (and noting $N(\mathbf{C}^\sharp,\mathbf{T}^\sharp) > 0$ to convert weak into strict in the composite) gives $\Phi(\mathbf{C}^\sharp,\mathbf{T}^\sharp) > \Phi(\mathbf{C}^\dagger,\mathbf{T}^\dagger)$ — contradicting $(\mathbf{C}^\dagger,\mathbf{T}^\dagger)\in\Gamma^\star$. ☐

Corollary 31. *Keep the notation and hypotheses of Theorem 30. Consider the set of optimisers* $\Delta^\star = \arg\max_{\mathbf{C}\subseteq\{\mathbf{S}=\mathbf{s}\}} \sum_{\mathbf{T}\subseteq\mathbf{W}} \Phi(\mathbf{C},\mathbf{T})$. *Assume* $\mathbf{C}^\dagger \in \Delta^\star$ *with* $\sum_{\mathbf{T}\subseteq\mathbf{W}} \Phi(\mathbf{C}^\dagger,\mathbf{T}) > 0$ *and that factivity is satisfied for* $\mathbf{C}^\dagger$*. Then* $\mathbf{C}^\dagger$ *is an actual cause of* φ *according to Definition 25.*

Proof. The summation hypothesis entails the existence of some $\mathbf{T}^\dagger \subseteq \mathbf{W}$ with $\Phi(\mathbf{C}^\dagger,\mathbf{T}^\dagger) > 0$. Because the slice $\{(\mathbf{C}^\dagger,\mathbf{T}) : \mathbf{T}\subseteq\mathbf{W}\}$ is contained in the joint search space $2^{\mathbf{S}}\times 2^{\mathbf{W}}$, we have $(\mathbf{C}^\dagger,\mathbf{T}^\dagger)\in\Gamma^\star$ as well. The conditions of Theorem 30 are satisfied, so $\mathbf{C}^\dagger$ is an actual cause of φ. ☐

When necessity dilution holds. The hypothesis is a property of the pair (M,Δ), not of Γ. It says that adding suspects to a candidate cause does not introduce new Δ-mass of flipping alternatives, and three settings make it automatic. *(i) Single-pivot models*: one variable $X^\star\in\mathbf{S}$ is pivotal under $(M,\mathbf{u})$ and the others are structurally idle, so N peaks at $\{X^\star\}$ and falls off as more suspects are added. *(ii) Factored* Δ *without Boolean-OR redundancy*: $\Delta_{\mathbf{C}}$ is a product of marginals $\bigotimes_{X\in\mathbf{C}}\nu_X$ with $\nu_X(\mathrm{dom}\,(X)) \leq 1$, and the flip event factorises coordinate-wise; $N(\mathbf{C})$ is then a product of factors each at most 1, decreasing in $|\mathbf{C}|$. *(iii) Conjunctive outcome events*: if $\neg\varphi$ requires flipping every variable in a required set $\mathbf{R}\subseteq\mathbf{S}$, the flip region's $\Delta_{\mathbf{C}}$-mass decreases as $\mathbf{C}$ grows beyond $\mathbf{R}$. The hypothesis fails in disjunctive overdetermination (e.g. $\varphi :\ A\vee B = 1$ with both A, B factual): no proper subset of $\mathbf{S}$ flips φ, so N is zero on subsets and positive on the full set — the opposite of dilution. In such cases no proper subset of $\mathbf{S}$ is an actual cause anyway, so Theorem 30 is not the appropriate tool; Proposition 33 is.

We can now return to the misalignment we mentioned earlier. One can imagine a reasoning mode in which one looks for causes of an outcome by inspecting the integrated probability $P(\mathbf{C}\rightsquigarrow\varphi) := \int\sum_{\mathbf{T}}\Phi(\mathbf{C},\mathbf{T})\,P_{\mathbf{U}}(d\mathbf{u})$, not knowing what the values of $\mathbf{u}$ are but knowing that φ occurred. It is worth pointing out that $\arg\max_{\mathbf{C}\subseteq\mathbf{S}} P(\mathbf{C}\rightsquigarrow\varphi)$ might fail to include some $\mathbf{C}$ that is an actual cause of φ for some setting $\mathbf{u}$.

Observation 32. *Define* $\Xi^{\max} = \arg\max_{\mathbf{C}\subseteq\mathbf{S}} P(\mathbf{C}\rightsquigarrow\varphi)$*. It is not the case that if* $\mathbf{C}^\dagger$ *is an actual cause in* $(M,\mathbf{u})$ *for some* $\mathbf{u}$*,* $\mathbf{C}^\dagger\in\Xi^{\max}$*.*

Proof. Consider a model with two binary parents A, B uniformly distributed and a binary child $Y = A\vee B$. Take $\varphi :\ Y=1$ and the suspect set $\mathbf{S}=\{A,B\}$ with $\mathbf{W}=\emptyset$. Compare $\mathbf{C}^\dagger=\{A,B\}$ at noise $\mathbf{u}_{11}$ $(A=B=1)$ and $\mathbf{C}^\sharp=\{A\}$ at noise $\mathbf{u}_{10}$ $(A=1, B=0)$.

- In $\mathbf{u}_{11}$, $\mathbf{C}^\dagger$ is an actual cause: setting both parents to 0 yields $Y = 0$, no proper subset achieves this, and the indicators in Φ are 1. With Γ_w uniform on $2^\emptyset$ (a single configuration of weight 1) and a strictly cardinality-decreasing Γ_s, $\Phi(\mathbf{C}^\dagger, \emptyset) \propto p^{\Gamma_s}(\{A, B\})$.

- In $\mathbf{u}_{10}$, $\mathbf{C}^\sharp$ is an actual cause: intervening to $A = 0$ flips Y to 0, the singleton is minimal, and again the indicators are 1. So $\Phi(\mathbf{C}^\sharp, \emptyset) \propto p^{\Gamma_s}(\{A\})$.

Integrating uniformly over the four noise realisations $\{\mathbf{u}_{ij}\}$, $P(\mathbf{C}^\dagger \rightsquigarrow \varphi) \propto p^{\Gamma_s}(\{A, B\})$ (only $\mathbf{u}_{11}$ contributes) while $P(\mathbf{C}^\sharp \rightsquigarrow \varphi) \propto 2\, p^{\Gamma_s}(\{A\})$ ($\mathbf{u}_{10}$ and $\mathbf{u}_{11}$ both contribute). Whether $\mathbf{C}^\dagger$ or $\mathbf{C}^\sharp$ ends up in $\Xi^{\max}$ depends on the precise shape of Γ_s: under sharply decreasing Γ_s, $\mathbf{C}^\sharp$ wins; under shallow decrease, $\mathbf{C}^\dagger$ may. Both cannot belong to $\Xi^{\max}$ simultaneously in general, so some actual causes are dropped. □

The reason is structural: actual cause sets may be subset-minimal without being cardinality-minimal, while integrated-probability maximisation is governed by Γ_s, which (under the cardinality-decreasing parameterisation that aligns with AC's minimality) penalises larger sets. The misalignment is therefore not a defect of Φ but a mismatch between AC's notion of minimality (subset relation) and Γ-weighted scoring.

The fix is to drop the maximisation step and use Φ directly to enumerate candidate causes, then filter for subset-minimality. The filter recovers exactly the actual causes in $\mathbf{S}$.

Proposition 33. *Under the hypotheses of Theorem 27, define*

$$\Xi \;:=\; \Big\{ \mathbf{C}^\dagger \subseteq \mathbf{S} \;:\; \sum_{\mathbf{T} \subseteq \mathbf{W}} \Phi(\mathbf{C}^\dagger, \mathbf{T}) > 0,\; \nexists\, \mathbf{C}' \subsetneq \mathbf{C}^\dagger \text{ with } \sum_{\mathbf{T} \subseteq \mathbf{W}} \Phi(\mathbf{C}', \mathbf{T}) > 0 \Big\}.$$

Then Ξ contains exactly the actual causes of φ in $(M, \mathbf{u})$ that are subsets of $\mathbf{S}$.

Proof. Ξ-membership $\Rightarrow$ actual cause. Let $\mathbf{C}^\dagger \in \Xi$. Theorem 28 applied to $\sum_{\mathbf{T}} \Phi(\mathbf{C}^\dagger, \mathbf{T}) > 0$ gives both factivity and context-sensitive necessity. For minimality, suppose $\mathbf{C}_{\text{sub}} \subsetneq \mathbf{C}^\dagger$ satisfied factivity and context-sensitive necessity. By the proof of Theorem 27 — which uses only factivity and necessity, not minimality — we would have $\sum_{\mathbf{T}} \Phi(\mathbf{C}_{\text{sub}}, \mathbf{T}) > 0$, contradicting the no-positive-proper-subset clause of Ξ.

Actual cause $\Rightarrow$ Ξ-membership. Let $\mathbf{C}^\dagger \subseteq \mathbf{S}$ be an actual cause of φ in $(M, \mathbf{u})$. Theorem 27 gives $\sum_{\mathbf{T}} \Phi(\mathbf{C}^\dagger, \mathbf{T}) > 0$. For the no-positive-proper-subset clause: suppose $\mathbf{C}' \subsetneq \mathbf{C}^\dagger$ had $\sum_{\mathbf{T}} \Phi(\mathbf{C}', \mathbf{T}) > 0$. Then by Theorem 28, both factivity and context-sensitive necessity would hold for $\mathbf{C}'$. So $\mathbf{C}'$ would witness a violation of $\mathbf{C}^\dagger$'s minimality, contradicting the assumption that $\mathbf{C}^\dagger$ is an actual cause. □

Proposition 33 converts the cardinality–subset misalignment into a procedure: enumerate $\{\mathbf{C} \subseteq \mathbf{S} : \sum_{\mathbf{T}} \Phi(\mathbf{C}, \mathbf{T}) > 0\}$ (which by Theorem 28 are exactly the cause sets satisfying context-sensitive necessity), filter for factivity and subset-minimality, and read off the actual causes. Some idiosyncratic features of the AC notion — in particular, the subset-minimality clause, which interacts poorly with Γ-weighted scoring — are part of what motivated the more scalable PCI formulation of Section 3 in the first place; the filter procedure here is the right tool when AC verdicts specifically are wanted, but expectation-based PCI attributions are what scale to ML-grade models.

Summary of guarantees. The section delivers, under explicit hypotheses on $\mathbf{S}$, Γ, and Δ: (i) a forward direction (Theorem 27) "actual cause $\Rightarrow$ $\Phi > 0$"; (ii) a backward direction (Theorem 28) "$\sum_{\mathbf{T}'} \Phi > 0 \Rightarrow$ context-sensitive necessity"; (iii) minimality preservation under cardinality-decreasing Γ_s (Theorem 30, Corollary 31); (iv) a counterexample (Observation 32) showing that integration over noise misaligns with subset-minimality; and (v) a corrective filter procedure (Proposition 33) that recovers AC verdicts exactly. The empirical counterpart of the correspondence — a benchmark of the necessity-only PCI estimator against exact subset enumeration on a scaled stone-throwing problem — is reported in Section 9.

8 Relation with the Probability of Actual Causality

Now we are in the position to disscuss the relation between PCI and Pearl's probability of actual causation. On Pearl's Definition 10.3.5 [Pearl, 2009, Ch. 10],

$$P(\text{caused}(x, y \mid e)) \;=\; \frac{P(U_{xy} \cap U_e)}{P(U_e)}, \qquad U_{xy} = \{\mathbf{u}\colon \mathrm{AC}(x, y; M, \mathbf{u})\}, \tag{33}$$

is the posterior probability, given the evidence, that we are in a noise state for which the actual-cause verdict holds, with $\mathrm{AC}(x, y; M, \mathbf{u})$ a chosen actual-causation predicate — the causal-beam definition of [Pearl, 2009, Ch. 10] or the Halpern–Pearl AC1–AC3 [Halpern, 2016].[21] The same construction underwrites Chockler and Halpern's degree of blame [Chockler and Halpern, 2004], which substitutes a graded responsibility share inside the expectation. Because (33) is the closest construction in the literature to PCI's expectation in Definition 17, we now ask in detail how it differs from PCI and how PCI improves on it.

We proceed in three steps. We replay Pearl's canonical desert-traveller illustration and Pearl's binary actual-cause verdicts; compute PCI on the same example and show it recovers Pearl's ranking with graded magnitudes; and introduce a weak-poison variant that exhibits three intuitive responsibility patterns — two of which Pearl's binary verdict cannot capture but PCI's graded reading does, and one (within-scenario ranking) Pearl preserves. Then we move to discuss other differences between PCI and PCI, which make the former more general and more scalable.

8.1 Pearl's worked example: the desert traveller

We start with the canonical illustration used by Pearl [Pearl, 2009, §10.3.3].

Example 34 (Desert traveller)**.** *A traveller has two enemies. The* shooter *(enemy 2) shoots and empties the canteen (X=1); the* poisoner *(enemy 1), unaware, puts cyanide in the water (P=1). The traveller dies (Y=1). Who is the actual cause of death?*

The example involves one source of uncertainty packed into a binary noise u:

- $u = 0$ — traveller drank the poisoned water before the canteen was emptied; cyanide killed him.
- $u = 1$ — canteen was empty before he could drink; dehydration killed him.

There are at least two intuitions here that we would like to recover. One, in any particular case, we want to be able to identify the actual cause. The other, perhaps a bit weaker, we still don't want to say that the dormant cause is completely blameless. Actual causality gives us the first half of the story, assigning full responsibility to one source here. Pearl's beam analysis [Pearl, 2009, Ch. 10] (equivalently Halpern–Pearl AC1–AC3 of Halpern, 2016) gives the actual-cause verdicts at each deterministic state:

- In $u = 0$: P=1 is the actual cause; X=1 is not.
- In $u = 1$: X=1 is the actual cause; P=1 is not.

So $U_{x,y} = \{u = 1\}$ and $U_{p,y} = \{u = 0\}$. Two natural forensic reports each push Pearl's posterior to certainty (Table 10): toxicology finding cyanide in the body is incompatible with $u = 1$, so $P(\text{caused poisoner} \mid e_A) = 1$; a forensic report of no cyanide is incompatible with $u = 0$, so $P(\text{caused shooter} \mid e_B) = 1$.

8.2 PCI on the basic desert traveller

To compare directly with Pearl, we run PCI under the same forensic conditioning: one structural model per scenario, with u pinned at the scenario's forensic realisation. The endogenous variables are $X, P, c, d, Y \in \{0, 1\}$ with

$$c = P\,(u' \vee X'), \qquad d = X\,(u \vee P'), \qquad y = c \vee d,$$

[21] All computations behind the comparison in this section — Pearl's original desert-traveller and the weak-poison variant — are collected in the companion notebook `docs/source/desert_traveler.ipynb`, which runs the framework's `ThinSearchSampler` on a scenario-specific pyro model for each forensic posterior and a parallel hand-rolled enumeration; the two agree within Monte Carlo noise on every value reported below.

Forensic posterior	P(caused shooter)	P(caused poisoner)
e_A: cyanide detected ($u = 0$)	0	**1**
e_B: no cyanide ($u = 1$)	**1**	0

Table 10: Pearl's probability of actual causality $P(\text{caused} \mid e)$ on the basic desert traveller. The verdict saturates at 1 for the cause whose path operated in the forensically identified scenario, and at 0 for the dormant one.

where c indicates the cyanide path operating (poisoned water drunk) and d the dehydration path operating (drank from empty canteen). The PCI components, all specified in advance:

- Suspects $\mathbf{S} = \{X, P\}$; candidate causes are the singletons $\{X\}$ and $\{P\}$.
- Witness pool $\mathbf{W} = \{c, d\}$ (the mediators); four subsets $\mathbf{T} \subseteq \mathbf{W}$, weighted by Γ cardinality-uniform (uniform on $|\mathbf{T}|$, then uniform within each cardinality class).
- Non-cause suspect (the one not in $\mathbf{C}$) is *not* intervened on or hard-coded; it follows the SCM under the noise prior on its own coin — consistent with the formal definition's requirement that only $\mathbf{C}$ and $\mathbf{T}$ are intervened (Definition 17).
- Alternative-value distribution Δ deterministically flips $1 \to 0$.
- Noise distribution $P_{\mathbf{U}}$ is the forensic posterior — a point mass at the scenario's u. We do *not* condition on the outcome Y itself; the forensic evidence sits above Y in the SCM, so conditioning on it is not the "fully conditional" anti-pattern of Section 3 (sec3:330–336).

For each candidate cause C we report three quantities, all in $[0, 1]$, higher meaning more responsibility, jointly averaged over the four sources of randomness (Γ, Δ, $P_{\mathbf{U}}$, and the SCM-driven non-cause suspect):

$$\begin{aligned} N_C &= \mathbb{E}\big[\mathbb{1}\{Y^n \neq y^\star\}\big] && \text{(necessity — removal flips } Y\text{)}, \\ S_C &= \mathbb{E}\big[\mathbb{1}\{Y^s = y^\star\}\big] && \text{(sufficiency — holding at factual sustains } Y\text{)}, \\ J_C &= \mathbb{E}\big[\mathbb{1}\{Y^n \neq y^\star\}\mathbb{1}\{Y^s = y^\star\}\big] && \text{(joint PNS-style).} \end{aligned} \tag{34}$$

Under the spec above (computation in the companion notebook), the results are:

Forensic posterior	Cause	N	S	J	Pearl P(caused)
e_A: $u = 0$ (cyanide)	shooter	$1/4$	$11/12$	$5/24$	0
	poisoner	$1/3$	1	$\mathbf{1/3}$	**1**
e_B: $u = 1$ (dehydration)	shooter	$1/3$	1	$\mathbf{1/3}$	**1**
	poisoner	$1/4$	$11/12$	$5/24$	0

Table 11: PCI and Pearl on the basic desert traveller, under the scenario's forensic posterior. PCI's joint score J ranks the same cause Pearl's P(caused) does in each scenario; PCI returns a graded reading where Pearl returns the binary $\{0, 1\}$.

Three points to read off Table 11:

(1) PCI and Pearl agree on the ranking. In each scenario, J is strictly larger for the cause Pearl identifies, and strictly smaller for the other. PCI's ordering matches the actual-cause verdict at each forensic posterior.

(2) PCI is graded; Pearl is binary. Pearl's P(caused) saturates at 0 or 1. PCI's J never reaches 1 here because the Γ-averaging over witness subsets dilutes each cause's score: the "best" witness subset for the operative cause does flip the outcome deterministically and gives $J = 1$ at that single configuration, but the average over the four subsets gives $1/3$. The cap $1/3$ is exactly $1/4 \times 1$ contribution from the one decisive configuration plus partial contributions from the others.

(3) The dormant cause is not at zero. PCI assigns $5/24 \approx 0.21$ to the dormant cause. This is not noise: in some witness configurations the dormant cause's removal does flip Y, because the non-cause suspect is marginalised and the model is not artificially pinned to "this scenario" beyond the noise realisation itself. Pearl's binary indicator throws this graded information away.

8.3 PCI expectation vs the AC machinery

The agreement above is reassuring, but the more interesting observation is what changes underneath. AC's necessity clause is an *existential*: *there exists* a witness set $\mathbf{T}$ and an alternative value c' such that the intervention flips Y. To verify the existential one in principle enumerates the candidate $(\mathbf{T}, c')$ pairs — this is the main source of AC's combinatorial hardness. Pearl's definition then averages the indicator of that already-enumerated predicate over noise states, inheriting the enumeration unchanged.

PCI collapses both moves into one expectation: Φ is averaged over the joint $(\mathbf{C}, \mathbf{T}, \mathbf{u})$ under $\Gamma \otimes P_{\mathbf{U}}$. The existential "does *some* witness work?" becomes the quantitative "*what fraction* of witnesses work?" — the information AC uses only one piece of, exposed as a graded score with sampling-based estimation.

In the desert traveller this is small change: four candidate witness sets, all trivially enumerable. The point is the scaling. In a model with n candidate witness variables, AC's existential demands checking up to 2^n subsets per noise state, and Pearl's posterior over the AC indicator inherits that cost per evaluation. PCI's expectation can be estimated at a Monte Carlo budget the user controls, with variance independent of the problem's combinatorial size. The desert-traveller agreement is the small-n end of a curve whose large-n end is the computational gap that motivates PCI in the first place.

Sso far we have two conceptual differences: PCI's graded reading and its Monte Carlo scaling. We have already seen that this helps to recover the intuiton that the shooter of a poisoned victim, while less blameworthy is nevertheless not completely blameless. Now we turn to another example, riding on the fact that actual causality is devoid of a sufficiency component, where a small change to the model makes Pearl's binary indicator structurally unable to track the responsibility patterns we think a responsibility attribution method should capture.

8.4 Weak-poison variant: two intuitions Pearl misses

We modify the desert traveller so the cyanide path is unreliable while the dehydration path stays deterministic. The change is small, but it surfaces three intuitions that a graded responsibility metric should capture and a binary actual-cause indicator structurally cannot.

Example 35 (Desert traveller, weak poison)**.** *Same enemies as in Example 34. The cyanide is a small dose that turns out fatal only with probability $\alpha = 0.1$, governed by an independent exogenous coin $\xi \sim \mathrm{Bern}(\alpha)$. The structural equations are*

$$c = p\,(u' \vee x'), \quad v_C = c \cdot \xi, \quad d = x\,(u \vee p'), \quad y = v_C \vee d,$$

where v_C is the new "cyanide-was-fatal" mediator: drinking the poison ($c = 1$) only kills when the fatality coin also operates ($\xi = 1$). $\alpha = 1$ recovers Pearl's original.

Three responsibility intuitions for an attribution method. Before any computation, the three intuitions that we expect an attribution method to recover in this case are:

1. **Adding noise to a path should lower its own cause's responsibility.** The cyanide path is now unreliable. In the scenario where cyanide killed (Scenario A, $u = 0, \xi = 1$), the poisoner is still the actual cause — but *less* responsible than in the basic deterministic version, because the killing depended on a coin landing favourably.
2. **Within a scenario, the operative cause should still win.** In Scenario A, even after softening, the poisoner should rank above the shooter. In Scenario B, the shooter should rank above the poisoner.
3. **A reliable cause in its own scenario should beat an unreliable cause in its own scenario.** Comparing across scenarios: the shooter in Scenario B (dehydration deterministic) should outrank the poisoner in Scenario A (cyanide stochastic). Their respective paths are operative in their own scenarios, but one is reliable and the other is not.

Pearl's $P(\text{caused})$ captures (2) — the within-scenario ranking — but misses (1) and (3). The AC predicate fires deterministically once we condition on the forensic posterior: in Scenario A, $u = 0$ and $\xi = 1$ jointly imply the cyanide path was the operative chain, so $P(\text{caused poisoner} \mid e_A) = 1$; in Scenario B, $u = 1$ implies the dehydration chain operated, so $P(\text{caused shooter} \mid e_B) = 1$. Both scenarios saturate at 1 for their respective operative cause, identical to the basic version (Table 12). Pearl's binary indicator is structurally indifferent to whether the cyanide path is deterministic or stochastic, and indifferent to a cross-scenario comparison.

Forensic posterior	$P(\text{caused shooter})$	$P(\text{caused poisoner})$
e_A: cyanide killed ($u = 0, \xi = 1$)	0	**1**
e_B: dehydration killed ($u = 1$)	**1**	0

Table 12: Pearl's $P(\text{caused})$ on the weak-poison variant. Identical to the basic version (Table 10); the binary AC indicator is insensitive to α.

What PCI does on the same model. We run PCI exactly as in Section 8.2, with one mechanical change: the witness pool becomes $\mathbf{W} = \{c, d, v_C\}$ (the new mediator v_C is included), and the forensic posterior in Scenario A pins both $u = 0$ and $\xi = 1$ (cyanide-was-fatal is directly forensically observable). Scenario B's posterior pins $u = 1$; ξ is structurally irrelevant since the cyanide path is dormant.

Table 13 reports PCI's three quantities alongside Pearl's verdict. The values are computed in the companion notebook and verified by a hand-rolled enumeration matching the framework.

Forensic posterior	Cause	N	S	J	Pearl $P(\text{caused})$
e_A: $u = 0, \xi = 1$ (cyanide killed)	shooter	≈ 0.167	≈ 0.958	≈ 0.146	0
	poisoner	≈ 0.208	1	$\approx \mathbf{0.208}$	**1**
e_B: $u = 1$ (dehydration killed)	shooter	$1/2$	1	$\mathbf{1/2}$	**1**
	poisoner	$1/4$	≈ 0.750	≈ 0.125	0

Table 13: PCI and Pearl on the weak-poison desert traveller, under the scenario's forensic posterior. PCI's J ranks the same cause Pearl's binary verdict does in each scenario; the magnitudes also reflect the asymmetry between the deterministic dehydration path and the stochastic cyanide path.

The three intuitions are all captured by J. **(1) Adding noise lowers the noisy cause's responsibility.** The poisoner in Scenario A drops from $J = 1/3$ in the basic version (Table 11) to $J \approx 0.208$ in the weak version (Table 13). The drop comes through the sufficiency factor: when we hold P at its factual value, the witness mechanism averages over configurations where v_C is not pinned and the cyanide step's ξ coin still acts. Even at $\xi = 1$ in the forensic posterior, the witness integration sees the fragility, and J falls. **(2) Within a scenario, the operative cause still wins.** In Scenario A, $J_{\text{poisoner}} \approx 0.208 > J_{\text{shooter}} \approx 0.146$. Cyanide was the operative path, and PCI tracks that. In Scenario B, $J_{\text{shooter}} = 1/2 > J_{\text{poisoner}} \approx 0.125$. **(3) Reliable beats unreliable, across scenarios.** The shooter in Scenario B ($J = 1/2$) outranks the poisoner in Scenario A ($J \approx 0.208$). The dehydration path is deterministic; conditional on $u = 1$, holding X at factual sustains Y with probability 1 regardless of the witness regime, so the shooter is a textbook responsible cause. The cyanide path is fragile; conditional on $u = 0, \xi = 1$, the sufficiency factor still sees the ξ coin under unpinned witness regimes, so the poisoner cannot hit the same ceiling. Pearl's PAC misses the desiderata for the following reasons. (1) requires sensitivity to mechanism noise, which the AC indicator does not have once the forensic posterior is applied; (3) requires cross-scenario comparison of *strength of responsibility*, which Pearl's $P(\text{caused})$ does not produce because each posterior independently saturates at 0 or 1;

8.5 Further differences between Pearl's probability of causation and PCI

Breadth of aggregation. Def. 10.3.5 aggregates over one dimension: the noise $\mathbf{u}$, conditional on evidence e, for a claim $(X = x, Y = y)$ already fixed. PCI aggregates over four: the suspect set $\mathbf{C}$ and witness set $\mathbf{T}$ via Γ, the alternative value $\mathbf{c}'$ via Δ, and the noise via $P_{\mathbf{U}}$. This is what supports per-feature comparison — the same PCI expectation, evaluated over varying $\mathbf{C}$ inside a fixed $\mathbf{S}$, returns the relative responsibility of each candidate cause. Def. 10.3.5 cannot perform that comparison without a separate evaluation per claim, each inheriting the AC-predicate cost discussed above.

Continuous variables. The AC predicates inside Def. 10.3.5 are event-shaped: they read off $\mathrm{AC}(x, y; M, \mathbf{u})$ for propositional x, y, and extensions to continuous variables require discretising the relevant event before the predicate applies. PCI's ci function is continuous-native (the absolute-difference impact score of Section 3), and the synthetic benchmark of Section 5 exercises that continuity directly.

Lineage in Pearl's PN/PS. Def. 10.3.5 takes a binary actual-causation predicate — itself a structural-equation construction that does not appeal to PN/PS at all — and lifts it to a probability through $P(\mathbf{u} \mid e)$; PN, PS, and PNS appear nowhere in the construction. PCI, by contrast, generalises PN and PS directly: the necessity factor Y^n is the counterfactual operation underlying PN, the sufficiency factor Y^s is the counterfactual operation underlying PS, both indexed by the cause–witness–alternative–noise tuple, with Γ and Δ exposing the design choices PN/PS makes implicitly. So Pearl's Def. 10.3.5 and PCI are not competing implementations of the same idea: they inherit from different parts of Pearl's own toolkit, with PCI playing the role of a context-sensitive, multi-variable extension of the PN/PS machinery rather than a probabilistic wrapper around AC.

Pearl's Def. 10.3.5 and PCI share the same explanatory target — probabilistic responsibility built on top of structural information — but instantiate it through different mechanisms. Pearl's definition averages a binary AC indicator against a posterior over noise; PCI integrates a continuous necessity–sufficiency kernel against a joint distribution over suspect sets, witness sets, alternative values, and noise. The basic desert traveller shows the two agree on ranking with PCI providing graded magnitudes; the weak-poison variant shows three intuitions that PCI's graded reading captures and Pearl's binary indicator structurally cannot. The cases where sufficiency does even more graded work — the continuous signal-mediation walkthrough of Section 4.3, the synthetic archetypes of Section 5, and the SIR benchmark of Section 10 — are where these constructions answer recognisably different questions, and the question PCI answers is the one we need for ML-grade causal attribution.

9 Scaling actual-cause computation: an empirical comparison

The correspondence in Theorems 27–28 and Proposition 33 is a statement about verdicts: under the stated hypotheses, AC actual causes coincide with the support of $\sum_{\mathbf{T}} \Phi$. This section reports an empirical companion to the correspondence. We ask whether the necessity-only PCI kernel of Definition 26, instantiated with a finite Monte Carlo estimator, recovers AC verdicts at problem sizes where exact subset enumeration is no longer tractable. The full implementation, raw outputs and plotting code are in the companion notebook `docs/source/actual_causality_benchmark.ipynb`.

Generalised throwing problem. For a size parameter n we replicate the canonical Sally/Billy structure n times in parallel. Binary suspects $A_1, \ldots, A_n, B_1, \ldots, B_n \in \{0, 1\}$ feed deterministic mediators $W_i^a = A_i, \quad W_i^b = B_i \wedge \neg A_i$ that capture overdetermination via preemption: B_i contributes only if A_i does not. The outcome is the conjunction $C = \bigwedge_{i=1}^n (W_i^a \vee W_i^b)$, and the model size grows as $4n+1$ variables. Ground-truth responsibility at site i is read off during the forward pass: A_i is responsible if $W_i^a = 1$, B_i is responsible if $W_i^b = 1$. The structure scales the canonical example without softening it — overdetermination, undercutting, and witness-mediated symmetry-breaking are present at every site — while keeping a closed-form ground truth available for evaluation.

The variable partition takes the roots $\{A_i, B_i\}_i$ as suspects and the deterministic mediators $\{W_i^a, W_i^b\}_i$ as witnesses, with no suspect a deterministic function of the others. This partition satisfies the closure condition required by Theorem 27 (see the Remark on deterministic intermediates and joint support, which addresses precisely the scenario in which both a root and its structural descendant are admitted as suspects). The benchmark is therefore within the scope of the formal correspondence; in particular, joint full support of the alternative-value distribution holds on the admissible suspect set $\{A_i, B_i\}_i$ even though the model itself is deterministic, because witnesses are not themselves part of the alternative-value distribution.

Exact subset enumeration (AC reference). The exact procedure mechanises Definition 25: for each non-empty cause subset $\mathbf{C} \subseteq \{A_i, B_i\}_i$ and each non-empty witness subset $\mathbf{T} \subseteq \{W_i^a, W_i^b\}_i$, we intervene $[\mathbf{C} \leftarrow \mathbf{1} - \mathbf{c}^\star, \mathbf{T} \leftarrow \mathbf{w}^\star]$, record whether the outcome flips, and accept $\mathbf{C}$ as actual cause if some witness $\mathbf{T}$ achieves a flip and no proper subset of $\mathbf{C}$ does. The search space has $\left(2^{2n} - 1\right)^2$ cause-witness pairs — $9, 225, 3969, 65025$ for $n = 1, 2, 3, 4$ — which already determines the ceiling reached by the exact method.

Approximate PCI estimator. The PCI estimator instantiates the kernel of Definition 26 via Monte Carlo using the `ThinSearchSampler` of Section 3. Algorithm 1 is the explicit procedure used to produce the curves in Figures 10–12.

Algorithm 1 Approximate PCI estimator on the scaled throwing problem.

Require: PSCM M; factual world $(\mathbf{s}^\star, \mathbf{w}^\star, c^\star)$; cardinality bounds n_s, n_w; sample budget N; witness flag wit $\in \{\text{on}, \text{off}\}$
Ensure: per-site verdicts $v_i \in \{A_i, B_i\}$ for $i = 1, \dots, n$
1: **if** wit $=$ off **then**
2: $\quad \mathbf{W} \leftarrow \emptyset$ ▷ no witness candidates
3: **end if**
4: **for** each $X \in \mathbf{S}$ **do**
5: $\quad L[X] \leftarrow [\,]$ ▷ per-suspect score list
6: **end for**
7: **for** $j = 1, \dots, N$ **do**
8: $\quad$ draw $\mathbf{u}_j \sim P_{\mathbf{U}}$
9: $\quad$ sample $k_s \sim \text{Unif}\{1, \dots, n_s\}, \quad k_w \sim \text{Unif}\{0, \dots, n_w\}$
10: $\quad$ sample $\mathbf{C}_j \subseteq \mathbf{S}$ uniformly with $|\mathbf{C}_j| = k_s$
11: $\quad$ sample $D_j \subseteq \mathbf{W}$ uniformly with $|D_j| = k_w$ ▷ witnesses dropped
12: $\quad \mathbf{T}_j \leftarrow \mathbf{W} \setminus D_j$ ▷ active witnesses
13: $\quad \mathbf{c}'_j \leftarrow \mathbf{1} - \mathbf{s}^\star|_{\mathbf{C}_j}$ ▷ bit-flip on $\{0, 1\}$
14: $\quad y_j^s \leftarrow M.\text{run}\big(\mathbf{u}_j,\, \text{do}(\mathbf{C}_j = \mathbf{s}^\star|_{\mathbf{C}_j},\, \mathbf{T}_j = \mathbf{w}^\star|_{\mathbf{T}_j})\big)$ ▷ sufficiency
15: $\quad y_j^n \leftarrow M.\text{run}\big(\mathbf{u}_j,\, \text{do}(\mathbf{C}_j = \mathbf{c}'_j,\, \mathbf{T}_j = \mathbf{w}^\star|_{\mathbf{T}_j})\big)$ ▷ necessity, shared $\mathbf{u}_j$
16: $\quad$ **for** each $X \in \mathbf{C}_j$ **do**
17: $\quad\quad$ append $|y_j^n - c^\star|$ to $L[X]$
18: $\quad$ **end for**
19: **end for**
20: **for** each $X \in \mathbf{S}$ **do**
21: $\quad \text{score}[X] \leftarrow \text{nanmean}(L[X])$
22: **end for**
23: **for** $i = 1, \dots, n$ **do**
24: $\quad v_i \leftarrow \arg\max\big(\text{score}[A_i], \text{score}[B_i]\big)$
25: **end for**
26: **return** $(v_1, \dots, v_n)$

The algorithm draws N intervention configurations under shared noise. The active suspect set $\mathbf{C}_j$ (line 6) is drawn at uniformly chosen cardinality $k_s \in \{1, \dots, n_s\}$; the dropped witness set D_j (line 7) is drawn the same way at $k_w \in \{0, \dots, n_w\}$, and the active witnesses on line 8 are the complement $\mathbf{W} \setminus D_j$ — so n_w bounds the number of witnesses *dropped*, not the number kept, and the active witness set has size $|\mathbf{W}| - n_w$ or larger. The witness ablation flag wit $=$ off is handled once on line 1 by emptying $\mathbf{W}$; this collapses the witness machinery on lines 7–8 and 11–12 to no-ops, producing the dotted accuracy curves of Figure 12.

Alternative values on line 9 are bit flips: on $\{0, 1\}$ the ϵ-excised alternative-value distribution has a single support point and ϵ plays no role; the explicit sampling form is preserved so that the same routine runs on continuous benchmarks. The two interventional worlds on lines 11–12 share the noise realisation $\mathbf{u}_j$ drawn on line 4 — the standard counterfactual coupling — which lets the per-suspect score on lines 14–16 be read off as the nan-mean of $|y_j^n - c^\star|$ over the draws in which $X \in \mathbf{C}_j$. The sufficiency outcome y_j^s is computed but not consumed: the AC verdict against which we benchmark has no sufficiency clause, so we read only the necessity world. The joint estimator with sufficiency active is exercised in Sections 5, 6, and 10, where AC has no comparable verdict to benchmark against. At each site $i = 1, \ldots, n$, line 17 returns the suspect — A_i or B_i — with the higher mean.

Benchmark protocol. For each problem size n we sample up to ten distinct factive worlds (with $C = 1$), instantiate the model, run both methods to verdicts, and compare to the analytic ground truth. Each method is granted a 60-minute global budget per run; if the budget is exhausted mid-loop, the largest completed n caps the curve. The approximate estimator is exercised in five configurations that vary along three axes identified in Section 3: (i) initial sample budget $N_0 \in \{250, 500\}$, scaled linearly with n by a factor $\alpha \in \{0.5, 1\}$ so that the total sample at size n is $\alpha \cdot N_0 \cdot n$; (ii) presence of witness interventions (wit $\in$ {on, off}), ablating $\mathbf{T} \leftarrow \mathbf{w}^\star$ to test whether witness pinning is load-bearing; (iii) cardinality bounds n_s, n_w, fixed at 4 in the default runs (answering "does some witness set of size $\geq |\mathbf{W}| - 4$ work?") and grown linearly as $4n$ in the dynamic-set-size configuration (recovering the unrestricted subset search at the cost of higher variance). The cardinality bound is a tunable approximation knob, not a hidden assumption: the dynamic-set-size run measures the empirical cost of bounding n_s, n_w at 4 on this problem, and the gap between the two configurations is what the ablation surfaces. The conceptual case for constraining cardinality at all — that uniform sampling over the full powerset biases attributions toward large-coalition interventions and is analogous to declining to enumerate all n-way regression interaction terms — is given in Section 3.

Results. Three observations summarise the experiment. First, runtime (Figure 10): the exact procedure times out at $4n + 1 = 17$ variables ($n = 4$) once the cause-witness search space crosses sixty-five thousand pairs, while the approximate methods continue to roughly 73–145 variables depending on sample budget, with the frugal α=0.5 run reaching the largest sizes. Second, search-space size (Figure 11, log scale): the exact method visits the full powerset, while the approximate methods visit at most a few hundred unique configurations per problem size — the gap is several orders of magnitude and widens with n. Third, accuracy (Figure 12): the exact method is correct by construction up to its early demise; the witness-using approximate methods stay above 0.85 correct-attribution rate up to roughly 60 variables and degrade gracefully beyond that, with the dynamic-set-size run sustaining among the highest accuracies at moderate problem sizes. Witness-free runs (dotted) underperform throughout, starting near 0.67 already at $n = 1$. The ablation confirms a structural claim of Section 3: pinning factual witness values is doing real work, not just regularising.

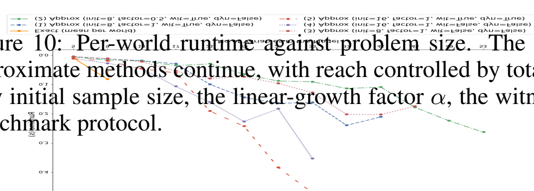


Figure 10: Per-world runtime against problem size. The exact method (orange) times out at 17 variables; approximate methods continue, with reach controlled by total sample budget. Five approximate configurations vary initial sample size, the linear-growth factor α, the witness ablation, and dynamic cardinality bounds; see Benchmark protocol.

Scope of the conclusion. The benchmark establishes that the necessity-only PCI estimator tracks AC verdicts on a faithful scaling of the canonical example with several orders of magnitude fewer evaluated configurations. The problem itself, however, is structured: responsibility attribution is independent across sites, exactly one of A_i, B_i is responsible at each site given C=1, supersets of any witness set witness the same attribution (so a small witness-cardinality bound costs little), and all variables are binary. These regularities account for some of the favourable scaling — in particular, the cardinality-bounded sampling losing little to dynamic-set-sizes sampling on this problem. Two extensions are natural follow-ups. First, a comparison against SHAP and causal SHAP on the same scaled-throwing problem is deferred to a companion benchmark (cf. Section 5 for the corresponding comparison on the continuous overdetermination example); the sufficiency component of PCI is what distinguishes it from those baselines, and a clean head-to-head requires the joint kernel rather than the necessity-only restriction used here. Second, benchmarks that break each of the structural regularities above — particularly continuous-variable extensions where AC's mechanical enumeration is no longer even well-defined — would test the estimator outside its current favourable regime. As a sanity check that the correspondence in Theorems 27–28 is empirically operational at scale, the present comparison is sufficient.

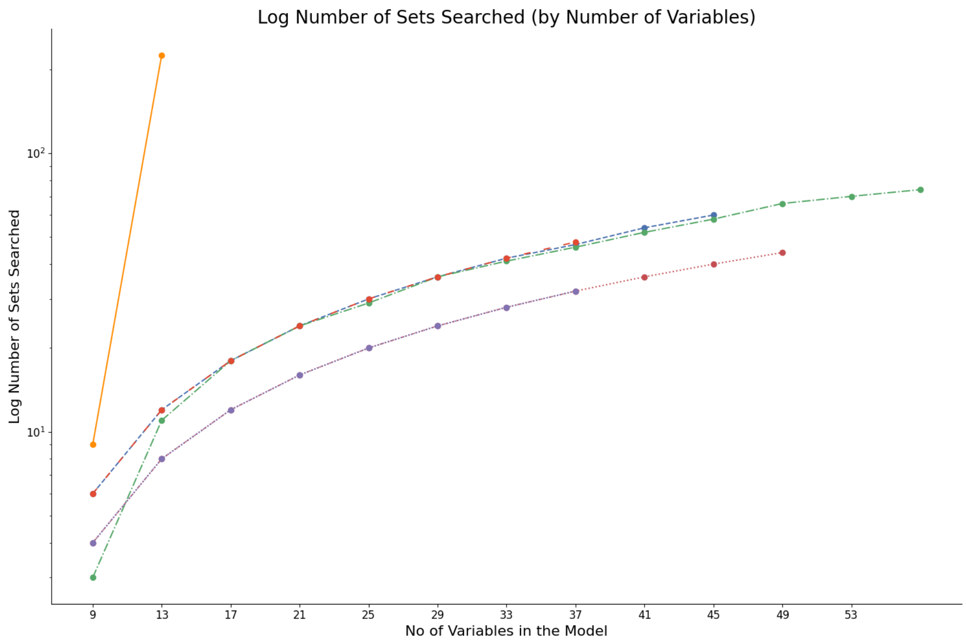


Figure 11: Number of cause-witness configurations explored per problem size, log scale. The exact method visits the full powerset; the approximate methods visit a few hundred unique configurations per problem size regardless of n — a gap of several orders of magnitude that grows with n.

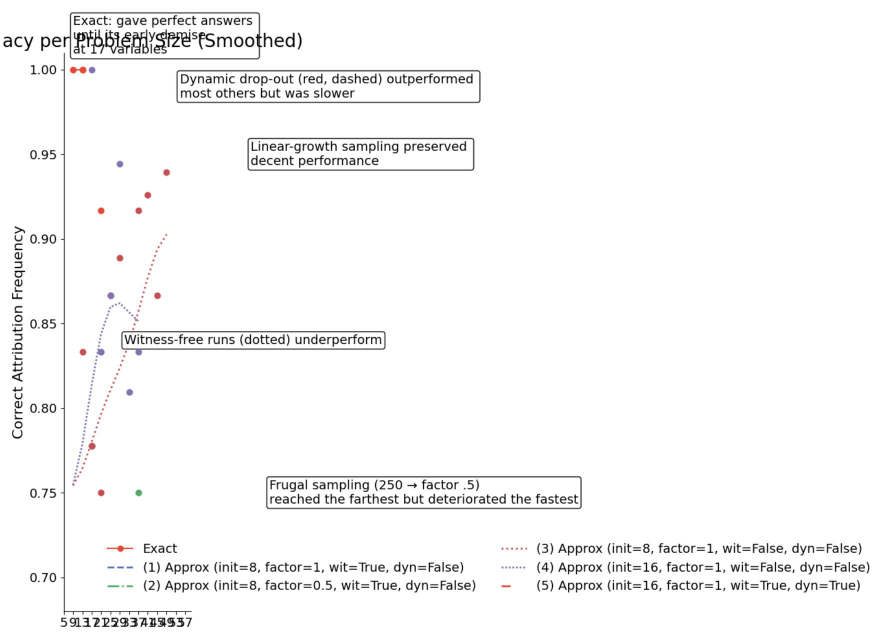


Figure 12: Correct-attribution rate against problem size, smoothed with a Gaussian filter. The exact method is correct up to 17 variables. Witness-using approximate runs stay above 0.85 up to roughly 60 variables and degrade gracefully thereafter; witness-free runs (dotted) underperform throughout, starting near 0.67. Dynamic-set-size sampling (dashed red) is among the highest at moderate problem sizes.

10 Dynamical SIR benchmark: PCI on continuous outcomes

The actual-causality benchmark of Section 9 exercises the necessity-only specialisation of PCI on a discrete outcome with a closed-form ground truth. This section shifts to the regime where AC has no verdict to compare against: a Bayesian dynamical model with a continuous outcome and asymmetrically interacting causes. The benchmark mirrors a public chirho tutorial[22] in setup and query, which uses chirho's `SearchForExplanation` handler to compute classical actual-cause probabilities. We keep the same model and the same query but replace the explanatory machinery with the PCI thin-search sampler. The full implementation, raw outputs and plotting code are in the companion notebook `docs/source/sir_benchmark.ipynb`.

Model. The dynamics are the standard SIR system $\dot{S} = -\beta SI$, $\dot{I} = \beta SI - \gamma I$, $\dot{R} = \gamma I$, with Bayesian priors $\beta \sim \mathrm{Beta}(18, 600)$ and $\gamma \sim \mathrm{Beta}(1600, 1600)$. Two non-pharmaceutical policies, each enacted with prior probability $1/2$, modulate the transmission rate via an intervention strength $l \in [0, 1]$ that scales β_0 to $(1 - l)\beta_0$. Lockdown alone has efficiency 0.6; masking efficiency depends on lockdown (0.1 under lockdown, 0.45 on its own); the joint efficiency is the clamped sum, capped at 0.95. Lockdown is enacted at $t = 1$, masking at $t = 1.5$. The outcome is the *overshoot* — the peak-to-final S gap of the simulated trajectory — and the undesirable event is $\mathrm{os\_too\_high} = \mathbb{1}[\mathrm{overshoot} > 24]$ on a 100-person population.

But-for analysis fails to separate the suspects. Conditioning on each pair of policy decisions and drawing 100 predictives yields $\Pr(\mathrm{overshoot} > 24) \approx 0.07$ with no intervention, 0.67 with both policies, but 0.82 with lockdown only and 0.87 with mask only. The mechanism is the asymmetric joint efficiency: combining the two policies caps the transmission reduction at 0.95, but neither policy alone reaches that cap, and mask-only is comparatively weak unless lockdown is absent (which shifts mask efficiency from 0.1 to 0.45). But-for cannot disentangle this context dependence: it treats each policy as a binary cause and reports an aggregate effect, with no way to express that lockdown's causal contribution depends on whether masking is in force. Figure 13 visualises the four interventional regimes: the trajectory bands collapse toward the unintervened baseline only under both policies, while either policy on its own leaves the infectious peak essentially intact, mirroring the near-equal but-for overshoot probabilities.

PCI thin-search procedure. The suspects are $\{\mathrm{lockdown}, \mathrm{mask}\}$. The witnesses are drawn from the policy efficiencies $\{\mathrm{lockdown\_efficiency}, \mathrm{mask\_efficiency}, \mathrm{joint\_efficiency}\}$, with the suspect itself excluded from the witness set on each draw. The factual world fixes both policies on ($\mathrm{lockdown} = 1, \mathrm{mask} = 1$), so the factual outcome $y^\star = \mathrm{overshoot}$ is read from the same trajectory. For each suspect the sampler draws 2,500 regimes, scoring each with the $|y^\star - y_{\mathrm{nec}}| - |y^\star - y_{\mathrm{suff}}|$ statistic and averaging across regimes. The nec term rewards regimes where removing the suspect moves the outcome *away* from $y^\star$; the suff term rewards regimes where fixing the suspect at factual keeps the outcome *close* to $y^\star$.

Single-world results. PCI assigns lockdown roughly $2.0\times$ the responsibility of mask:

suspect	$\overline{\lvert y^\star - y_{\mathrm{nec}}\rvert}$	$-\overline{\lvert y^\star - y_{\mathrm{suff}}\rvert}$	total
lockdown	8.12	−6.58	**1.54**
mask	7.43	−6.67	**0.76**

The gap is driven entirely by the necessity term: removing lockdown moves the overshoot further from $y^\star$ than removing mask does (8.12 vs 7.43 in mean absolute distance), while the sufficiency terms agree to within 0.1. This is consistent with the but-for finding that either policy *alone* is roughly equally bad. What distinguishes the suspects is that lockdown's removal *under context* (with witnesses pinned to factual) pulls the world further from $y^\star$ than mask's removal does. Figure 14 makes the necessity-term gap visible. Each point is one sampled regime, with y_{nec} on the horizontal axis (focal suspect intervened to alternative, other active suspects in $\mathbf{C} \setminus \{X\}$ intervened, witnesses at factual) and y_{suff} on the vertical axis (focal suspect at factual, other active suspects intervened, witnesses at factual). The dotted red cross-hair marks the factual outcome $y^\star$ at the both-policies-on world ($\mathrm{lockdown} = \mathrm{mask} = 1$); because that world is the high-overshoot scenario, $y^\star$ sits near the

[22] `https://basisresearch.github.io/chirho/explainable_sir.html`

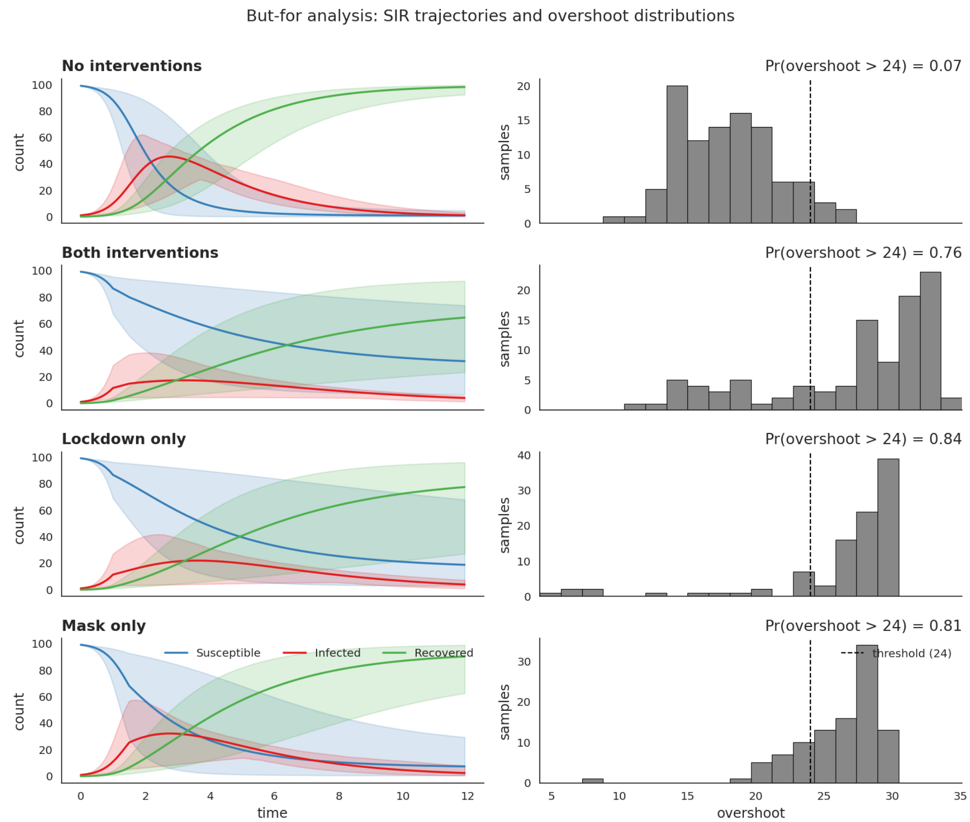


Figure 13: SIR trajectories and overshoot distributions under the four interventional regimes (no interventions, both policies, lockdown only, mask only). Left: posterior bands for susceptible (blue), infectious (red) and recovered (green) over time, with the lockdown enacted at $t = 1$ and the mask mandate at $t = 1.5$. Right: per-regime histograms of the overshoot, with the 24-overshoot threshold marked. Either policy on its own leaves the infectious peak essentially intact; only the joint regime suppresses it appreciably, yet the lockdown-only and mask-only overshoot probabilities are near-indistinguishable — the asymmetric joint efficiency that PCI disentangles.

upper-right of the cloud rather than at its centre, and a regime contributes positively to the score iff its point lies further from the cross-hair along x than along y. Under that geometry the lockdown density is shifted noticeably toward smaller y_{nec} at fixed y_{suff} — removing lockdown pulls y_{nec} well below $y^\star$ while sufficiency-side overshoot stays close to $y^\star$ — whereas the mask density sits closer to the $y_{\text{nec}} = y_{\text{suff}}$ diagonal, indicating that removing mask does not pull the world as far from $y^\star$.

Robustness across factual worlds. We replicate the analysis across 10 factual worlds drawn from the prior (rather than the single hand-set world above), running thin search on each and averaging. Lockdown's mean total exceeds mask's in the majority of worlds; the verdict is not an artifact of the chosen factual.

Effect of context. The chirho tutorial isolates one downstream variable at a time, asking whether the verdict survives when that variable is held fixed at factual versus left free. We replicate the experiment by partitioning the PCI regimes by whether the partner-policy efficiency variable was sampled into the witness set (Figure 15). For lockdown, with mask_efficiency *fixed* as a witness the score is 0.81; *free*, it rises to 2.67. For mask, with $\text{lockdown\_efficiency}$ *fixed* the score is -0.48 — mask fails to register as a cause at all under that context regime — and *free* it rises to 2.72. Lockdown therefore registers as a cause in both context regimes, while mask's status as a cause is contingent on its partner being allowed to adapt. PCI thus expresses not just *which* policy is the dominant cause but *how robustly* that causal claim holds across counterfactual contexts — a finer-grained verdict than the binary "actual cause / not" that classical AC produces.

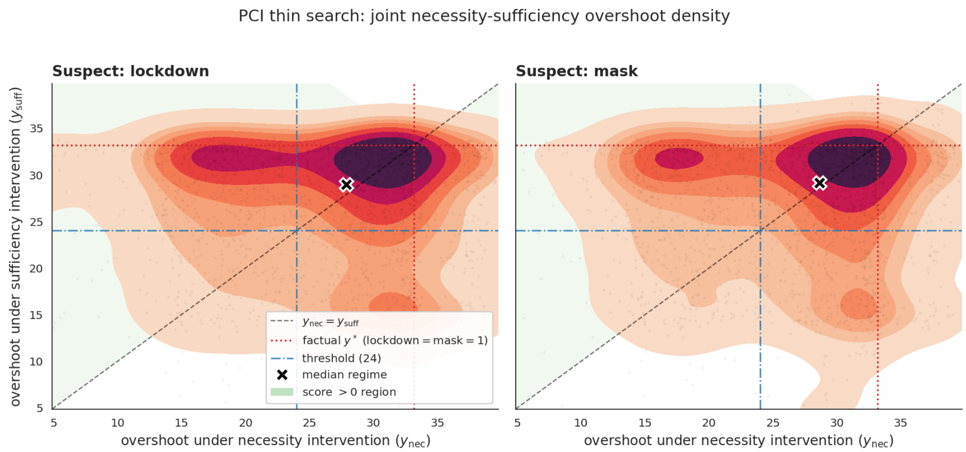


Figure 14: Joint necessity-sufficiency overshoot density per suspect. Each point is one sampled regime. Dotted red cross-hair: factual $y^\star$ at $\text{lockdown} = \text{mask} = 1$ (the high-overshoot anchor); dash-dot blue: the 24-overshoot threshold; dashed black: $y_{\rm nec} = y_{\rm suff}$. A regime contributes positively to the PCI total iff it sits further from the red cross-hair along the necessity axis than along the sufficiency axis. The lockdown density is shifted toward smaller $y_{\rm nec}$ relative to $y_{\rm suff}$ (the cause signature on this geometry); the mask density sits closer to the diagonal.

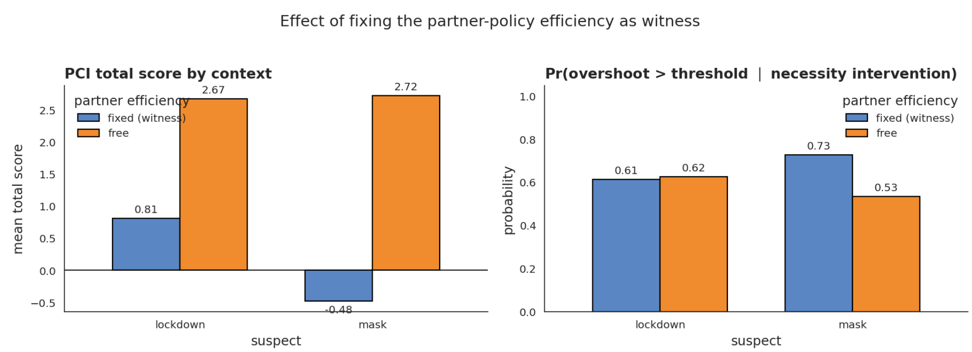


Figure 15: Different-contexts experiment. Bars compare PCI total score (left) and $\Pr(\text{overshoot} > 24 \mid \text{necessity})$ (right) when the partner-policy efficiency variable is fixed as a witness versus left free. Lockdown's score is positive in both regimes; mask's score flips sign across regimes, indicating that mask is a context-sensitive rather than a robust cause.

Takeaway. On the dynamical SIR setting, PCI recovers the qualitative diagnosis of the chirho tutorial — lockdown is the proximate cause of excessive overshoot, despite both policies appearing similar under but-for analysis — and expresses it on a continuous, decomposable scale. The necessity-vs-sufficiency decomposition distinguishes suspects whose isolation effects are similar but whose removal-under-context effects diverge; the contexts experiment further distinguishes robust from context-sensitive causes. The same machinery applies to any chirho probabilistic program with a continuous outcome and a tractable enumeration over interventional regimes.

11 Scaling PCI to a real-world automated valuation model

In this section, we apply PCI to a valuation model for residential housing. Disclosure constraints on the trained model and its training data prevent us from publishing a quantitative accuracy comparison against ground truth on this system; the quantitative validation of PCI therefore lives entirely in the controlled synthetic benchmark of Section 5, against an analytic ground truth that the production model does not admit. The role of this section is correspondingly narrow: to report (i) that PCI is computationally feasible at production scale on a model with millions of training points, and (ii) that the qualitative SHAP-vs-PCI divergence observed in Section 5 persists when the model is a real trained one. What is reportable here is the shape of the two attribution distributions and the structural pattern of where attribution mass lands, not the substantive content of the features they assign to.

Setting. The unit of analysis is a residential property transaction. PCI is run against the full country-level production AVM; what we show in this section, however, is an illustrative prototype based on the state of New York rather than the full national model structure, which we cannot disclose in detail. The NY prototype shares the architecture and the causal-kernel construction described below with the full system but is smaller in scope and DAG complexity. The computational-feasibility claim refers to the full national system; the qualitative SHAP-vs-PCI behaviour we illustrate is a property of the shared architecture and carries over from the prototype. The outcome variable y is the (log-epsilon-standardized) sale price. Transactions are described by a set of categorical variables $x_{cat} = x_1, \ldots, x_k$ and continuous (transformed) variables[23] $x_{con} = x_{k+1}, \ldots, x_n$.

Causal model. A causal DAG Γ expresses our causal assumptions about the relationships between variables. It was written manually in collaboration with domain experts and refined by adding edges to break implied conditional independencies that are not upheld by the data. The NY-prototype DAG has 25 nodes and 99 edges; topology with anonymised node labels is shown in Figure 16; the full national DAG is larger and is not shown. As demonstrated in Section 9, even the prototype is already large enough that exact actual-causality enumeration is computationally infeasible, and the same conclusion applies a fortiori to the national system.

Variables x_i are drawn from distributions Dist_i parameterized by functions $f_i(\mathrm{pa}_i, \varphi_i)$ of the parent variables pa_i[24]: $x_i \sim \mathrm{Dist}_i(f_i(\mathrm{pa}_i, \varphi_i))$. We call these distributions *causal kernels*. The distributions are either (truncated) normal or categorical, chosen based on the domain of the variable that they model. The functions f_i are deep neural networks with parameters φ_i. The parameters φ_i are optimized using cross-entropy loss for categorical distributions and average negative log likelihood for continuous distributions. The outcome is modeled by a sparse variational Gaussian Process (SVGP; Titsias, 2009, Hensman et al., 2013), $x_{gp} = (x_{con}, \eta(x_{cat}, \varphi_{cat}))$, $y \sim \mathcal{SVGP}(\mu(x_{gp}, \varphi_\mu), \kappa(x_{gp}, \varphi_\kappa))$, where η is a linear embedding of the categorical variables into the continuous space, jointly optimized with the GP.

Feasibility at production scale. To apply PCI we transform the model as described in Section 3 and evaluate the impact of features from a list of potential causal suspects, using the absolute difference impact score of Example 20. We draw Monte Carlo samples from the expanded model. Conditioning on a given suspect X_i being active gives us a sample of the scores corresponding to that feature, which we use to estimate the expected value thereof. The estimation converges at around $25{,}000$ samples; for a batch of 50 data points this takes around 60 minutes on commodity hardware. Sampling needs to be performed only once; the samples can then be reused to compute the attribution score of any suspect. This establishes that the method is computationally tractable at production scale — not a foregone conclusion, since the expanded-model construction in Section 3 introduces additional latent dimensions per suspect and per witness.

Qualitative comparison to SHAP. We compare PCI attributions to SHAP scores [Lundberg and Lee, 2017] on the same trained model. SHAP heavily weights variables that are downstream of many other variables in the causal graph, effectively assigning attribution to what amount to summary statistics of the outcome. PCI distributes attribution more evenly across the upstream causal structure. The pattern is consistent with what Section 5 establishes against ground truth: SHAP's observational marginalisation allocates weight differently from intervention-and-witness-based PCI scores, and on a deep DAG this difference manifests as concentration on summary-statistic-like nodes. Figure 17 shows the two attribution distributions with anonymised feature labels; what is reportable here is the shape of the two distributions, not which features they assign to. We therefore do not make a quantitative claim in this section — the quantitative claim is the synthetic benchmark of Section 5, and the role of this section is to show that the same qualitative shape persists when the model is a real trained one operating on millions of data points.

[23] Usually log-epsilon-standardized, standardized-truncated, or standardized.

[24] For root nodes, pa_i is empty.

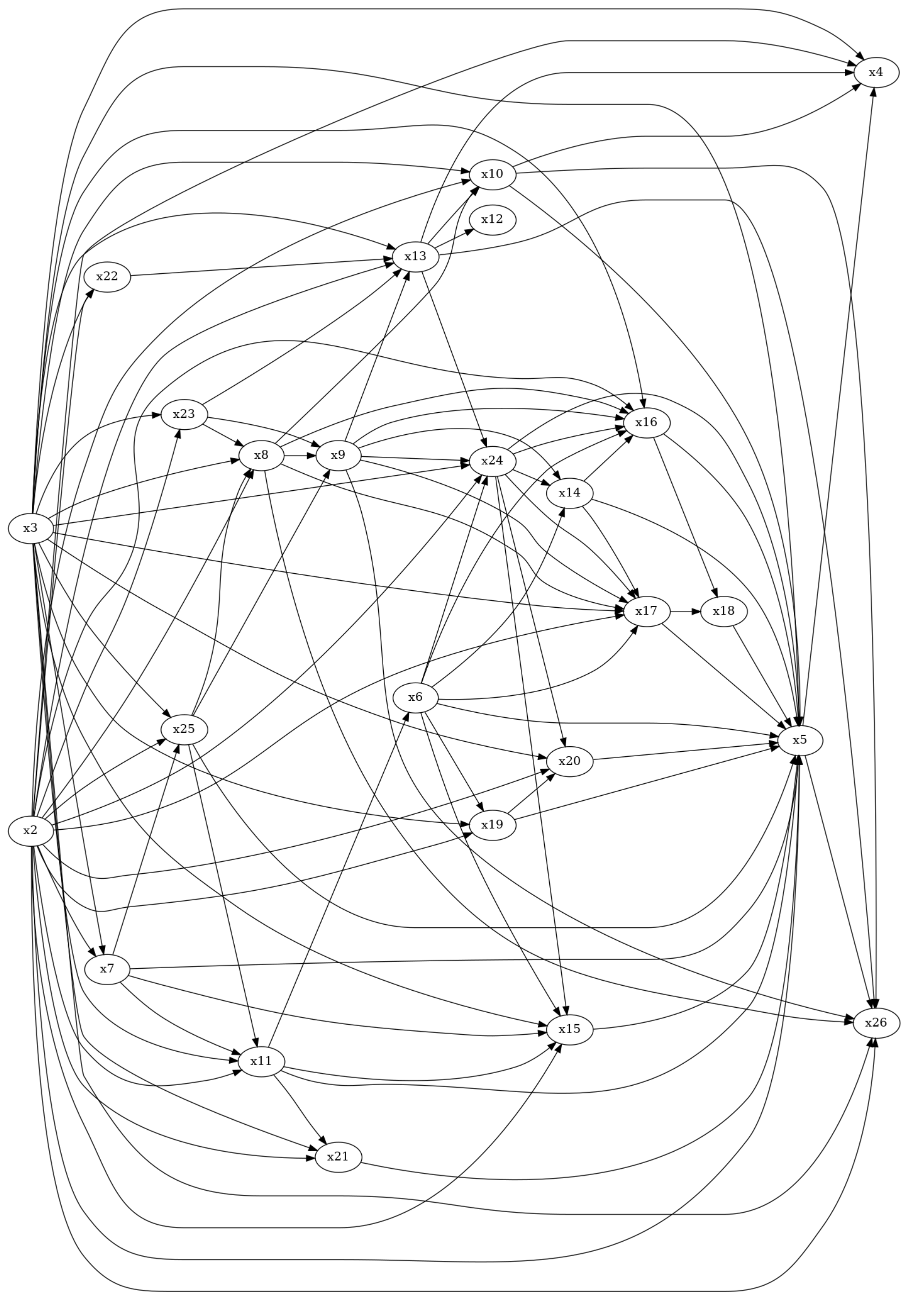
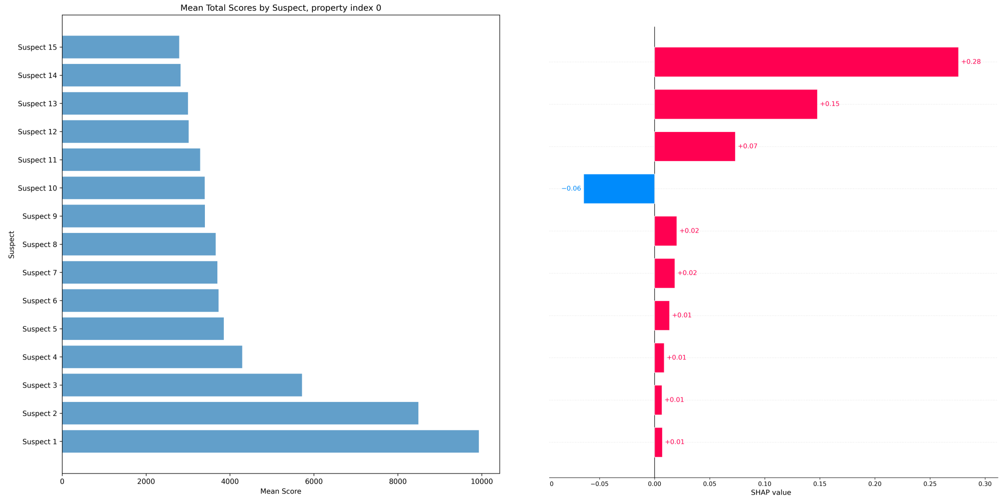


Figure 16: Topology of the AVM causal DAG (NY-prototype version), with node labels anonymised. The graph has 25 nodes and 99 edges; the full national DAG is larger and is not shown. Several variables sit deep in the DAG with high in-degree; these are the nodes towards which SHAP allocates disproportionate attribution mass (see Figure 17).



(a) PCI attribution mass.

(b) SHAP attribution mass.

Figure 17: Attribution distributions on the AVM, with feature labels anonymised. **Left:** PCI attribution mass spread across upstream causal variables (suspects 1–15, ranked by mean total score). **Right:** SHAP attribution mass concentrated on a small number of downstream variables. The qualitative pattern matches the controlled-setting comparison of Section 5.

12 Discussion and Conclusions

We are now in a position to articulate the main differences between PCI and the two approaches that inspired it, against the background of what the rest of the paper has shown. The main lesson from the literature on actual causality is that intuitive causal attribution often requires holding some endogenous variables — members of a "witness set" — fixed at their factual values. This ensures context sensitivity by controlling complicating causal paths. We treat witnesses somewhat differently: instead of looking for a single witness set that reveals a causal relationship, we use a distribution over witness sets and integrate the causal impact score across them. As in many probabilistic generalisations, this focuses on distributions rather than point estimates, and it lets PCI account for the number of witness sets and the strength of within-witness-set effects. Moreover, the construction admits off-the-shelf probabilistic inference rather than worst-case subset enumeration. The relationship between the two views is now formal rather than gestural: under matching choices of Γ and Δ, AC verdicts coincide with the support of $\sum_{\mathbf{T}} \Phi$ (Theorems 27–28 and Proposition 33), and the empirical companion in Section 9 confirms this at scales where AC enumeration is intractable.

The main concept borrowed from Pearl's probability of necessity and sufficiency is the joint distribution over outcomes from two interventional regimes, pushed forward through a user-chosen causal impact function. The key difference is that we use witness sets to allow for context-sensitivity and to break symmetries present in otherwise pathological cases like overdetermination and undercutting. Section 4 works through the binary instantiation in which the necessity and sufficiency components coincide with PN and PS, and Section 5 reads the two components apart on a continuous synthetic model where each is needed to surface a different causal archetype.

The further generalisations are mostly bookkeeping in the same spirit. Instead of fixing a single alternative value for a potential cause, we integrate over a factual-excised alternative-value distribution; this generalises immediately to continuous variables. Instead of asking whether a particular set is an actual cause, we sample suspect sets and gauge a feature's responsibility in terms of its participation in sets with causal impact. Instead of asking whether the outcome differs, a customisable causal impact function ci measures the differences across counterfactual worlds, including between continuous variables. The variable-selection distribution Γ, the alternative-value distribution Δ, and the impact function ci become explicit, user-facing components of the framework rather than implicit defaults.

Against SHAP-family attributions, the framework's central commitment is to *intervention with witnesses* rather than *observational marginalisation*. The controlled comparisons of Section 4 show that this commitment is load-bearing on canonical overdetermination and continuous-mediation examples, where SHAP and Causal SHAP fail in characteristic ways that PCI does not. The synthetic benchmark of Section 5 confirms this on a model with known ground truth, and the case study of Section 11 reports the same qualitative pattern on a production-grade trained model with millions of data points. A separate comparison to gradient-based attribution — where the contrast is differentiability and locality rather than the causal/non-causal axis — is reported in Section 6. The thread connecting these results is structural: the witness mechanism that satisfies D-A-rank for Alice in the binary OBCB walkthrough of Section 4 is the same machinery driving the upstream-versus-downstream attribution split on the AVM in Section 11.

A broader test is how PCI fares against the desiderata that the XAI literature typically asks counterfactual-explanation methods to meet, of which Verma et al. [2024] list five. *Sparsity* — the requirement that explanations change as few features as possible — follows directly from the cardinality cap on suspect sets in Γ, which biases the search toward small $\mathbf{C}$. *Multiple counterfactuals* — the ability to return several alternative explanations rather than commit to one — falls out of the Monte Carlo construction, since a single sweep over Γ and Δ produces a population of $(\mathbf{C}, \mathbf{c}', \mathbf{T})$ triples rather than a single optimum. *Causal-constraint satisfaction* — the demand that explanations respect downstream propagation rather than treat features as independently editable — is automatic once suspect interventions are routed through the causal model: changing $\mathbf{C}$ propagates to descendants by construction. *Model-agnosticism* — applicability to arbitrary predictors — PCI satisfies only partially: it can wrap a black-box predictor, but at the cost of the causal sensitivity that motivates the framework in the first place.

Actionability — the requirement that explanations not recommend changes to features users cannot in practice change — is where we part company with the literature. The standard recommendation is to fold ease-of-change into the cause-identification procedure itself, weighting features by mutability so that immutable ones are never returned as causes (Verma et al. [2024]; cf. Karimi et al. [2021], who shift the focus from explanations altogether to interventions). We think this conflates two distinct questions. A procedure that suppresses gender as a possible cause whenever a lease decision is at stake will help applicants game the existing system, but it will also prevent them from noticing the kind of structural problem that responsibility attribution can in principle surface. Cost-of-action layers can sit downstream of PCI for users who need actionable recommendations, but they should compose with responsibility attribution rather than replace it.

Several limitations are worth flagging. PCI presumes access to a trained probabilistic causal model from which counterfactual outcomes can be sampled — a stronger requirement than methods that work from observational data and a causal graph alone. The Monte Carlo variance of the estimator grows with model size and with the cardinalities of suspect and witness sets; Section 9 reports empirical scaling and an ablation that isolates the contribution of witness pinning, but sample-efficient estimators for high-cardinality settings remain an open direction. Finally, the AVM case study is deliberately limited in disclosure: the apples-to-apples SHAP-vs-PCI comparison rests on the synthetic benchmark of Section 5, with the production model serving as a feasibility witness rather than as quantitative evidence.

Where PCI falls short. Three caveats about PCI's reach deserve explicit discussion. *(i) AC mismatch under noise integration.* Observation 32 shows that integrating the kernel against $P_{\mathbf{U}}$ before maximisation can drop AC causes that are subset-minimal but not cardinality-minimal — the Γ-weighted ranking is governed by cardinality, while AC's minimality is governed by the subset relation. Proposition 33 provides the operational route to AC verdicts in such cases; Theorem 30 closes the gap only under *necessity dilution* (Definition 29), a regularity that fails in disjunctive overdetermination. *(ii) Cardinality knob versus sample efficiency.* The cardinality cap is a tunable hyperparameter rather than a hidden default, but it trades precision for variance: tight bounds shrink the search space and reduce Monte Carlo variance per iteration but exclude joint-cause sets that AC's exhaustive enumeration would surface; loose bounds restore generality at the cost of slower convergence. Problem families with diffuse joint causation need more generous settings than the canonical AC benchmark uses, and the cap is therefore worth reporting alongside results. *(iii) Cost of the PSCM-availability assumption.* Pointwise verdicts at the configuration level require a sampleable structural model. Methods that bound PN/PS/PNS from observational data and a causal graph alone — e.g. Tian and

Pearl's bounds — apply in settings where PCI cannot, and the stronger PSCM commitment should be weighed against this when only a graph and a dataset are available.

Taken together, PCI provides a scalable, context-sensitive, causal explanation method for causal probabilistic machine-learning models. A natural probabilistic generalisation of ideas already in the literature leads to a tractable estimator that can be paired with standard inference algorithms. We benchmarked the approach against exact actual-causality computations, characterised its qualitative behaviour on synthetic models with known ground truth and on canonical SHAP failure cases, and showed its computational feasibility on a production-grade model.

Responsibility attribution has not yet been effectively mechanised in software for large, complicated models, which makes the exact notions present in the literature infeasible to apply. PCI develops approximate causal attribution workable for users of models of non-trivial size, with the design choices — which variables are suspects, what counts as an alternative value, what counts as a difference — exposed to the user rather than baked in.

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